

Molecular Simulation on Quantum Computing Systems

Variational Ground-State Energy Surfaces and Their Reconstruction by Polynomial and Spline Interpolation

CSE 402: Numerical Analysis, Simulation and Modeling • Group 09 • Section A • Supervisor: Ishrat Jahan

 <https://github.com/QuitttCat/CSE-402-Numerical-Analysis-Simulation-and-Modeling-Project>

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Abstract

This report simulates the ground-state potential energy of eleven molecules with the Variational Quantum Eigensolver (VQE) and then asks a numerical-analysis question: given a few expensive samples of a smooth function, how well can it be filled in between them? We ran 23 scans, 692 independent variational optimisations over active spaces of six to ten qubits, and checked every energy against exact diagonalisation of the same active-space Hamiltonian. Fourteen of the 23 scans fall inside the 1.6 mHa chemical-accuracy threshold; the rest miss it because the optimiser ran out of budget on the deepest circuits or the ansatz could not represent the state, not because the variational principle failed. Each scan was reconstructed by piecewise-linear interpolation, a single global polynomial, and cubic or bicubic splines, all scored out of sample: leave-one-out cross-validation in one dimension, a half-grid holdout in two. Splines reduce the error of the linear baseline by a factor of about 2 to 15 on most scans and place equilibrium bond lengths within 0.001–0.09 Å of the reference from grids spaced 0.075–0.25 Å apart, roughly an order of magnitude of resolution for no extra quantum cost. One result ran against expectation: on LiH the global polynomial, which textbook theory says should collapse into Runge’s phenomenon, beats the cubic spline by a factor of 38. We show why, find the node spacing at which the failure does appear, and use the full campaign to decide which reconstruction method to trust and when.

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CCS Concepts

• **Computing methodologies** → Quantum mechanic simulation; • **Mathematics of computing** → Interpolation; • **Applied computing** → Chemistry; • **Hardware** → Quantum computation.

Keywords

Variational quantum eigensolver, potential energy surface, polynomial interpolation, cubic spline, Runge’s phenomenon, cross-validation, numerical simulation, quantum chemistry

1 Introduction and Base Paper Review

1.1 Background and motivation

The central quantity in computational chemistry is the ground-state electronic energy of a molecule as a function of where its nuclei sit. That function, the *potential energy surface* (PES), decides equilibrium geometries, reaction barriers and vibrational frequencies. Almost everything a chemist wants to predict comes from the shape of this surface, not from any single point on it.

Two facts make the PES a clean subject for a numerical-analysis project, and this report leans on both.

The first is that computing one point on the surface is a well-posed eigenvalue problem. Under the Born–Oppenheimer approximation the nuclei are held fixed, and the electronic Hamiltonian becomes a finite Hermitian matrix whose smallest eigenvalue is the energy we want. For the small active spaces used here, that matrix can be diagonalised exactly, so every approximate result in this report can be checked against ground truth, a comparison simulation work rarely gets to make.

The second is that each point is expensive, and gets more expensive as the system grows. The matrix has dimension $2^n \times 2^n$ in the qubit representation, so exact diagonalisation scales exponentially with the number of spin-orbitals kept. The Variational

Quantum Eigensolver (VQE) [10, 13] sidesteps that exponential storage cost by replacing diagonalisation with minimisation of a Rayleigh quotient over a family of trial states. It is approximate, it is why “quantum computing for chemistry” is an active research area, and, seen from the outside, it is simply an expensive black-box function $\theta \mapsto E(\theta)$ handed to a classical optimiser.

That expense is what turns building a PES into an interpolation problem. A one-dimensional scan of a bond length at 0.1 Å resolution over a 1.5 Å window costs sixteen variational optimisations. A two-dimensional surface at the same resolution on both axes costs the square of that. The most expensive single scan in this project, the BeH₂ surface, needed 110 optimisations; halving the grid spacing on both axes would have needed 440. Meanwhile, what a chemist actually wants from the surface, the location of the minimum, the curvature at the bottom of the well, the height of a barrier, is a property of the continuous function, not of the sampled points. Interpolation bridges that gap, and how well it bridges it is a numerical-analysis question with a measurable answer.

1.2 Base paper review

Citation. The base paper for this project is A. Peruzzo, J. McClean, P. Shadbolt, M.-H. Yung, X.-Q. Zhou, P. J. Love, A. Aspuru-Guzik and J. L. O’Brien, “A variational eigenvalue solver on a photonic quantum processor,” *Nature Communications* 5, 4213 (2014) [13], the paper that introduced the Variational Quantum Eigensolver.

Problem addressed. The standard quantum route to molecular ground-state energies, quantum phase estimation, needs long, deep, fully coherent circuits, far beyond what near-term devices can run. The base paper asks whether the same quantity can be obtained with only short quantum circuits, by moving most of the work to a classical computer.

Method. The paper splits the calculation into a hybrid loop. The quantum processor prepares a parametrised trial state and estimates the expectation value of each Pauli term of the qubit Hamiltonian; the classical computer sums the terms into an energy and hands it to a classical optimiser (Nelder–Mead in the paper), which proposes new parameters. The variational principle guarantees that every energy the loop reports is an upper bound on the true ground-state energy, so minimising it drives the trial state towards the ground state. Each energy evaluation needs only a short state preparation, at the price of many repeated measurements.

Demonstration and findings. The algorithm was run on a two-qubit photonic chip to compute the ground-state energy of the helium hydride ion HeH⁺ in a minimal basis over a range of bond lengths. The paper’s central claim is that variational hybrid algorithms trade circuit depth for repetitions, which makes molecular simulation a realistic target for small, noisy quantum hardware. McClean *et al.* [10] later formalised the theory, and Kandala *et al.* [8] introduced the hardware-efficient ansatz used in this report.

How this work builds on it. We implement the same loop (Algorithm 1) and rely on the same variational bound as a correctness check. We extend the base paper in four directions: from one two-qubit molecule to eleven molecules on six to ten qubits; from a single bond coordinate to bends, a dihedral twist and five two-dimensional

surfaces; from a device-specific parametrisation to UCCSD and hardware-efficient ansatzes driven by classical optimisers, including an exact parameter-shift gradient; and, most importantly for this course, from sampling a curve point by point to asking how accurately the curve can be reconstructed *between* the sampled points. The base paper samples the dissociation curve but never asks how to fill it in; that interpolation question is the numerical-analysis core of this report.

Related work. Our project plan also surveyed the broader VQE literature. The review of Tilly *et al.* [21] covers methods and best practices for the algorithm. Two open difficulties it highlights, the choice of a good ansatz and optimisation landscapes that flatten into barren plateaus [9] where gradient variance vanishes, motivated generative approaches such as the Generative Quantum Eigensolver [11], which trains a GPT-2 style model to propose circuits and was demonstrated on H₂, LiH, BeH₂ and N₂. This report does not pursue those directions. It takes the standard VQE as given and studies a question all of them leave untouched: what to do with the expensive samples once they exist.

1.3 Core objective

The core objective of this work is twofold. First, to implement the Variational Quantum Eigensolver as a numerical algorithm and verify it, point by point, against exact diagonalisation of the same active-space Hamiltonian across a broad campaign of molecules and geometric coordinates. Second, to treat the resulting potential energy surfaces as a reconstruction problem, a smooth function known only at a few expensive samples, and to measure strictly out of sample how well piecewise-linear, global polynomial and spline interpolation recover it, so as to decide which reconstruction method to trust, and when.

1.4 What this report covers

This report documents a full simulation campaign and the numerical analysis built on top of it.

Simulation. We implement VQE in two forms. A core library (src/) is built from scratch on CUDA-Q [12] and PySCF [19]: a hardware-efficient ansatz, a UCCSD ansatz, seven SciPy optimisers and a hand-written gradient-descent routine that computes its gradients exactly through the parameter-shift rule [17] rather than through finite differences. The 23-scan campaign itself is produced by per-molecule drivers that use Qiskit Nature’s UCCSD ansatz [15], an exact statevector estimator and SciPy’s SLSQP or COBYLA. Section 2.1 develops the theory and Section 3 the implementation.

Campaign. We run 23 scans over 11 molecules, H₂, LiH, BeH₂, H₂O, NH₃, N₂, O₂, F₂, H₂O₂, NO₂ and N₂O, covering bond stretches, bend angles, a dihedral twist, and five two-dimensional surfaces. Table 1 lists all of them. Every energy is checked against exact diagonalisation of the same active-space Hamiltonian (Table 38).

Reconstruction. We reconstruct every scan with piecewise-linear interpolation, a global polynomial, and a cubic or bicubic spline, and score each one out of sample, leave-one-out in 1-D, half-grid holdout in 2-D. Results are collected in Table 39 and discussed per molecule in Section 4.1.

Geometry recovery. We use the reconstructions to find equilibrium geometries at sub-grid resolution and compare them against experimental and model reference values (Table 40).

An honest negative result. On LiH the global polynomial does not show Runge’s phenomenon over the scanned interval, and beats the cubic spline by a factor of 38. We report this, explain why, and find the node density at which the expected failure does show up (Section 4.1.2).

1.5 What this project is not

Two things are worth clearing up before the rest of the report. First, none of the main results need physical quantum hardware: the quantum circuits are simulated exactly on a classical CPU by a statevector simulator. This is a numerical simulation project, and VQE appears here as a numerical algorithm, an ansatz, a cost function and an optimiser, not as a piece of hardware engineering. A supplementary demonstration that runs the same calculation on real IBM Quantum hardware is discussed in Section 4.4.5, mainly to show the gap between noiseless simulation and present-day devices.

Second, the absolute energies reported here are not meant to be chemically predictive. Every calculation uses a minimal STO-3G basis and a small complete active space. The energies are internally consistent and exactly checkable, but they are not converged to the basis-set limit, and the equilibrium geometries we recover deviate from experiment by amounts that belong to the *model*, not to the solver. We say explicitly, for every number, which reference it is being measured against.

1.6 How the rest of the report is organised

Section 2 is the methodology: the quantum-chemical and variational theory behind VQE (Section 2.1), then the interpolation methods and how we validate them, with pseudocode for the main algorithms (Section 2.2). Section 3 covers implementation and experiments: the software, with code excerpts (Section 3.1), the simulation setup and computational protocol (Section 3.2), and the parameter choices, datasets and benchmarks (Section 3.3). Section 4 presents the results molecule by molecule, with every figure placed next to the discussion it belongs to (Section 4.1), then the campaign-wide accuracy and reconstruction results (Sections 4.2 and 4.3) and the discussion that only makes sense across molecules, including a comparison with the base paper (Section 4.4). Section 5 gives the key takeaways, the limitations and the future work, and Section 6 explains how to obtain and run the code.

2 Methodology

2.1 Background: quantum chemistry and the variational method

2.1.1 The electronic structure problem. For a molecule with M nuclei of charge Z_A at positions $\mathbf{R}_A$ and N electrons at positions $\mathbf{r}_i$, the non-relativistic Hamiltonian in atomic units [5, 20] is

$$\hat{H} = -\sum_{i=1}^N \frac{\nabla_i^2}{2} - \sum_{i=1}^N \sum_{A=1}^M \frac{Z_A}{|\mathbf{r}_i - \mathbf{R}_A|} + \sum_{i<j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}, \quad (1)$$

where nuclear kinetic energy has been dropped and nuclear repulsion enters as an additive constant. This is the Born–Oppenheimer

approximation: a proton is roughly 1836 times heavier than an electron, so the electrons settle into equilibrium essentially instantly at any nuclear configuration, and the electronic energy can be computed at a fixed $\{\mathbf{R}_A\}$. The map

$$\{\mathbf{R}_A\} \mapsto E_0(\{\mathbf{R}_A\}) \quad (2)$$

is the potential energy surface this project samples and reconstructs.

The difficulty sits entirely in the last term of (1). Drop electron–electron repulsion and the problem separates into N independent one-electron problems. Keep it, and the exact wavefunction lives in a space whose dimension grows combinatorially with the number of electrons and basis functions.

2.1.2 Second quantisation and the active space. Expanding the wavefunction in a finite one-particle basis $\{\phi_p\}_{p=1}^K$ (here the minimal STO-3G set) and moving to second quantisation turns (1) into

$$\hat{H} = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{2} \sum_{pqrs} h_{pqrs} a_p^\dagger a_q^\dagger a_r a_s + E_{\text{nuc}}, \quad (3)$$

where $a_p^\dagger, a_p$ create and annihilate an electron in spin-orbital p , and the one- and two-body integrals h_{pq}, h_{pqrs} are computed classically by PySCF. The Hilbert space now has dimension 2^{2K} before any symmetry restriction, still exponential.

The standard remedy, used throughout this work, is the *complete active space* (CAS) approximation. Orbitals split into a set held doubly occupied, a set held empty, and an active set of n_{act} orbitals holding n_{el} electrons, where full configuration interaction is carried out. We write this as $(n_{\text{el}}e, n_{\text{act}}o)$. The choice trades accuracy for tractability; Table 1 records the active space used for every scan, ranging from (2e, 5o) for LiH to (10e, 6o) for F₂.

This is the single most consequential approximation in the report. It is why the computed equilibrium geometries differ from experiment by tenths of an Ångström in some cases, and it is also why exact diagonalisation stays feasible, which is why every variational result here can be checked.

2.1.3 Fermion-to-qubit mapping. To write (3) in terms of qubits, the fermionic operators must be mapped onto Pauli operators in a way that preserves the anticommutation relations $\{a_p, a_q^\dagger\} = \delta_{pq}$. The Jordan–Wigner transformation [7] does this with

$$a_p^\dagger = \left(\bigotimes_{k<p} Z_k \right) \otimes \frac{X_p - iY_p}{2}, \quad (4)$$

where the leading string of Z operators enforces antisymmetry under exchange. The Bravyi–Kitaev encoding [1] does the same job with $O(\log K)$ rather than $O(K)$ operator weight. Either way, the result is a qubit Hamiltonian

$$\hat{H} = \sum_j c_j \hat{P}_j, \quad \hat{P}_j \in \{I, X, Y, Z\}^{\otimes n_q}, \quad (5)$$

a weighted sum of Pauli strings with real coefficients c_j . Every energy evaluation in this project is an expectation value of the form (5), and the qubit counts n_q in Table 1 (six, eight or ten) are twice the number of active spatial orbitals minus two: the parity mapping used by the experiment drivers removes two qubits by exploiting the conserved particle-number parities.

2.1.4 The variational principle. The whole VQE algorithm rests on one fact from linear algebra, which a numerical-analysis reader already knows as the Rayleigh quotient bound. For any normalised state $|\psi\rangle$,

$$\langle\psi|\hat{H}|\psi\rangle \geq E_0 = \min \text{spec}(\hat{H}), \quad (6)$$

with equality only if $|\psi\rangle$ is a ground eigenvector. The proof is short: expand $|\psi\rangle = \sum_k c_k |\phi_k\rangle$ in the eigenbasis, so $\langle\psi|\hat{H}|\psi\rangle = \sum_k |c_k|^2 E_k \geq E_0 \sum_k |c_k|^2 = E_0$.

Equation (6) turns an eigenvalue problem into a minimisation problem, and minimisation is something numerical analysis knows how to handle. Given a family of trial states $|\psi(\theta)\rangle$ that we can cheaply prepare and measure,

$$E(\theta) = \langle\psi(\theta)|\hat{H}|\psi(\theta)\rangle, \quad (7)$$

minimising E over θ gives the best estimate of E_0 available inside that family. Equation (6) also guarantees the estimate is an *upper bound*: the optimiser can never report an energy below the true ground state, so any energy that comes out lower than the exact reference signals a bug, not a discovery. We use this one-sided error as a correctness check throughout the report.

2.1.5 Ansatz design. The trial-state family is the *ansatz*. Choosing one is the same trade-off as choosing a polynomial degree in Section 2.2: expressive power against optimisation difficulty. We implement two ansatz families.

Hardware-efficient ansatz. The hardware-efficient ansatz [8] prepares the Hartree–Fock reference, then applies d layers of an independent R_y rotation on every qubit followed by a CNOT ladder:

$$|\psi(\theta)\rangle = \prod_{\ell=1}^d \left[U_{\text{CNOT}} \prod_{q=1}^{n_q} R_y(\theta_{\ell q}) \right] |\psi_{\text{HF}}\rangle. \quad (8)$$

The parameter count is exactly $p = n_q d$. The rotations give it flexibility; the entangling ladder gives it the correlation a Hartree–Fock product state cannot represent on its own. This family knows nothing about the chemistry it is fitting: it is cheap and shallow, and it comes with no guarantee of containing a good approximation to the true ground state.

UCCSD ansatz. The unitary coupled-cluster singles-and-doubles ansatz is chemically motivated:

$$|\psi(\theta)\rangle = e^{\hat{T}(\theta) - \hat{T}^\dagger(\theta)} |\psi_{\text{HF}}\rangle, \quad \hat{T} = \hat{T}_1 + \hat{T}_2, \quad (9)$$

where $\hat{T}_1$ and $\hat{T}_2$ generate single and double excitations out of the Hartree–Fock determinant. Its parameter count is not a free choice: it is fixed by the number of symmetry-allowed excitations, which is why the p column of Table 1 ranges so widely, from 20 parameters for NO_2 in a (5e, 4o) space to 117 for N_2 in (6e, 6o). UCCSD is far more expressive than the hardware-efficient ansatz, but its circuits are deeper and its optimisation landscape harder, which shows up directly in the accuracy results of Section 3.2.

2.1.6 Classical optimisation and the parameter-shift rule. From the optimiser’s point of view, $E(\theta)$ is just an expensive black-box scalar function. The codebase wires up eight optimisers in total. Three are gradient-free methods from `scipy.optimize` [23], COBYLA [14], Nelder–Mead and Powell; four more are gradient-based or quasi-Newton methods from the same library, BFGS, L-BFGS-B, CG and

SLSQP, which by default estimate their gradients by finite differences. Which one was used for each scan is recorded in Table 1.

The eighth optimiser is a gradient-descent routine we wrote from scratch, for a reason specific to this course: the gradient of E can be computed *exactly*, with no finite-difference truncation error. Because each θ_i enters the circuit only through a rotation $\exp(-i\theta_i \hat{P}/2)$ with $\hat{P}^2 = I$, we have the *parameter-shift rule* [17]

$$\frac{\partial E}{\partial \theta_i} = \frac{1}{2} \left[E(\theta + \frac{\pi}{2} \mathbf{e}_i) - E(\theta - \frac{\pi}{2} \mathbf{e}_i) \right]. \quad (10)$$

This looks like a central difference, but the step is a fixed, macroscopic $\pi/2$, and the result is *exact*, not $\mathcal{O}(h^2)$. There is no truncation error to trade off against floating-point cancellation, so the usual optimal-step-size analysis simply does not apply. It is a rare case in which a numerical derivative carries no discretisation error at all. Its cost is $2p$ energy evaluations per gradient, which is why COBYLA remains the practical default for production scans.

Putting the last three subsections together gives the full VQE loop that every scan in this report runs at every geometry, summarised in Algorithm 1. Steps 3 and 4 are the only quantum part; everything else is classical bookkeeping around a scalar minimisation.

Algorithm 1 Variational Quantum Eigensolver at one fixed geometry

Require: qubit Hamiltonian $\hat{H} = \sum_j c_j \hat{P}_j$, ansatz $|\psi(\theta)\rangle$ with p parameters, optimiser, tolerance ε

Ensure: estimate $E^* \geq E_0$ of the ground-state energy

- 1: $\theta \leftarrow$ initial guess (Hartree–Fock reference, all rotation angles zero)
- 2: **repeat**
- 3: prepare $|\psi(\theta)\rangle$ on the (simulated) circuit
- 4: measure $E(\theta) = \sum_j c_j \langle\psi(\theta)|\hat{P}_j|\psi(\theta)\rangle$
- 5: **if** optimiser needs a gradient **then**
- 6: **for** $i = 1, \dots, p$ **do**
- 7: $g_i \leftarrow \frac{1}{2} [E(\theta + \frac{\pi}{2} \mathbf{e}_i) - E(\theta - \frac{\pi}{2} \mathbf{e}_i)]$ ▷ parameter-shift rule, exact
- 8: **end for**
- 9: **end if**
- 10: $\theta \leftarrow$ optimiser step using $E(\theta)$ (and g , if used)
- 11: **until** $|E(\theta) - E_{\text{prev}}| < \varepsilon$ or the evaluation budget is spent
- 12: **return** $E^* \leftarrow E(\theta)$

A full scan repeats Algorithm 1 once per sampled geometry, building up exactly the small, expensive, unevenly reliable data set that Section 2.2 then has to reconstruct.

Optimiser choice interacts with circuit depth in a way that shows up throughout the campaign. `NEW_MOLECULE_PROMPT.md`, the project’s own notes for extending the pipeline to a new molecule, records a concrete comparison on O_2 ’s hardest geometry: tripling COBYLA’s evaluation budget improved the energy by 0.6 mHa (1.7%), while warm-starting the optimiser from the previous geometry’s converged parameters improved it by 4.0 mHa, roughly *seven times* more, for no extra cost at all. More budget helps a stuck optimiser a little; a better starting point helps it a lot. Both tools appear in the codebase, and Section 4.4 returns to this trade-off when it explains which scans do and do not reach chemical accuracy.

2.1.7 The reference calculations. Three energies are recorded at every geometry, and keeping them apart matters for reading every table in this report.

- E_{HF} , the Hartree–Fock mean-field energy. Each electron moves in the averaged field of the others, and correlation is absent entirely. This is the starting point the variational calculation is meant to improve on.
- E_{VQE} , the variational result: the quantity under test.
- E_{exact} , exact diagonalisation of the identical active-space Hamiltonian, computed with `numpy.linalg.eigvalsh`. This is ground truth for the model, not for the molecule.

Two error measures follow:

$$\delta = E_{\text{VQE}} - E_{\text{exact}} \quad (\text{variational error, } \geq 0), \quad (11)$$

$$E_{\text{corr}} = E_{\text{HF}} - E_{\text{exact}} \quad (\text{correlation energy}). \quad (12)$$

E_{corr} is what the whole exercise is trying to recover; δ is how much of it VQE missed. A run is only doing useful work when $\delta \ll E_{\text{corr}}$, and Table 38 reports both side by side so that ratio can be read off directly. The conventional target is *chemical accuracy*, $\delta < 1.6 \text{ mHa}$ (1 kcal mol^{-1}), the threshold below which errors in predicted reaction rates become tolerable.

2.2 Numerical methods for surface reconstruction

Once we have n expensive samples $\{(x_i, y_i)\}_{i=1}^n$ of a smooth function, y_i an energy and x_i a geometric coordinate, the rest of the project is a classical reconstruction problem: fill in the function between the samples. This section covers the three methods we use and, just as important, how we grade them.

2.2.1 The piecewise-linear baseline. Every raw scan plot in this project joins neighbouring points with straight lines. That default is itself an interpolation scheme, and the crudest one available:

$$L(x) = y_i + \frac{y_{i+1} - y_i}{x_{i+1} - x_i} (x - x_i), \quad x \in [x_i, x_{i+1}]. \quad (13)$$

Its weaknesses matter because they motivate everything that follows. L is continuous but not differentiable at the nodes; its second derivative is zero on every interval and undefined at every node, so it carries no curvature information; and the minimum it implies is always exactly one of the sampled points. A piecewise-linear reconstruction can never locate an equilibrium geometry more finely than the grid spacing, which for these scans is $0.1\text{--}0.25 \text{ \AA}$, one to two orders of magnitude coarser than the precision chemists usually quote. For equally spaced nodes with spacing h , the classical bound is

$$\|f - L\|_{\infty} \leq \frac{h^2}{8} \|f''\|_{\infty}, \quad (14)$$

so halving the grid quarters the error, but doubles a cost that is already quadratic on a surface. That is exactly the wrong direction, and it is why a better interpolant is worth having. The *Linear* columns in Table 39 report this baseline throughout.

2.2.2 Global polynomial interpolation. Given n distinct nodes, exactly one polynomial of degree at most $n - 1$ passes through all of them [2, 22]. In Lagrange form,

$$P_{n-1}(x) = \sum_{i=1}^n y_i \ell_i(x), \quad \ell_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}. \quad (15)$$

On paper this looks like the ideal answer: one smooth closed-form curve, infinitely differentiable, exact at every node, with no piecewise bookkeeping. The classical error theorem gives, for $f \in C^n$,

$$f(x) - P_{n-1}(x) = \frac{f^{(n)}(\xi)}{n!} \prod_{i=1}^n (x - x_i) \quad (16)$$

for some ξ in the interval. The $1/n!$ factor is reassuring; the factors $f^{(n)}(\xi)$ and the node polynomial are not. For equally spaced nodes the node polynomial grows sharply near the ends of the interval, and $f^{(n)}$ can grow faster than $n!$ shrinks.

Runge’s phenomenon. The resulting failure is Runge’s phenomenon [16]: on equally spaced nodes, a high-degree interpolant can oscillate violently between nodes, and the oscillation gets worse as more nodes are added. Runge’s own example, $f(x) = 1/(1 + 25x^2)$ on $[-1, 1]$, diverges in the supremum norm as $n \rightarrow \infty$.

Two points matter for reading our results correctly, and Section 4.1.2 comes back to both of them.

First, a zero fit residual is not evidence of accuracy. The interpolant passes through every node by construction, so the in-sample residual is zero to machine precision, our runs record maxima as low as $1.4 \times 10^{-11} \text{ mHa}$, while the interpolant may be doing anything at all between nodes. Any honest check has to be out of sample.

Second, Runge’s phenomenon is a property of the function being interpolated, not of the degree by itself. It shows up when the function has a singularity in the complex plane close to the real interval; Runge’s example has poles at $\pm i/5$. A function that stays analytic over a wide neighbourhood of the interval is approximated very well by high-degree polynomials. Whether a given PES falls into the good case or the bad case is an empirical question, and we answer it separately for each scan.

Degree sweeps. Because the best degree is not known in advance, each 1-D scan is fitted at every degree from 1 up to $n - 2$ and scored out of sample. The resulting curves, the `*_degree_sweep.png` figures throughout Section 4.1, tabulated in the `tab:deg:*` tables, show error falling as the model gains the flexibility to follow the function, then, in the Runge regime, rising again as it gains the flexibility to oscillate. The minimum of that curve is the degree we actually use.

2.2.3 Cubic spline interpolation. The spline resolves the same tension differently: instead of one high-degree polynomial over the whole interval, fit a separate cubic on each subinterval and force the pieces to join smoothly. On $[x_i, x_{i+1}]$,

$$S_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3, \quad (17)$$

subject to the interpolation conditions $S_i(x_i) = y_i$, $S_i(x_{i+1}) = y_{i+1}$ and, at every interior node, the continuity conditions

$$S'_{i-1}(x_i) = S'_i(x_i), \quad S''_{i-1}(x_i) = S''_i(x_i), \quad (18)$$

[3]. These constraints fix all coefficients up to two boundary conditions (here the not-a-knot condition, SciPy’s default). The resulting linear system is tridiagonal and diagonally dominant, so it solves in $O(n)$ operations and is unconditionally stable.

The structural difference from (15) that matters is *locality*: each cubic piece is determined mostly by nearby data, so a perturbation at one node decays geometrically along the spline rather than

propagating globally. There is no mechanism for a Runge-style blow-up, and the classical bound

$$\|f - S\|_\infty \leq \frac{5}{384} h^4 \|f^{(4)}\|_\infty \quad (19)$$

is $O(h^4)$ rather than the $O(h^2)$ of (14): halving the grid cuts the error by sixteen. Because the spline is C^2 , it also has a meaningful second derivative, which is what makes estimating vibrational frequencies and barrier curvature possible at all.

2.2.4 Two-dimensional reconstruction. The five surface scans need 2-D versions of all three methods.

Bilinear. The straight-line baseline generalises to bilinear interpolation on a rectangular grid: interpolate along one axis, then along the other. It inherits every weakness of its 1-D counterpart.

Bicubic spline. The tensor-product cubic spline

$$S(x, y) = \sum_{i=0}^3 \sum_{j=0}^3 c_{ij} (x - x_k)^i (y - y_l)^j \quad (20)$$

is fitted on the rectangular grid, cubic in each direction, with SciPy’s RectBivariateSpline. The spline order on each axis drops automatically when that axis has too few points to support a cubic, a real constraint on the 10×5 H₂O₂ grid.

Total-degree polynomial. The 2-D analogue of the polynomial fits

$$P(x, y) = \sum_{i+j \leq d} c_{ij} x^i y^j, \quad (21)$$

which has $\binom{d+2}{2}$ coefficients. Unlike the 1-D case this is generally a *least-squares* fit rather than an exact interpolant, because a rectangular grid of N points does not in general admit an exact total-degree interpolant. The in-sample residual is therefore nonzero, and we report it alongside the out-of-sample error, a distinction that turns out to matter for N₂O and NO₂.

2.2.5 Validation methodology. In-sample residuals tell us nothing about an interpolant, so every reconstruction error in this report is measured out of sample.

Leave-one-out cross-validation (1-D). For each interior node i , we rebuild the interpolant on the remaining $n - 1$ points and evaluate it at the withheld x_i :

$$e_i = \mathcal{I}_{\setminus i}(x_i) - y_i. \quad (22)$$

Algorithm 2 states this precisely. Endpoints are excluded: removing an endpoint turns interpolation into extrapolation, a different and harder question. Nodes where the remaining data cannot support the requested degree are excluded too and marked n/a. This is why the fully interpolating degree $n - 1$ often cannot be scored at all, and why the tables report a *best* degree from the sweep instead.

Half-grid holdout (2-D). Leave-one-out does not work on a grid interpolant: remove one cell and the grid stops being rectangular, so RectBivariateSpline cannot be built on what remains. Instead, we keep every other value on each axis, which keeps the subgrid rectangular while withholding about three quarters of the points. The interpolant is built on the coarse subgrid and scored on everything held out, as in Algorithm 3.

Algorithm 2 Leave-one-out cross-validation at the interior nodes

Require: nodes $x_1, \dots, x_n$, values $y_1, \dots, y_n$, polynomial degree d

Ensure: error e_i at every interior node $i = 2, \dots, n - 1$

```

1: for  $i = 2, \dots, n - 1$  do
2:    $X \leftarrow \{x_1, \dots, x_n\} \setminus \{x_i\}$ ,  $Y \leftarrow \{y_1, \dots, y_n\} \setminus \{y_i\}$ 
3:   if  $|X| \geq d + 1$  then
4:     fit degree- $d$  polynomial  $P$  to  $(X, Y)$ 
5:      $e_i^{\text{poly}} \leftarrow P(x_i) - y_i$ 
6:   else
7:      $e_i^{\text{poly}} \leftarrow \text{n/a}$ 
8:   end if
9:    $e_i^{\text{linear}} \leftarrow \text{linear\_interp}(X, Y, x_i) - y_i$ 
10: end for
```

Algorithm 3 Half-grid holdout for a 2-D surface

Require: full grid $Z[i, j]$ over axes $x_1, \dots, x_m$ and $y_1, \dots, y_k$

Ensure: error at every held-out grid point

```

1:  $B \leftarrow \{1, 3, 5, \dots\}$ ,  $A \leftarrow \{1, 3, 5, \dots\}$   $\triangleright$  every other index on each axis
2: build coarse grid from  $x[B]$ ,  $y[A]$ ,  $Z[B, A]$ 
3: fit bicubic spline  $S$  and bilinear interpolant  $\ell$  on the coarse grid
4: for each  $(i, j)$  not in  $B \times A$  do
5:    $e_{ij}^{\text{spline}} \leftarrow S(x_i, y_j) - Z[i, j]$ 
6:    $e_{ij}^{\text{linear}} \leftarrow \ell(x_i, y_j) - Z[i, j]$ 
7: end for
```

This holdout answers the question that actually matters for an N^2 -cost campaign: *if we had run only half the grid on each axis, at a quarter of the quantum cost, how much would we have lost?* A good answer is what licenses coarse grids on molecules too expensive to sample densely, and it is the single most practically useful measurement in this report.

Error metrics. All errors are reported in milli-Hartree over the withheld set S :

$$\max = \max_{i \in S} |e_i| \times 10^3, \quad (23)$$

$$\text{RMS} = \left(\frac{1}{|S|} \sum_{i \in S} e_i^2 \right)^{1/2} \times 10^3, \quad (24)$$

$$\text{MAE} = \frac{1}{|S|} \sum_{i \in S} |e_i| \times 10^3. \quad (25)$$

We rank by RMS because it penalises the isolated large excursions typical of polynomial failure, which MAE would water down.

2.2.6 Locating the minimum. We extract the equilibrium geometry from a reconstruction by dense sampling to bracket the minimum, then a bounded scalar minimiser. The search is kept inside the data range on purpose: outside it, a high-degree polynomial diverges quickly, and any minimum found there would just be an artefact of extrapolation. We also count the number of interior minima of the fitted polynomial, since a physically sensible curve over these intervals should have exactly one; a count above one is direct evidence of spurious oscillation. Results are in Table 40.

3 Implementation and Experiments

3.1 Software architecture

The codebase is roughly 24,400 lines of Python in three layers: a molecular-modelling and VQE core, a per-molecule experiment

layer, and a numerical post-processing layer. This section covers the design decisions that affect the results we report.

3.1.1 Layer 1: core (src/). molecule.py (1887 lines) wraps NVIDIA CUDA-Q Solvers' `create_molecule` [12]. It runs the classical pre-computation, Hartree-Fock, MP2, CASCI, CASSCF, CCSD and FCI through PySCF [19], and exposes the qubit Hamiltonian together with energies, one- and two-body integrals, qubit counts, the Pauli decomposition, an exact-diagonalisation reference, the Hartree-Fock reference state and UCCSD operator pools. It also renders molecules three different ways: through py3Dmol, through Plotly, and through a dependency-free 3Dmol.js HTML page that needs nothing but a browser. A `visualize()` helper picks a backend automatically, py3Dmol inside a notebook, Plotly otherwise, or a standalone HTML file if neither is available, and these are the renderings shown alongside the energy curves in the live scan windows. The four generated files in `out/` (`caffeine_ball_and_stick.html`, `benzene_spacefill.html`, `h2o_ball_and_stick.html` and `nh3_plotly.html`) are checked-in examples of exactly this: two standalone 3Dmol.js pages, one HTML export from py3Dmol, and one Plotly export, produced by `examples/molecule_demo.py`.

`vqe.py` (740 lines) implements the algorithm of Section 2.1, including both ansatzes, all eight optimisers, and the parameter-shift gradient. Listing 1 shows the hardware-efficient ansatz: the Hartree-Fock reference is prepared by flipping the first n_{el} qubits, after which each layer applies one independent R_y rotation per qubit and a nearest-neighbour CNOT ladder.

```
1 def build_hardware_efficient_ansatz(
2     n_qubits, n_electrons, depth=2):
3     n_params = n_qubits * depth
4
5     @cudaq.kernel
6     def kernel(thetas: list[float]):
7         qubits = cudaq.qvector(n_qubits)
8         # (a) Hartree-Fock reference
9         for i in range(n_electrons):
10             x(qubits[i])
11         # (b) 'depth' layers of rotations + entanglers
12         idx = 0
13         for _layer in range(depth):
14             for q in range(n_qubits):
15                 ry(thetas[idx], qubits[q])
16                 idx += 1
17             for q in range(n_qubits - 1):
18                 x.ctrl(qubits[q], qubits[q + 1])
19
20     return kernel, n_params
```

Listing 1: Hardware-efficient ansatz. The parameter count is exactly $n_{\text{qubits}} * \text{depth}$; increasing depth enlarges the trial-state family at the cost of a harder optimisation, the same flexibility-versus-conditioning trade-off as raising a polynomial degree.

Listing 2 shows the hand-written optimiser implementing the parameter-shift rule of (10) and Algorithm 1. We kept it in the codebase on purpose, so the gradient computation is visible rather than hidden inside a library call.

```
1 theta = np.array(theta0, dtype=float)
2 p = len(theta)
3 prev_energy = cost(theta)
4 for k in range(max_iterations):
```

```
5     grad = np.empty(p)
6     for i in range(p):
7         shift_vec = np.zeros(p)
8         shift_vec[i] = shift # pi/2, exact
9         grad[i] = 0.5 * (cost(theta + shift_vec)
10                        - cost(theta - shift_vec))
11     theta = theta - learning_rate * grad
12     energy = cost(theta)
13     if abs(prev_energy - energy) < tol:
14         prev_energy = energy
15         break
16     prev_energy = energy
```

Listing 2: Gradient descent with parameter-shift gradients. The inner loop costs $2p$ energy evaluations per gradient, which is why COBYLA is the practical default for production scans.

Two short scripts under `examples/` exercise this layer end to end without touching any single molecule's experiment folder: `molecule_demo.py` builds caffeine, benzene, water and ammonia and renders each one through all three visualisation backends, and `vqe_demo.py` runs H_2 three ways back to back, UCCSD with COBYLA, UCCSD with the hand-written gradient descent, and the hardware-efficient ansatz with COBYLA, plotting the live loss curves and printing a side-by-side comparison. They are the fastest way to see every ansatz, optimiser and visualisation backend work without running a full molecular scan.

One platform note affects reproducibility: `cudaq-solvers` is declared in `requirements.txt` as a Linux/WSL-only dependency (`sys_platform != "win32"`), so the `src/` core and the demo scripts need WSL or a Linux machine. The campaign drivers use Qiskit and PySCF instead (Section 3.3) and do not depend on it.

3.1.2 Layer 2: experiments (experiment/). Each molecule has a directory with a `*_ground_state_estimation.py` driver. A run sweeps a geometric coordinate, solves the variational problem at every point, and plots the curve live next to a rendering of the deforming molecule. When the sweep finishes, the window switches to *review mode*: arrow keys or a slider walk back and forth along the completed curve while the molecular rendering redraws to match the selected geometry.

The chemistry engine of the drivers. Every driver follows the same recipe. PySCF builds the Hamiltonian for the geometry, restricted to a complete active space by Qiskit Nature's `ActiveSpaceTransformer` where the full problem is too large. The parity mapper, with two-qubit reduction, maps it onto qubits. A UCCSD circuit built on the Hartree-Fock reference state is transpiled once, outside the optimiser loop, and its energy is evaluated by Qiskit's exact `StatevectorEstimator`, with the nuclear-repulsion and inactive-orbital constants added back. SciPy's `minimize` then drives the parameters, warm-started from the previous geometry. The `src/` core of Layer 1 implements the same algorithm on CUDA-Q instead and is exercised by the demo scripts.

Persistence and the scan_io contract. VQE points are expensive: a fraction of a second each for H_2 , but two to four minutes each for the larger active spaces, so a single H_2O stretch scan runs for roughly forty minutes and the BeH_2 surface considerably longer. No

result should ever be paid for twice. Every completed scan therefore saves itself automatically to two files:

- *_scan.json, energies plus full metadata (coordinate, fixed values, active space, optimiser, range). This is what `-load` reads, and what every table in this report was generated from.
- *_scan.csv, the same energies as a flat table, for spreadsheets and external plotting.

The format is shared across all molecules through `experiment/scan_io.py` (140 lines), so the drivers cannot drift into writing files each other cannot read. Each file records a `format_version` and is refused outright on a mismatch instead of half-loading and failing confusingly later, a small decision that saves real time on a project with several people touching the same code. Reloading a saved scan re-enters the interactive review window in seconds, with no chemistry re-executed.

Animating a surface. Three scripts, `beh2_surface_gif.py`, `h2o2_surface_gif.py` and `h2o2_twist_gif.py`, turn a saved scan into an animated GIF showing the surface rising out of the grid point by point (the animation lives in `assets/`). The three *_anim_still.png figures in this report, marked as animation frames wherever they appear, are single frames pulled from those GIFs, since a printed page cannot show motion.

Extending the pipeline. `NEW_MOLECULE_PROMPT.md` is the project’s internal playbook for adding a new molecule: how to decide whether a coordinate needs a curve or a surface, which code patterns to reuse (the ParityMapper, transpiling once instead of per point, warm-starting from the previous geometry, a serpentine traversal order for surface grids), and a list of concrete traps found the hard way, among them an open-shell molecule silently defaulting to a restricted reference, an active space too small to keep the energy curve monotonic, and the cost of SLSQP’s finite-difference gradient scaling with the number of parameters. It functioned as the project’s shared debugging notebook as much as a template for new code.

3.1.3 Layer 3: numerical post-processing. Each molecule additionally has *_polynomial/ and *_spline/ directories. These run no chemistry at all: they load a saved scan and do pure numerical analysis on it, so they finish in seconds and can be re-run freely. Listing 3 shows the leave-one-out driver behind Algorithm 2, the single most important measurement in the post-processing layer.

```
1 def leave_one_out(x, y, degree):
2     n = len(x)
3     poly_err = np.full(n, np.nan)
4     linear_err = np.full(n, np.nan)
5
6     for i in range(1, n - 1):
7         keep = np.ones(n, dtype=bool)
8         keep[i] = False
9         xk, yk = x[keep], y[keep]
10        # fitting degree d needs at least d+1 points
11        if len(xk) >= degree + 1:
12            poly_err[i] = float(
13                Polynomial.fit(xk, yk, degree)(x[i]) - y[i]
14            )
15            linear_err[i] = float(np.interp(x[i], xk, yk) - y[i])
```

```
16 return {"polynomial": poly_err, "linear": linear_err}
```

Listing 3: Leave-one-out cross-validation at the interior nodes. Endpoints return NaN because withholding one turns the problem into extrapolation; nodes where the remaining data cannot support the requested degree are excluded the same way.

Listing 4 shows the 2-D half-grid holdout of Algorithm 3, including the automatic spline-order reduction needed when a coarse axis has fewer than four points.

```
1 bi = np.arange(0, len(bond), 2)
2 aj = np.arange(0, len(angle), 2)
3 coarse_bond, coarse_angle = bond[bi], angle[aj]
4 coarse_Z = Z[np.ix_(bi, aj)]
5
6 # spline order must stay below the number of points per axis
7 kx = min(3, len(coarse_bond) - 1)
8 ky = min(3, len(coarse_angle) - 1)
9 spline = RectBivariateSpline(coarse_bond, coarse_angle,
10                             coarse_Z, kx=kx, ky=ky)
11
12 held_out = np.ones_like(Z, dtype=bool)
13 held_out[np.ix_(bi, aj)] = False
14 for i in range(len(bond)):
15     for j in range(len(angle)):
16         if not held_out[i, j] or not np.isfinite(Z[i, j]):
17             continue
18         spline_err[i, j] = float(
19             spline(bond[i], angle[j])[0, 0] - Z[i, j]
20         )
21         linear_err[i, j] = _bilinear(
22             coarse_bond, coarse_angle, coarse_Z,
23             bond[i], angle[j]) - Z[i, j]
```

Listing 4: Half-grid holdout for surfaces. Taking every other value on each axis keeps the subgrid rectangular, which leave-one-out cannot do, while withholding roughly three quarters of the points.

3.1.4 Reproducibility. Because every scan is saved with its metadata, every number and every figure in this report can be regenerated without re-running any quantum simulation. The tables in this document were generated programmatically from the stored JSON files rather than typed in by hand, which rules out an entire class of transcription error. Section 6.1 lists the commands.

3.2 Simulation setup and computational protocol

All calculations use the minimal STO-3G basis with an active space chosen per molecule. Geometries are built analytically from the scanned internal coordinates, with every other coordinate held at the fixed value recorded in the scan metadata. For each geometry the driver builds the active-space Hamiltonian with PySCF, maps it to qubits, starts the ansatz parameters at the Hartree–Fock reference, minimises the energy with the recorded optimiser, and stores E_{HF} , E_{VQE} , E_{exact} , the qubit count, the parameter count, the converged parameter vector, and, where the optimiser reports it, the number of energy evaluations used.

Table 1 is the full inventory: 23 scans over 11 molecules, 692 variational optimisations in total. Qubit counts range from six to ten, and ansatz parameter counts from 20 to 117. Table 2 gives the detail on the five two-dimensional grids.

Table 1: Complete computational campaign. Every VQE scan stored in the repository, with the active space, qubit count and ansatz parameter count actually used. N is the number of geometries at which the variational problem was solved. Bond lengths are in Å and angles in degrees.

Molecule	Coordinate	Type	Range / step	N	n_q	p	Active space	Optimizer	Basis
H ₂	bond length	curve	0.3–3 / 0.05	55	–	–	full	default	sto3g
LiH	bond length	curve	0.9–3.9 / 0.25	13	8	24	(2e, 5o)	slsqp	sto3g
BeH ₂	bend angle	curve	90–180 / 5	19	6	26	(4e, 4o)	default	sto3g
BeH ₂	symmetric stretch	curve	0.9–2.4 / 0.1	16	6	26	(4e, 4o)	default	sto3g
BeH ₂	2-D surface	surface	11×10 grid	110	6	26	(4e, 4o)	default	sto3g
H ₂ O	bend angle	curve	70–160 / 5	19	6	26	(4e, 4o)	default	sto3g
NH ₃	bend angle	curve	96–120 / 3	9	8	54	(6e, 5o)	slsqp	sto3g
NH ₃	symmetric stretch	curve	0.85–1.6 / 0.075	11	8	54	(6e, 5o)	slsqp	sto3g
NH ₃	2-D surface	surface	7×7 grid	49	8	54	(6e, 5o)	slsqp	sto3g
N ₂	bond length	curve	0.8–2.4 / 0.2	9	10	117	(6e, 6o)	cobyla	sto3g
O ₂	bond length	curve	0.9–2.5 / 0.2	9	10	68	(8e, 6o)	cobyla	sto3g
F ₂	bond length	curve	0.9–2.5 / 0.1	17	10	35	(10e, 6o)	default	sto3g
H ₂ O ₂	bend angle	curve	80–140 / 5	13	10	92	(8e, 6o)	cobyla	sto3g
H ₂ O ₂	symmetric stretch	curve	1–3 / 0.1	21	10	92	(8e, 6o)	cobyla	sto3g
H ₂ O ₂	2-D surface	surface	10×5 grid	50	10	92	(8e, 6o)	cobyla	sto3g
H ₂ O ₂	dihedral twist	curve	0–180 / 10	19	10	92	(8e, 6o)	cobyla	sto3g
NO ₂	bend angle	curve	90–180 / 5	19	6	20	(5e, 4o)	cobyla	sto3g
NO ₂	symmetric stretch	curve	0.9–2.4 / 0.1	16	8	37	(7e, 5o)	cobyla	sto3g
NO ₂	2-D surface	surface	11×10 grid	110	6	20	(5e, 4o)	cobyla	sto3g
N ₂ O	bend angle	curve	90–180 / 5	19	6	26	(4e, 4o)	default	sto3g
N ₂ O	N–N stretch	curve	0.9–2.2 / 0.1	14	8	54	(6e, 5o)	default	sto3g
N ₂ O	N–O stretch	curve	0.9–2.3 / 0.1	15	8	54	(6e, 5o)	default	sto3g
N ₂ O	2-D surface	surface	10×6 grid	60	6	26	(4e, 4o)	default	sto3g

Table 2: The five two-dimensional scans. N is the total number of variational solves, which grows as the product of the two axis lengths; this quadratic cost is the reason the holdout experiment of Section 2.2.5 matters.

Molecule	Grid (axis × axis)	N	n_q
BeH ₂	bond: 11 pts 0.9–2.4, angle: 10 pts 90–180	110	6
NH ₃	bond: 7 pts 0.85–1.6, angle: 7 pts 96–120	49	8
H ₂ O ₂	dihedral: 10 pts 0–180, angle: 5 pts 80–140	50	10
NO ₂	bond: 11 pts 0.9–2.4, angle: 10 pts 90–180	110	6
N ₂ O	rnn: 10 pts 0.9–2.25, rno: 6 pts 0.95–1.7	60	6

3.3 Parameter choices, datasets and benchmarks

Chemistry model. Every calculation uses the minimal STO-3G basis. PySCF supplies the orbitals, from restricted Hartree–Fock for the closed-shell molecules and restricted open-shell Hartree–Fock for the open-shell O₂ and NO₂. An active space ($n_{\text{el}}e, n_{\text{act}}o$) chosen per molecule, from (2e, 5o) for LiH to (10e, 6o) for F₂, keeps the problem at six to ten qubits (Table 1). Where a bond breaks, the larger active space is used: our own tests showed that (4e, 4o) is too small to describe a breaking bond, and yields a curve that dips back down past its minimum instead of rising towards dissociation. This is why the N₂O stretches use (6e, 5o) while its bend and surface use (4e, 4o).

Ansatz, optimiser and starting point. Every campaign scan uses a UCCSD ansatz. Its parameter count is fixed by the active space, from 20 for NO₂ to 117 for N₂ (the p column of Table 1). The first geometry of a scan starts from the Hartree–Fock reference with

all parameters zero; every later geometry warm-starts from the previous geometry’s converged parameters, falling back to zeros if the parameter count changes. SciPy’s SLSQP is used for the smaller ansätze (H₂, LiH, BeH₂, H₂O, NH₃, F₂, N₂O) and COBYLA for the deepest ones (N₂, O₂, H₂O₂, NO₂), each with a ceiling of 1000 iterations. The *Optimizer* column of Table 1 records the choice.

Scan grids. The grids are a deliberate budget decision, because coarse sampling is exactly what the reconstruction experiment is about. Bond lengths are sampled from 0.05Å steps for H₂ up to 0.25Å for LiH; bend angles at 3° or 5°; the H₂O₂ dihedral at 10°; and the five surfaces on grids of between 7 × 7 and 11 × 10 points (Table 2).

Datasets. No external dataset is used. The data are the 23 scans produced by this project, stored in the repository as *_scan.json (energies plus full metadata) and *_scan.csv. Together they hold the 692 VQE energies, each with its Hartree–Fock and exact reference energy and the converged parameter vector. Every table and figure in this report is generated from those files.

Benchmarks and metrics. Three energies are recorded at every geometry, E_{HF} , E_{VQE} and E_{exact} (exact diagonalisation of the identical active-space Hamiltonian), and the variational error is judged against the chemical-accuracy threshold of 1.6 mHa. Reconstructions are scored strictly out of sample, by leave-one-out cross-validation in one dimension and the half-grid holdout in two (Section 2.2.5), using max, RMS and MAE errors, and are always compared with the piecewise-linear baseline. Recovered equilibrium

geometries are compared with the experimental value where the scan metadata records one, and otherwise with the model’s own optimised geometry (Table 40).

4 Results and Discussion

4.1 Results by molecule

Molecules appear in order of increasing electron count, from H_2 to N_2O . Each subsection states the model setup, gives the variational results, and then discusses the reconstruction, with every figure placed immediately next to the paragraph that discusses it. All figures are reproduced unmodified from the repository; the three marked *animation frame* are single frames pulled from animated GIFs, which a printed page cannot show moving.

4.1.1 H_2 : hydrogen.

Setup. H_2 has two electrons in two spin-orbitals, so it needs no active-space truncation at all: the STO-3G calculation is the full configuration interaction. The bond was scanned from 0.30 to 3.00 Å in 0.05 Å steps, 55 geometries, by a wide margin the densest sampling in the campaign, affordable only because each point costs a fraction of a second.

Variational results. Table 3 is the complete scan. The variational solution is essentially exact: the largest deviation from exact diagonalisation anywhere on the curve is 3×10^{-4} mHa, roughly 5000 times below the chemical-accuracy threshold, and the mean rounds to zero at four decimal places. The energy minimum falls at $r = 0.75$ Å with $E = -1.137117$ Ha, matching the accepted STO-3G result.

The correlation column is the interesting one pedagogically. Near equilibrium, Hartree–Fock misses only a few tens of milli-Hartree, but by $r = 3.0$ Å the deficit has grown to 277.6 mHa. This is the textbook failure of restricted Hartree–Fock at dissociation: the single-determinant reference forces both electrons to share a delocalised molecular orbital even as the atoms separate, which describes two isolated hydrogen atoms poorly. The variational treatment tracks the exact curve throughout, exactly the regime where correlated methods earn their cost.

Reconstruction. None was done. The repository has no `h2_polynomial/` or `h2_spline/` directory, so H_2 contributes no rows to Table 39 and no figures to this report. The gap is defensible on its own terms, with 55 nodes at 0.05 Å spacing there is not much for an interpolant to recover, and the piecewise-linear reconstruction is already adequate, but it does mean the densest scan in the campaign is the one scan with no interpolation evidence behind it. We flag it as a gap rather than paper over it.

4.1.2 LiH : lithium hydride, and a negative result.

Setup. LiH has four electrons, two of which are correlated in a (2e, 5o) active space over eight qubits with a 24-parameter ansatz. The bond was scanned from 0.9 to 3.9 Å in 0.25 Å steps, 13 nodes, a deliberately coarse grid, and optimised with SLSQP, which converged in an average of 167 energy evaluations.

Variational results. Table 5 gives the scan. Accuracy is excellent: mean variational error 0.0001 mHa, maximum 0.0006 mHa, against

a correlation energy that reaches 153.2 mHa at the stretched end. The lowest computed node is -7.880947 Ha at $r = 1.65$ Å. The efficiency is notable too: 167 evaluations on a 24-parameter landscape converges comfortably, and it is the reason this system is accurate where the deeper-circuit molecules of Sections 4.1.6 and 4.1.7 are not.

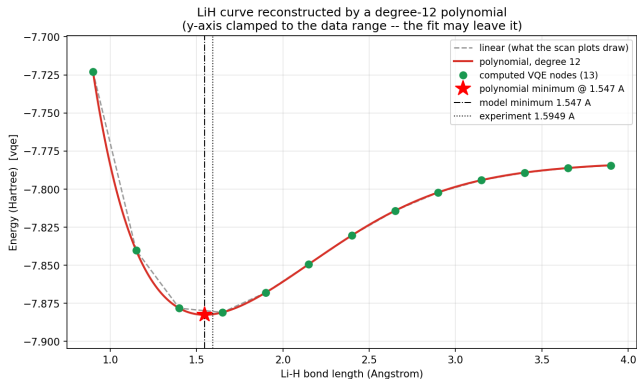


Figure 1: LiH dissociation curve reconstructed by a single global polynomial through all 13 nodes (degree 12). The fit passes through every computed point by construction, with an in-sample residual of 1.4×10^{-11} mHa, machine precision. That residual is *not* evidence of accuracy, which is the whole point of the out-of-sample analysis in Fig. 3.

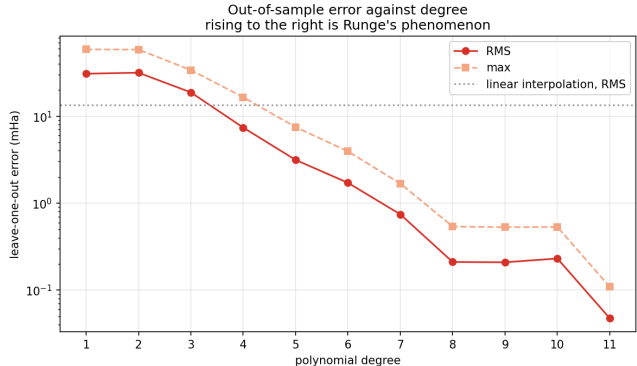


Figure 2: LiH polynomial degree sweep: leave-one-out error against fitted degree, tabulated in Table 4. The Runge signature we expected, a minimum followed by a rise, does not appear. Error falls essentially monotonically to the highest testable degree.

The result the experiment did not expect. The LiH polynomial experiment was written to reproduce Runge’s phenomenon. It did not, and saying so plainly is more useful than shipping the story we expected to tell.

Table 4 and Fig. 2 show leave-one-out RMS falling from 31.64 mHa at degree 2 to 0.047 mHa at degree 11, the *highest degree the 13-node scan can test*, with no rise anywhere. The full degree-12 interpolant has exactly one interior minimum, so it

Table 3: H₂ bond length scan: Hartree–Fock, VQE and exact energies (Ha) at every geometry; $\delta = (E_{\text{VQE}} - E_{\text{exact}}) \times 10^3$ (mHa).

r (Å)	E_{HF}	E_{VQE}	E_{exact}	δ	r (Å)	E_{HF}	E_{VQE}	E_{exact}	δ
0.3	-0.593828	-0.601804	-0.601804	0.000	1.7	-0.854338	-0.971427	-0.971427	0.000
0.35	-0.780455	-0.789269	-0.789269	0.000	1.75	-0.841349	-0.966335	-0.966335	0.000
0.4	-0.904361	-0.914150	-0.914150	0.000	1.8	-0.828848	-0.961817	-0.961817	0.000
0.45	-0.987513	-0.998416	-0.998416	0.000	1.85	-0.816842	-0.957833	-0.957833	0.000
0.5	-1.042996	-1.055160	-1.055160	0.000	1.9	-0.805333	-0.954339	-0.954339	0.000
0.55	-1.079051	-1.092630	-1.092630	0.000	1.95	-0.794318	-0.951290	-0.951290	0.000
0.6	-1.101128	-1.116286	-1.116286	0.000	2	-0.783793	-0.948641	-0.948641	0.000
0.65	-1.112997	-1.129905	-1.129905	0.000	2.05	-0.773749	-0.946350	-0.946350	0.000
0.7	-1.117349	-1.136189	-1.136189	0.000	2.1	-0.764178	-0.944375	-0.944375	0.000
0.75	-1.116151	-1.137117	-1.137117	0.000	2.15	-0.755066	-0.942678	-0.942678	0.000
0.8	-1.110850	-1.134148	-1.134148	0.000	2.2	-0.746401	-0.941224	-0.941224	0.000
0.85	-1.102511	-1.128362	-1.128362	0.000	2.25	-0.738169	-0.939982	-0.939982	0.000
0.9	-1.091914	-1.120560	-1.120560	0.000	2.3	-0.730353	-0.938922	-0.938922	0.000
0.95	-1.079637	-1.111339	-1.111339	0.000	2.35	-0.722939	-0.938021	-0.938021	0.000
1	-1.066109	-1.101150	-1.101150	0.000	2.4	-0.715910	-0.937255	-0.937255	0.000
1.05	-1.051657	-1.090342	-1.090342	0.000	2.45	-0.709250	-0.936605	-0.936605	0.000
1.1	-1.036539	-1.079193	-1.079193	0.000	2.5	-0.702944	-0.936055	-0.936055	0.000
1.15	-1.020964	-1.067930	-1.067930	0.000	2.55	-0.696975	-0.935589	-0.935589	0.000
1.2	-1.005107	-1.056741	-1.056741	0.000	2.6	-0.691328	-0.935196	-0.935196	0.000
1.25	-0.989114	-1.045783	-1.045783	0.000	2.65	-0.685988	-0.934864	-0.934864	0.000
1.3	-0.973111	-1.035186	-1.035186	0.000	2.7	-0.680941	-0.934584	-0.934584	0.000
1.35	-0.957203	-1.025054	-1.025054	0.000	2.75	-0.676172	-0.934349	-0.934349	0.000
1.4	-0.941481	-1.015468	-1.015468	0.000	2.8	-0.671669	-0.934151	-0.934151	0.000
1.45	-0.926017	-1.006487	-1.006487	0.000	2.85	-0.667417	-0.933985	-0.933985	0.000
1.5	-0.910874	-0.998149	-0.998149	0.000	2.9	-0.663405	-0.933846	-0.933846	0.000
1.55	-0.896099	-0.990476	-0.990476	0.000	2.95	-0.659619	-0.933729	-0.933729	0.000
1.6	-0.881732	-0.983473	-0.983473	0.000	3	-0.656048	-0.933632	-0.933632	0.000
1.65	-0.867804	-0.977130	-0.977130	0.000					

Table 4: LiH bond length polynomial degree sweep: leave-one-out error (mHa) against fitted degree d ; the best degree is in bold.

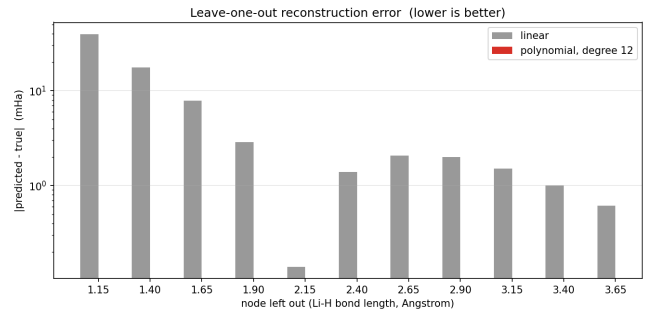
Degree d	max	RMS	MAE
1	58.6150	30.7998	24.0411
2	58.4160	31.6399	27.4945
3	33.8929	18.7903	16.6363
4	16.5561	7.3950	6.3092
5	7.4935	3.1427	2.4570
6	3.9421	1.7138	1.3773
7	1.6812	0.7436	0.5334
8	0.5390	0.2104	0.1416
9	0.5286	0.2088	0.1290
10	0.5323	0.2307	0.1321
11	0.1091	0.0474	0.0254

invents no spurious features at all. The headline comparison comes out backwards from what the textbook predicts:

Method	LOO RMS
Piecewise linear	13.373 mHa
Cubic spline	1.815 mHa
Global polynomial ($d = 11$)	0.047 mHa

The global polynomial beats the cubic spline by a factor of 38, and the linear baseline by a factor of 282.

Why. Runge’s phenomenon is a property of the *function being interpolated*, not an automatic consequence of high degree. It bites

**Figure 3: LiH leave-one-out error at each interior node for the polynomial and for the piecewise-linear baseline. The polynomial sits below the baseline everywhere, by two to three orders of magnitude.**

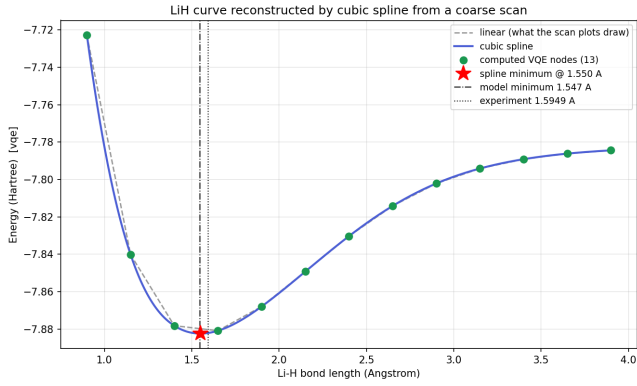
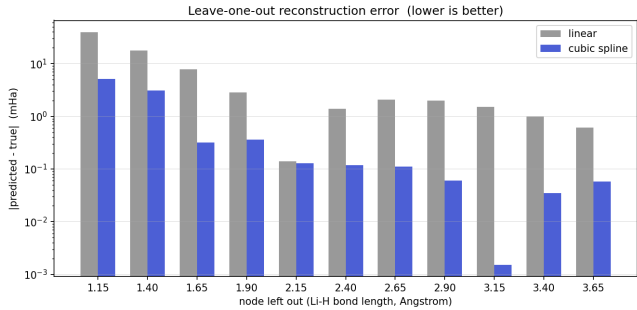
when the function has a singularity in the complex plane close to the real interval, Runge’s own $1/(1 + 25x^2)$ has poles at $\pm i/5$, right next to $[-1, 1]$. The LiH CAS(2, 5) curve over $[0.9, 3.9]\text{\AA}$ is smooth and gently varying with no nearby singularity, so a high-degree polynomial approximates it very well. Equation (16) agrees: when $f^{(n)}$ stays bounded, the $1/n!$ factor wins.

Where the trap does start. The trap is real, it just begins at a higher node density. Re-running the same interval at $0.125\text{\AA}$ spacing, 25 nodes instead of 13, gives best degree 13 at RMS 0.0004 mHa, degree

Table 5: LiH bond length scan: Hartree–Fock, VQE and exact energies (Ha) at every geometry; $\delta = (E_{\text{VQE}} - E_{\text{exact}}) \times 10^3$ (mHa).

r (Å)	E_{HF}	E_{VQE}	E_{exact}	δ
0.9	-7.705753	-7.722835	-7.722835	0.000
1.15	-7.823711	-7.840175	-7.840175	0.000
1.4	-7.860539	-7.878230	-7.878231	0.001
1.65	-7.859852	-7.880947	-7.880947	0.000
1.9	-7.841112	-7.867989	-7.867989	0.000
2.15	-7.813947	-7.849305	-7.849305	0.000
2.4	-7.783382	-7.830343	-7.830343	0.000
2.65	-7.752218	-7.814172	-7.814172	0.000
2.9	-7.722219	-7.802147	-7.802147	0.000
3.15	-7.694586	-7.794129	-7.794129	0.000
3.4	-7.670060	-7.789145	-7.789145	0.000
3.65	-7.648946	-7.786160	-7.786160	0.000
3.9	-7.631193	-7.784406	-7.784406	0.000

18 at 0.0005 mHa, and degree 23 at 0.0065 mHa: error bottoms out and then climbs by a factor of 16. That is the Runge signature. The scale is worth noting, though: the whole effect plays out below one hundredth of a milli-Hartree, a clean textbook trend rather than the catastrophe the phenomenon is usually illustrated with. Contrast this with the N_2O surface (Section 4.1.11), where the polynomial failure is genuinely catastrophic.


Figure 4: Cubic spline reconstruction of the LiH curve.

Figure 5: Spline and linear leave-one-out error. The spline improves on the linear baseline by 7.4 $\times$ in RMS, a solid result the polynomial nonetheless beats.

Geometry. Both reconstructions locate the minimum in essentially the same place: 1.5474Å (polynomial) and 1.5496Å (spline), against a best computed node of 1.65Å. The polynomial value reproduces the model’s own optimised bond length of 1.5474Å to four decimal places. From a 0.25Å grid, the interpolation recovers a geometry accurate to the model’s own reference, roughly two orders of magnitude of resolution gained for zero extra quantum cost. Both sit 0.045–0.048Å below the experimental 1.5949Å, and that gap belongs to the STO-3G/CAS(2, 5) model, not to the interpolation.

4.1.3 BeH_2 : beryllium hydride.

Setup. BeH_2 has six electrons in a (4e, 4o) active space, six qubits, 26 parameters. It is the most thoroughly explored system in the campaign: a symmetric stretch (16 nodes, 0.9–2.4Å), a bend (19 nodes, 90–180°) and a full two-dimensional surface on an 11 $\times$ 10 grid, 110 variational solves, the joint-largest surface here.

Variational results. The bend scan is excellent: mean error 0.0044 mHa, maximum 0.0062 mHa, four orders of magnitude inside chemical accuracy (Table 6). The stretch scan is noticeably worse, mean 0.5595 mHa and maximum 3.4778 mHa (Table 7), and the surface inherits this at mean 0.4541 mHa, maximum 4.6537 mHa. The pattern makes physical sense: bending at a fixed bond length barely touches the electronic structure, while stretching towards dissociation drives the system multireference, where a 26-parameter ansatz started at Hartree–Fock struggles most. The correlation energy tells the same story, rising from 5.9 mHa across the bend to 68.1 mHa across the stretch.

Table 6: BeH_2 bend angle scan: Hartree–Fock, VQE and exact energies (Ha) at every geometry; $\delta = (E_{\text{VQE}} - E_{\text{exact}}) \times 10^3$ (mHa).

θ (deg)	E_{HF}	E_{VQE}	E_{exact}	δ
90	-15.471340	-15.475542	-15.475547	0.004
95	-15.482637	-15.487013	-15.487018	0.005
100	-15.492809	-15.497363	-15.497368	0.005
105	-15.501972	-15.506701	-15.506707	0.006
110	-15.510230	-15.515127	-15.515133	0.006
115	-15.517671	-15.522726	-15.522732	0.006
120	-15.524368	-15.529570	-15.529576	0.006
125	-15.530383	-15.535720	-15.535726	0.006
130	-15.535765	-15.541223	-15.541229	0.006
135	-15.540554	-15.546118	-15.546124	0.006
140	-15.544781	-15.550437	-15.550442	0.005
145	-15.548470	-15.554201	-15.554207	0.005
150	-15.551640	-15.557431	-15.557436	0.004
155	-15.554305	-15.560140	-15.560144	0.004
160	-15.556475	-15.562341	-15.562343	0.003
165	-15.558157	-15.564041	-15.564042	0.002
170	-15.559355	-15.565249	-15.565250	0.001
175	-15.560073	-15.565971	-15.565972	0.001
180	-15.560312	-15.566211	-15.566212	0.001

Bend reconstruction. The bend is the easiest reconstruction problem in the campaign. The curve is smooth and nearly harmonic, and every method does well (Table 39): linear 0.344 mHa RMS, the polynomial at best degree 9 reaching 0.0001 mHa, the spline 0.0008 mHa. The degree sweep (Fig. 9, Table 8) shows the classic bowl shape, with a worst-case RMS of 7.59 mHa at low degree, underfitting rather than oscillation.

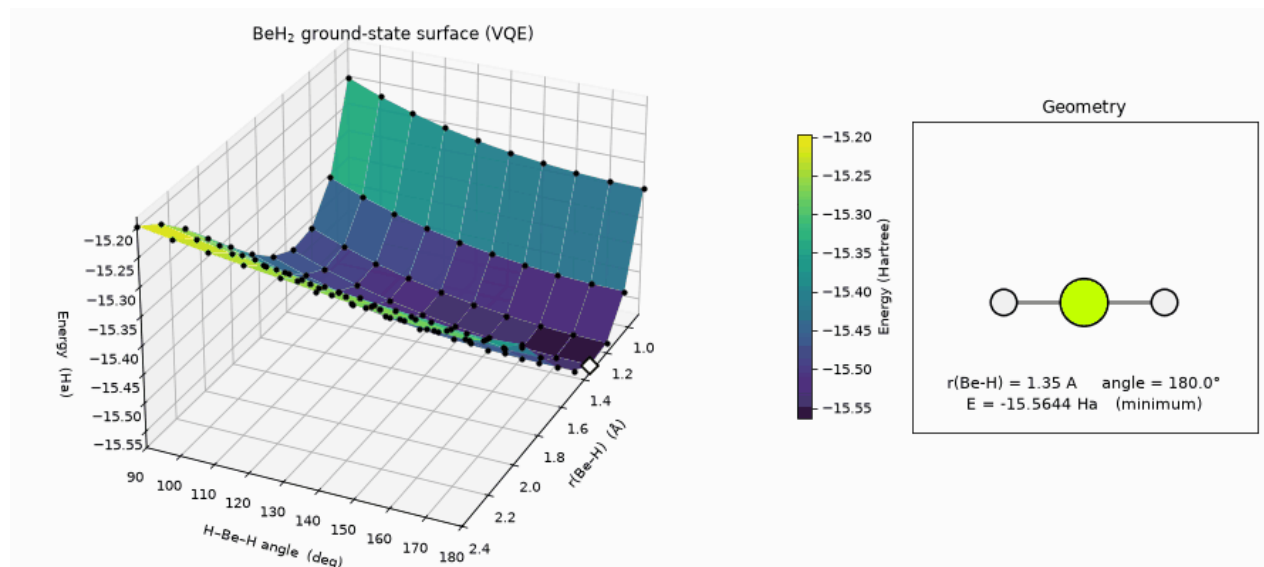


Figure 6: BeH₂ ground-state energy surface (animation frame). 110 VQE energies rise out of a bond-length $\times$ bond-angle grid in scan order while the molecular rendering redraws to match. The trough runs along the angle axis: bending is soft, stretching is stiff. The reconstruction analysis below turns this into a number.

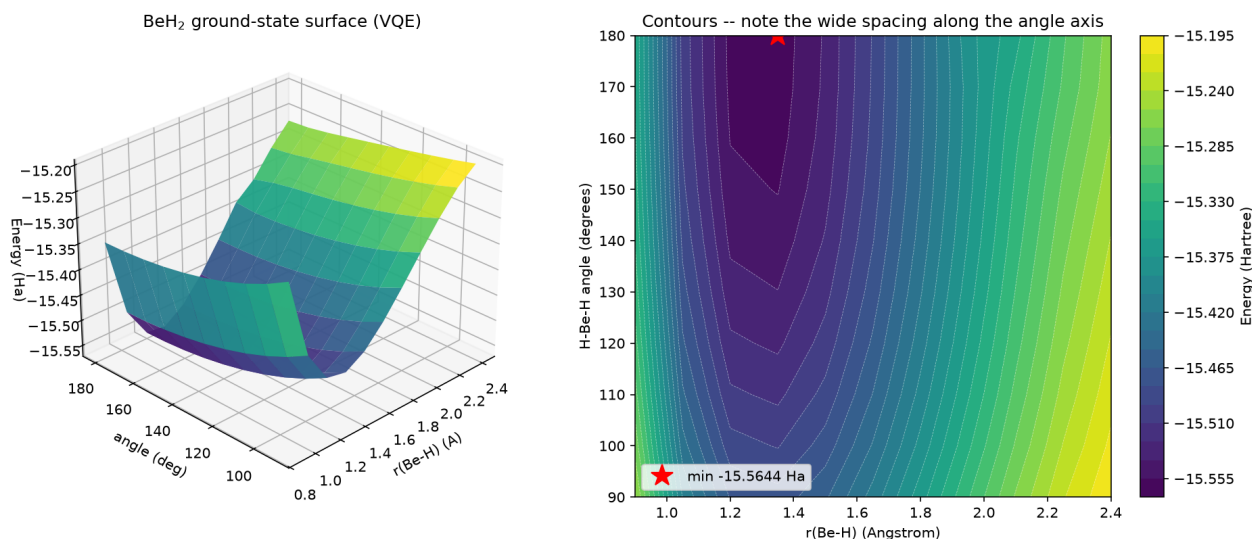


Figure 7: BeH₂ completed energy surface, static render. The minimum sits at the linear geometry $\theta = 180^\circ$, as expected for a $D_{\infty h}$ molecule.

Stretch reconstruction, and a spurious minimum. The stretch is harder, and it gives the campaign's clearest example of why a good fit statistic does not mean a usable curve. Polynomial and spline are almost tied statistically, 4.496 mHa versus 4.647 mHa RMS, both roughly $2.2\times$ better than the 10.12 mHa linear baseline. But the recovered geometries disagree completely: the spline places the minimum at 1.289 Å, close to the reference 1.3264 Å, while the full-degree polynomial places it at 0.9305 Å, at the extreme left edge of the scanned interval, 0.40 Å off, and physically meaningless.

This is Runge's phenomenon caught in the act. The degree-15 interpolant oscillates near the interval boundary, and the oscillation dips below the true curve at the edge, creating a spurious global minimum. The degree sweep (Table 9) confirms the mechanism: worst-case RMS reaches 572.5 mHa, four orders of magnitude above the best degree. The lesson generalises to every polynomial row in Table 40: an aggregate error metric can look fine while the interpolant is qualitatively wrong exactly where someone would read it.

Table 7: BeH₂ symmetric stretch scan: Hartree–Fock, VQE and exact energies (Ha) at every geometry; $\delta = (E_{\text{VQE}} - E_{\text{exact}}) \times 10^3$ (mHa).

r (Å)	E_{HF}	E_{VQE}	E_{exact}	δ
0.9	-15.338455	-15.346488	-15.346490	0.001
1	-15.455668	-15.462870	-15.462871	0.001
1.1	-15.521999	-15.528632	-15.528633	0.001
1.2	-15.553586	-15.559818	-15.559819	0.001
1.3	-15.561278	-15.567233	-15.567233	0.001
1.4	-15.552460	-15.558235	-15.558237	0.002
1.5	-15.532213	-15.537897	-15.537897	0.001
1.6	-15.504086	-15.509753	-15.509754	0.001
1.7	-15.470606	-15.476333	-15.476334	0.001
1.8	-15.433626	-15.439488	-15.439488	0.001
1.9	-15.394538	-15.400613	-15.400614	0.001
2	-15.354417	-15.384666	-15.385345	0.678
2.1	-15.314113	-15.350025	-15.351020	0.996
2.2	-15.274312	-15.317412	-15.318911	1.499
2.3	-15.235575	-15.287969	-15.290260	2.291
2.4	-15.198367	-15.262967	-15.266445	3.478

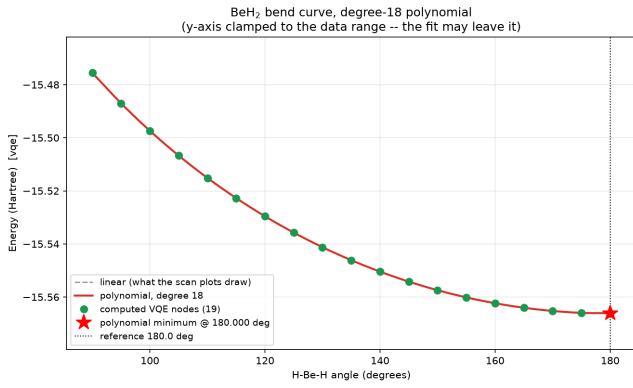


Figure 8: BeH₂ bend, polynomial reconstruction.

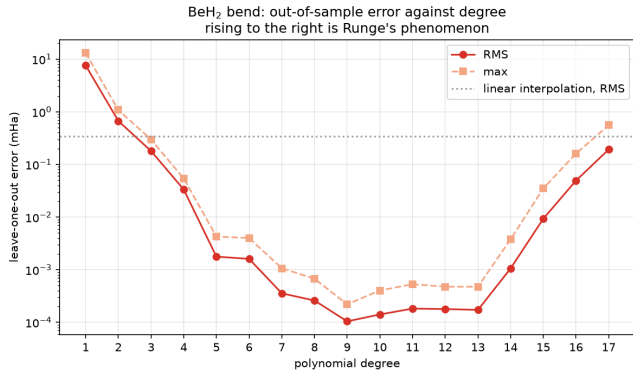


Figure 9: BeH₂ bend degree sweep. Error falls to 0.0001 mHa at degree 9.

Table 8: BeH₂ bend angle polynomial degree sweep: leave-one-out error (mHa) against fitted degree d ; the best degree is in bold.

Degree d	max	RMS	MAE
1	13.1025	7.5881	6.7476
2	1.0822	0.6700	0.6044
3	0.2973	0.1796	0.1579
4	0.0540	0.0334	0.0296
5	0.0043	0.0018	0.0015
6	0.0040	0.0016	0.0013
7	0.0011	0.0004	0.0003
8	0.0007	0.0003	0.0002
9	0.0002	0.0001	0.0001
10	0.0004	0.0001	0.0001
11	0.0005	0.0002	0.0001
12	0.0005	0.0002	0.0001
13	0.0005	0.0002	0.0001
14	0.0038	0.0011	0.0005
15	0.0351	0.0093	0.0036
16	0.1607	0.0493	0.0196
17	0.5580	0.1928	0.0755

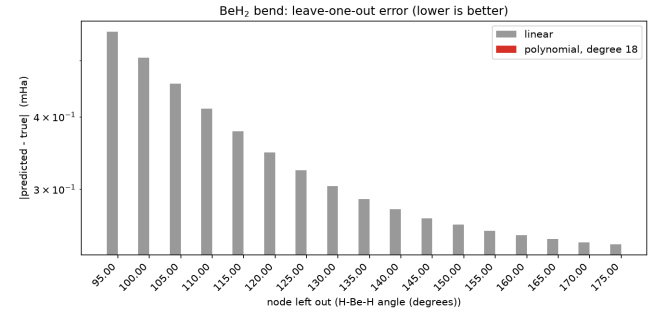


Figure 10: BeH₂ bend, polynomial versus linear leave-one-out error.

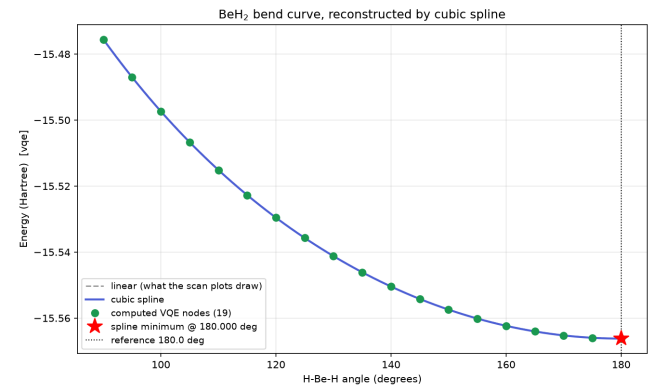


Figure 11: BeH₂ bend, cubic spline reconstruction.

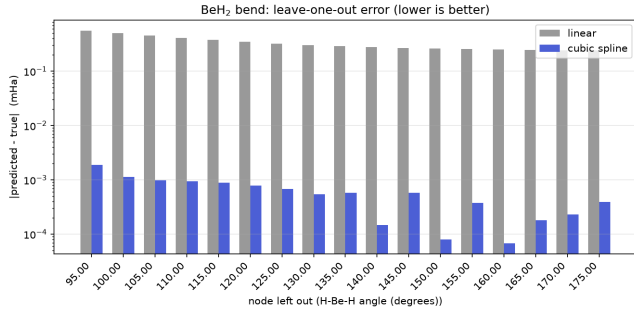


Figure 12: BeH₂ bend, spline leave-one-out error. RMS 0.0008 mHa against a linear baseline of 0.344 mHa, a 430× improvement on an already easy problem.

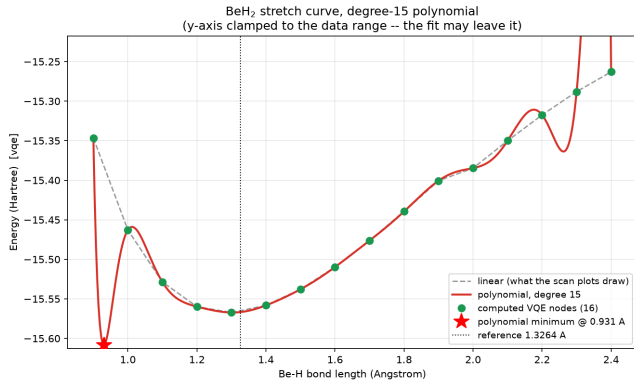


Figure 13: BeH₂ symmetric stretch, polynomial reconstruction. Note the behaviour at the left edge, which produces the spurious minimum discussed in the text.

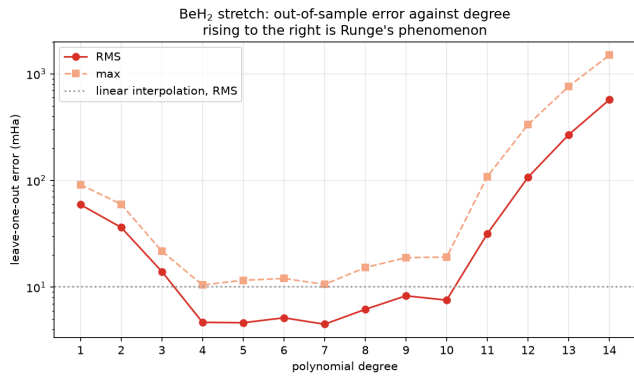


Figure 14: BeH₂ stretch degree sweep. Worst-case RMS reaches 572.5 mHa.

Table 9: BeH₂ symmetric stretch polynomial degree sweep: leave-one-out error (mHa) against fitted degree d ; the best degree is in bold.

Degree d	max	RMS	MAE
1	91.4818	59.4448	50.2601
2	59.9675	36.3232	31.4312
3	21.7420	14.0259	12.8937
4	10.4896	4.6810	3.7688
5	11.6111	4.6444	3.4286
6	12.0656	5.1597	3.8667
7	10.6220	4.4964	3.2178
8	15.2741	6.1943	3.8921
9	18.9213	8.2993	5.8991
10	19.1465	7.5641	5.7692
11	109.0775	31.7178	15.6726
12	334.6386	107.3525	51.8664
13	763.7332	268.9519	124.8257
14	1498.5750	572.5081	256.2855

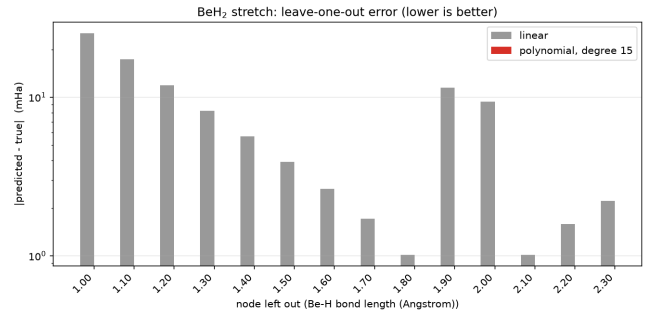


Figure 15: BeH₂ stretch, polynomial versus linear leave-one-out error.

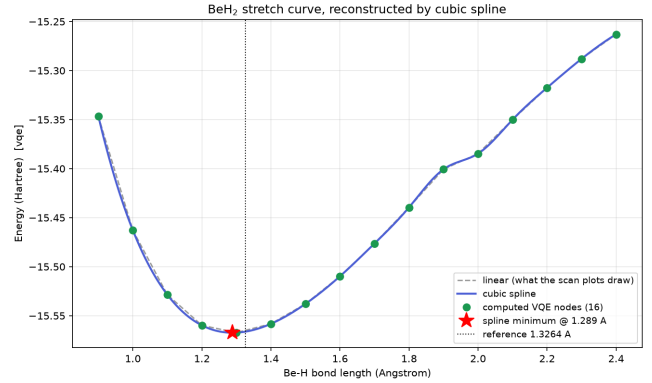


Figure 16: BeH₂ stretch, cubic spline reconstruction. The spline stays monotone at the edges and places the minimum at 1.289 Å.

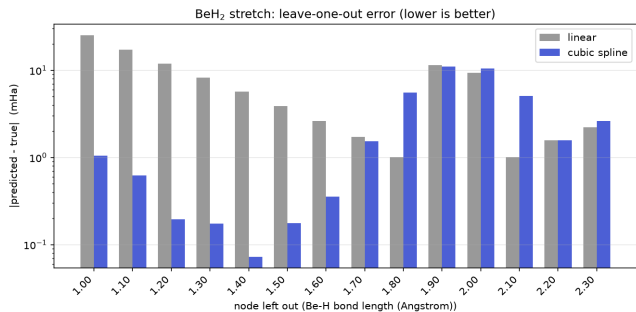


Figure 17: BeH₂ stretch, spline leave-one-out error, RMS 4.647 mHa.

Surface reconstruction. The half-grid holdout earns its keep here. Fitting on a 6×5 subgrid, one quarter of the 110 computed points, and scoring on the remaining three quarters gives bicubic spline RMS 3.098 mHa against a bilinear baseline of 17.97 mHa, a $5.8\times$ improvement. The total-degree polynomial reaches 3.966 mHa but, unlike the 1-D case, carries a nonzero in-sample residual of 8.58 mHa: it cannot represent the surface exactly even on the points it was fitted to.

The practical reading is direct: *we could have run a quarter of this surface and reconstructed the rest to within about 3 mHa RMS.* At two to four minutes per point, that saves several hours, and it is the result that justifies coarse grids on more expensive molecules.

Geometry. The spline surface locates the minimum at $(1.292\text{\AA}, 180.0^\circ)$ against the reference $(1.3264\text{\AA}, 180.0^\circ)$: the angle exactly right, the bond $0.034\text{\AA}$ short, recovered from a grid whose bond spacing is $0.15\text{\AA}$. The polynomial surface gives $(1.2995\text{\AA}, 176.38^\circ)$, missing the linear geometry by nearly four degrees, a qualitative error for a molecule whose linearity is a symmetry requirement.

4.1.4 H₂O: water.

Setup. H₂O has ten electrons in a (4e, 4o) active space, six qubits, 26 parameters. The repository holds one scan: the H–O–H bend from 70 to 160° in 5° steps at a fixed bond length, 19 geometries. A symmetric-stretch coordinate is implemented in the driver and documented in the project README, but no saved stretch scan is present, so we cannot report on it here.

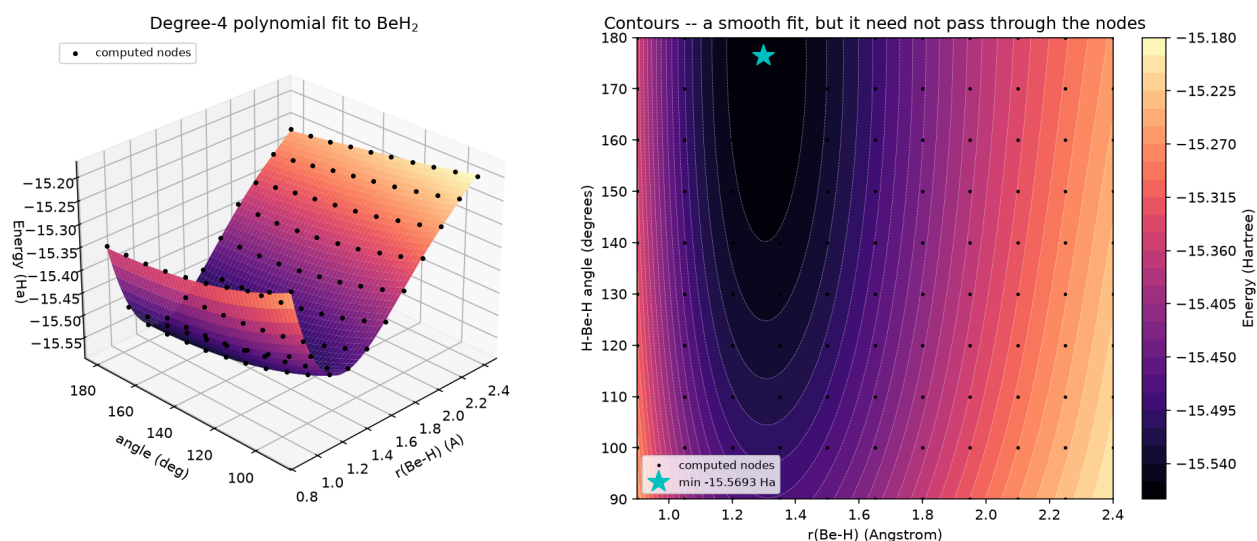
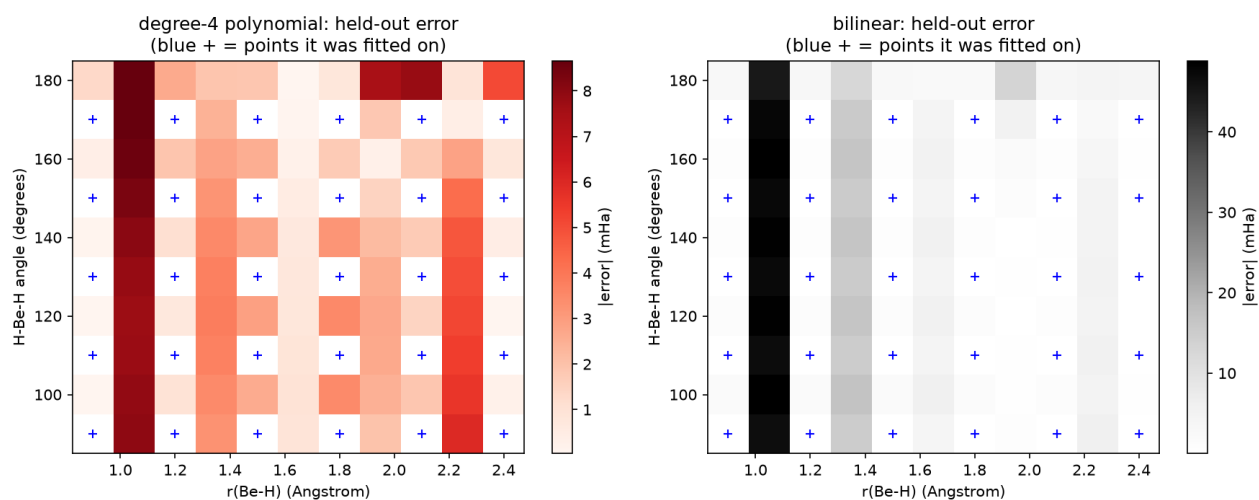
Variational results. Table 10 gives the complete scan. Accuracy is among the best in the campaign: mean error 0.0007 mHa, maximum 0.0015 mHa, three orders of magnitude inside chemical accuracy. The minimum falls at $\theta = 100^\circ$ with $E = -74.971353$ Ha. The experimental H–O–H angle is 104.5° ; the 4.5° gap belongs to the minimal basis and small active space, and no improvement to the solver would remove it. Correlation energy across the scan reaches 11.1 mHa, small, because bending at a fixed bond length is a gentle perturbation of the electronic structure, exactly as we saw for BeH₂.

Reconstruction. As with H₂, none was done: the repository has no h2o_polynomial/ or h2o_spline/ directory. H₂O therefore contributes no rows to Table 39 and no figures. Since H₂O is the molecule named in the project README as the worked example,

Table 10: H₂O bend angle scan: Hartree–Fock, VQE and exact energies (Ha) at every geometry; $\delta = (E_{\text{VQE}} - E_{\text{exact}}) \times 10^3$ (mHa).

θ (deg)	E_{HF}	E_{VQE}	E_{exact}	δ
70	-74.912380	-74.923506	-74.923507	0.000
75	-74.928523	-74.938913	-74.938913	0.000
80	-74.941332	-74.951073	-74.951073	0.000
85	-74.951034	-74.960204	-74.960204	0.000
90	-74.957841	-74.966501	-74.966502	0.000
95	-74.961953	-74.970155	-74.970156	0.000
100	-74.963568	-74.971353	-74.971353	0.000
105	-74.962882	-74.970280	-74.970280	0.000
110	-74.960090	-74.967125	-74.967126	0.001
115	-74.955392	-74.962083	-74.962084	0.001
120	-74.948995	-74.955358	-74.955359	0.001
125	-74.941120	-74.947170	-74.947170	0.001
130	-74.932011	-74.937762	-74.937763	0.001
135	-74.921941	-74.927413	-74.927413	0.001
140	-74.911219	-74.916437	-74.916438	0.001
145	-74.900201	-74.905201	-74.905202	0.001
150	-74.889289	-74.894117	-74.894119	0.001
155	-74.878927	-74.883640	-74.883641	0.001
160	-74.869584	-74.874243	-74.874244	0.001

this is the most conspicuous gap in the campaign, and we flag it rather than let it pass quietly.

Figure 18: BeH₂ surface, total-degree polynomial reconstruction.Figure 19: BeH₂ surface, polynomial holdout error distribution.

4.1.5 NH₃: ammonia.

Setup. NH₃ has ten electrons in a (6e, 5o) active space, eight qubits, 54 parameters, C_{3v} symmetry with two free coordinates. SLSQP converged in 255–348 evaluations depending on the scan. Three scans were run: an umbrella bend (9 nodes, 96–120°), a symmetric N–H stretch (11 nodes, 0.85–1.6 Å) and a 7 × 7 surface (49 points).

NH₃ is the one molecule whose raw scan plots survive in the repository alongside the interpolation figures, so Figs. 22 to 24 show the campaign’s output in its original form.

Variational results. All three scans reach chemical accuracy. The bend attains mean 0.0158 mHa (maximum 0.0494), the stretch mean 0.1937 mHa (maximum 0.7605), and the surface mean 0.3343 mHa,

although the surface maximum, 4.5362 mHa, exceeds the threshold at its worst point, presumably the most stretched corner of the grid. The stretch-versus-bend contrast mirrors BeH₂ exactly.

The inversion barrier. NH₃ carries a real physical observable that only a C² reconstruction can deliver. The molecule tunnels between two pyramidal configurations through a planar transition state at 120°, and the height of that barrier is a classic spectroscopic quantity. From the spline reconstruction of the bend curve, the barrier is

$$\Delta E_{\text{inv}} = 17.06 \text{ mHa} \approx 10.7 \text{ kcal mol}^{-1}, \quad (26)$$

measured from the interpolated minimum at 104.38° to the planar geometry. The experimental value is roughly 5.8 kcal mol^{−1}; the overestimate belongs to the minimal basis and restricted active space rather than to the reconstruction. The methodological point

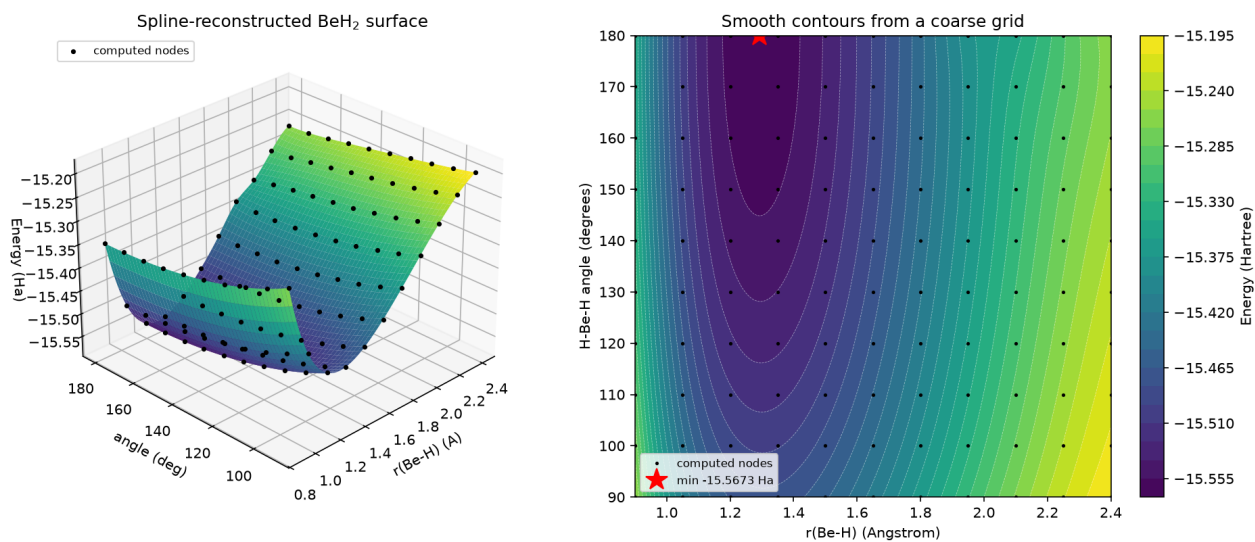


Figure 20: BeH₂ surface, bicubic spline reconstruction from the half grid.

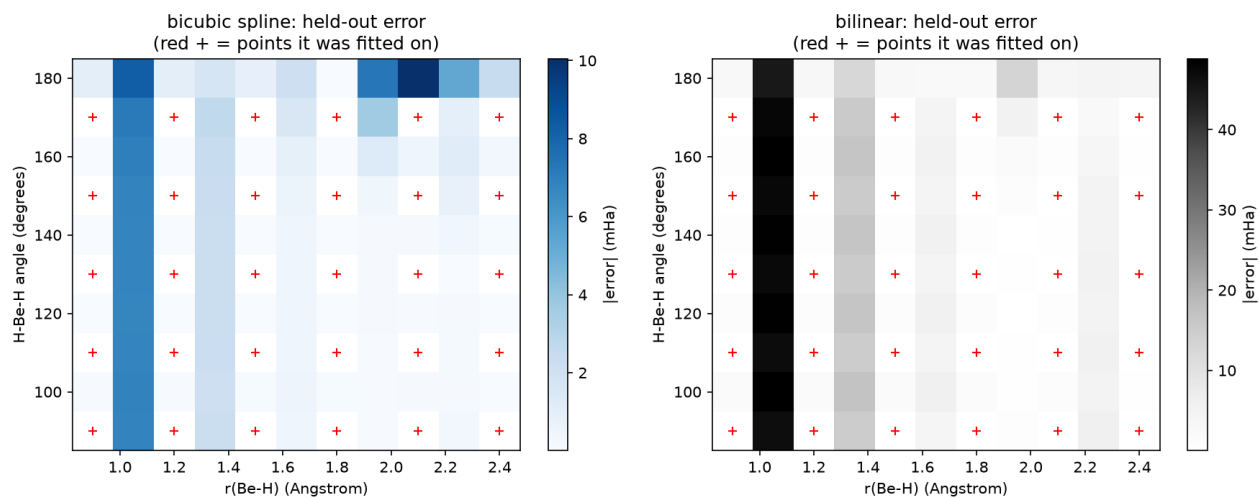


Figure 21: BeH₂ surface, spline holdout error. RMS 3.098 mHa against 17.97 mHa bilinear.

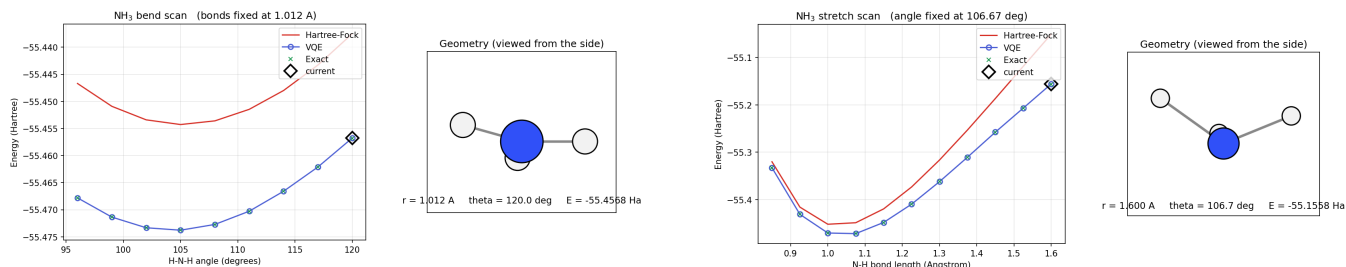


Figure 22: NH₃ umbrella bend, raw VQE scan.

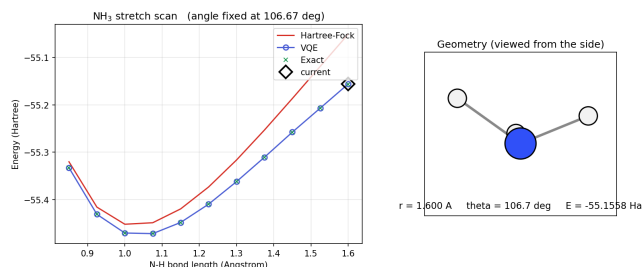
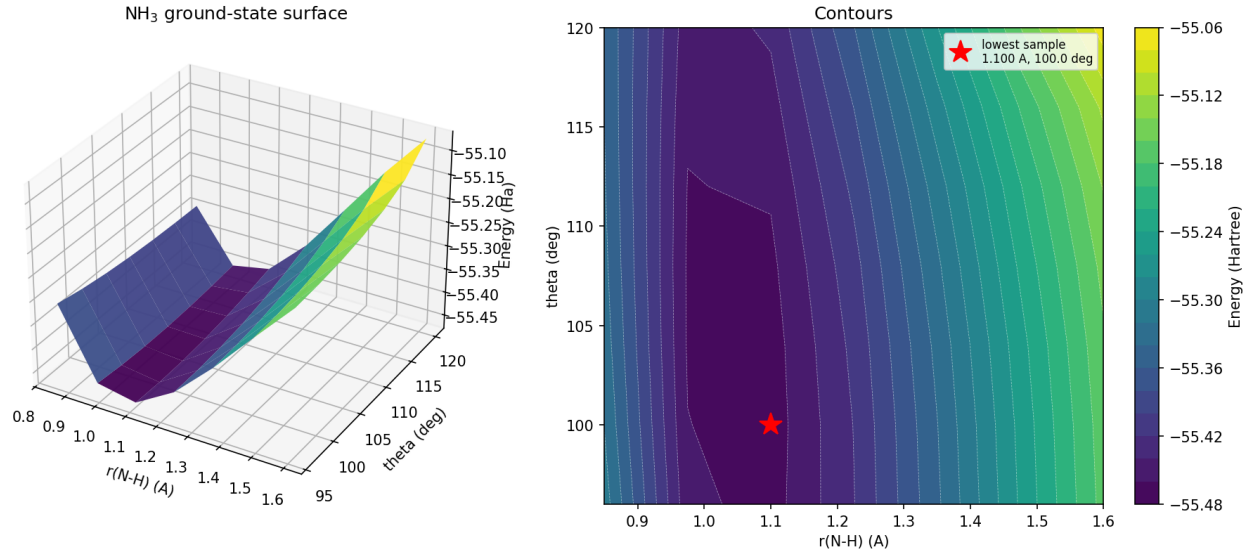


Figure 23: NH₃ symmetric N-H stretch, raw VQE scan.

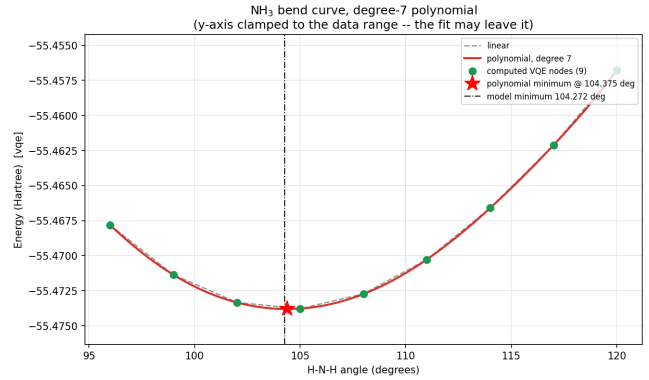
Figure 24: NH₃ raw energy surface over the bond-length $\times$ umbrella-angle grid, 49 VQE solves.Table 11: NH₃ bend angle scan: Hartree-Fock, VQE and exact energies (Ha) at every geometry; $\delta = (E_{\text{VQE}} - E_{\text{exact}}) \times 10^3$ (mHa).

θ (deg)	E_{HF}	E_{VQE}	E_{exact}	δ
96	-55.446703	-55.467822	-55.467829	0.007
99	-55.450917	-55.471373	-55.471381	0.008
102	-55.453414	-55.473341	-55.473349	0.008
105	-55.454280	-55.473791	-55.473799	0.008
108	-55.453602	-55.472728	-55.472736	0.008
111	-55.451472	-55.470284	-55.470295	0.011
114	-55.447996	-55.466601	-55.466619	0.018
117	-55.443295	-55.462119	-55.462143	0.024
120	-55.437509	-55.456768	-55.456818	0.049

Table 12: NH₃ symmetric stretch scan: Hartree-Fock, VQE and exact energies (Ha) at every geometry; $\delta = (E_{\text{VQE}} - E_{\text{exact}}) \times 10^3$ (mHa).

r (Å)	E_{HF}	E_{VQE}	E_{exact}	δ
0.85	-55.320316	-55.332929	-55.332933	0.004
0.925	-55.416129	-55.431383	-55.431387	0.004
1	-55.452322	-55.470863	-55.470873	0.009
1.075	-55.449123	-55.472133	-55.472148	0.015
1.15	-55.419851	-55.448498	-55.448528	0.030
1.225	-55.373570	-55.409923	-55.409975	0.052
1.3	-55.316579	-55.362264	-55.362374	0.110
1.375	-55.253251	-55.310751	-55.310961	0.210
1.45	-55.186578	-55.257680	-55.258050	0.370
1.525	-55.118575	-55.206832	-55.207397	0.565
1.6	-55.050603	-55.155814	-55.156574	0.760

is what matters here: the barrier is read off the *interpolated* curve, because the planar geometry 120° is an endpoint of a 3° grid and the minimum at 104.38° falls between nodes. A piecewise-linear reconstruction would have snapped the minimum to 105° and distorted the barrier along with it.

Figure 25: NH₃ bend, polynomial reconstruction.Table 13: NH₃ bend angle polynomial degree sweep: leave-one-out error (mHa) against fitted degree d ; the best degree is in bold.

Degree d	max	RMS	MAE
1	4.9654	3.3600	3.1111
2	0.6015	0.3645	0.3319
3	0.1545	0.0973	0.0895
4	0.1172	0.0605	0.0431
5	0.1596	0.0836	0.0682
6	0.1489	0.0738	0.0593
7	0.2826	0.1590	0.1199

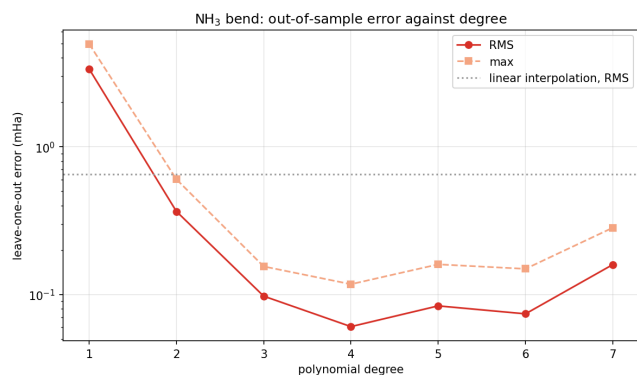


Figure 26: NH₃ bend degree sweep; best degree 4 at RMS 0.061 mHa.



Figure 27: NH₃ bend, polynomial versus linear leave-one-out error.

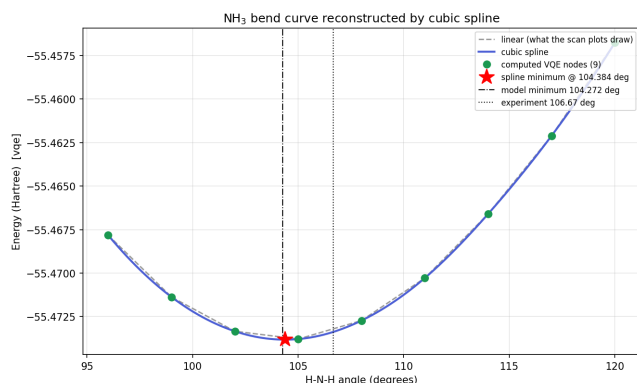


Figure 28: NH₃ bend, cubic spline reconstruction, the curve the inversion barrier is measured from.

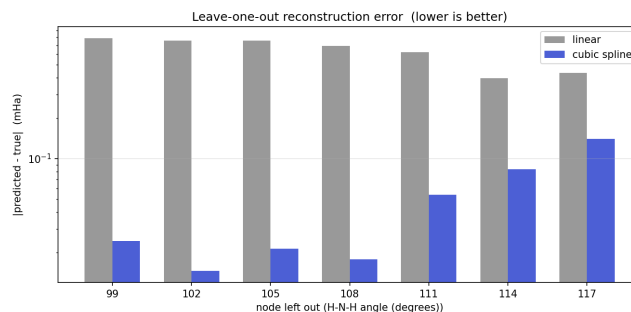


Figure 29: NH₃ bend, spline leave-one-out error. RMS 0.067 mHa against a linear baseline of 0.653 mHa.

Reconstruction. With only 9 and 11 nodes respectively, the NH₃ curves are the sparsest 1-D data in the campaign, and the results are correspondingly instructive. On the bend, polynomial (0.061 mHa at degree 4) and spline (0.067 mHa) are statistically indistinguishable, and both beat linear (0.653 mHa) by an order of magnitude. On the stretch the polynomial wins more clearly, 0.466 mHa at degree 6 against 0.920 mHa for the spline and 12.79 mHa for linear, a 27× improvement over the baseline.

The surface is the hardest case. With a 7×7 grid, the half-grid holdout leaves a 4×4 subgrid, the bare minimum that supports a bicubic spline on both axes. Spline RMS is 8.28 mHa and total-degree polynomial 8.49 mHa, against a bilinear baseline of 30.11 mHa: a 3.6× improvement, but well above chemical accuracy in absolute terms. This is the clearest evidence in the campaign that the half-grid experiment has a floor: below roughly five points per axis there is not enough information left to reconstruct a surface accurately, whichever method is used. We also did not generate a half-grid error map for the NH₃ surface the way we did for BeH₂, NO₂ and N₂O, so only the reconstructed surfaces themselves are shown below, not an error plot.

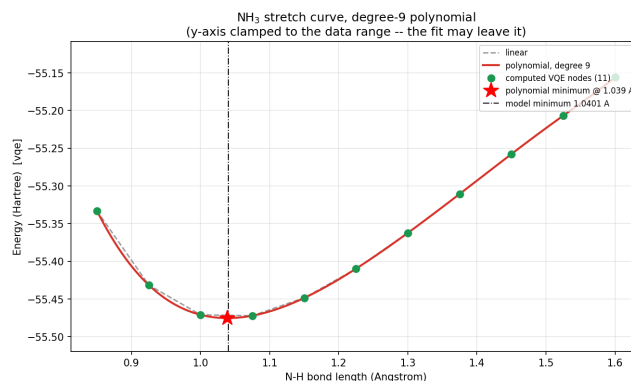
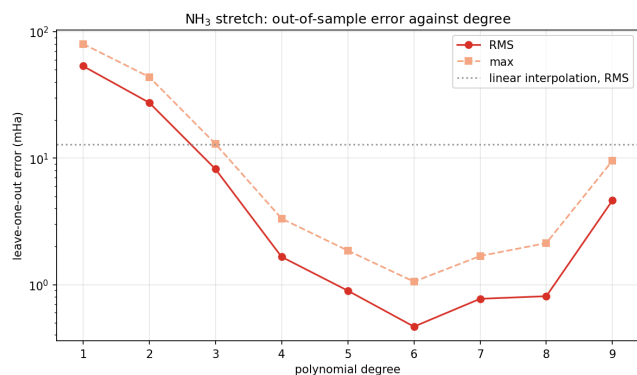
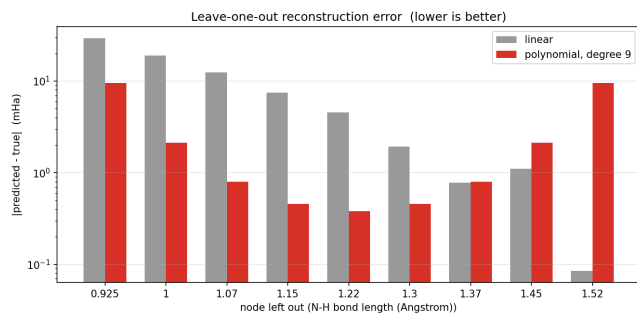


Figure 30: NH₃ stretch, polynomial reconstruction.

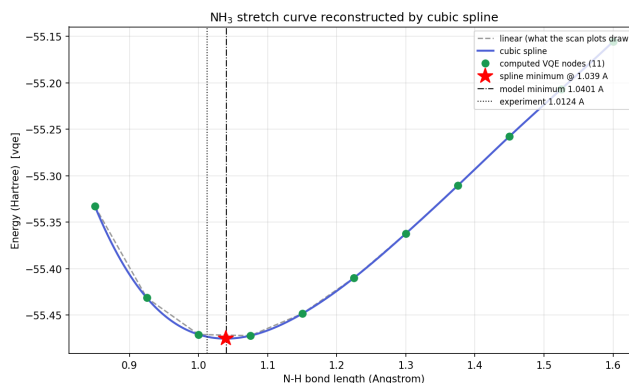
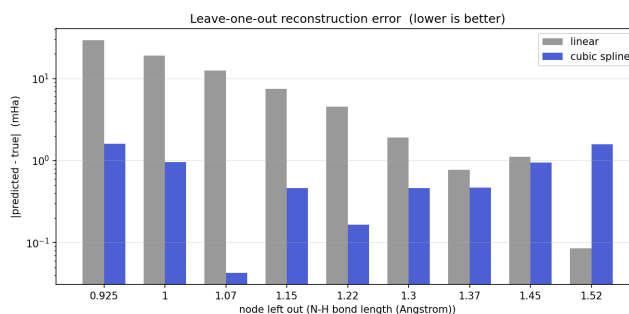
Geometry. The reconstructions recover the model's own optimised geometry almost exactly. The bend minimum sits at 104.38° (both methods) against a model reference of 104.27°, an error of 0.11° from a grid spaced at 3°. The stretch minimum sits at 1.0391 (polynomial) and 1.0394Å (spline) against a model reference of

Figure 31: NH₃ stretch degree sweep; best degree 6.Table 14: NH₃ symmetric stretch polynomial degree sweep: leave-one-out error (mHa) against fitted degree d ; the best degree is in bold.

Degree d	max	RMS	MAE
1	79.9923	53.4177	48.1491
2	43.8140	27.4471	24.3178
3	12.9864	8.1997	7.2974
4	3.3268	1.6579	1.4349
5	1.8593	0.8965	0.6895
6	1.0562	0.4660	0.3318
7	1.6842	0.7732	0.5741
8	2.1305	0.8105	0.5427
9	9.5652	4.6410	2.9185

Figure 32: NH₃ stretch, polynomial versus linear leave-one-out error.Table 15: NH₃ 2-D surface polynomial degree sweep: leave-one-out error (mHa) against fitted degree d ; the best degree is in bold.

Degree d	max	RMS	MAE
1	119.2770	74.4766	65.5689
2	60.6062	34.3305	29.5216
3	16.0243	8.4926	7.0075

Figure 33: NH₃ stretch, cubic spline reconstruction.Figure 34: NH₃ stretch, spline leave-one-out error, RMS 0.920 mHa.

1.0401Å: errors of 0.0010 and 0.0007Å from a grid spaced at 0.075Å, a hundredfold gain in effective resolution and the single most convincing demonstration in the campaign of what interpolation buys. The surface agrees, placing the minimum at (1.0454Å, 102.49°) against the model surface reference (1.05Å, 102.05°).

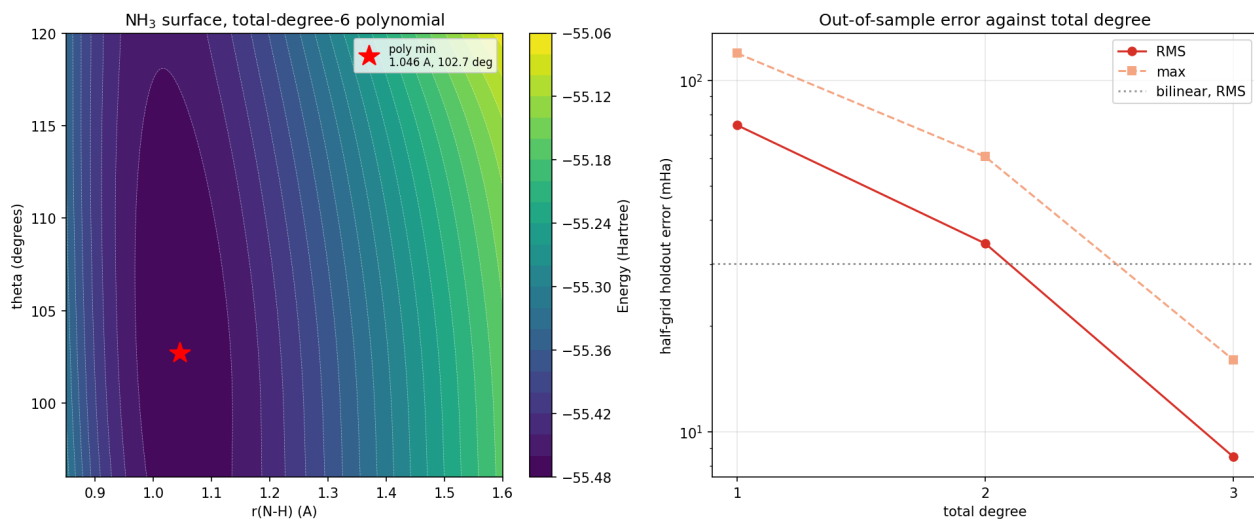
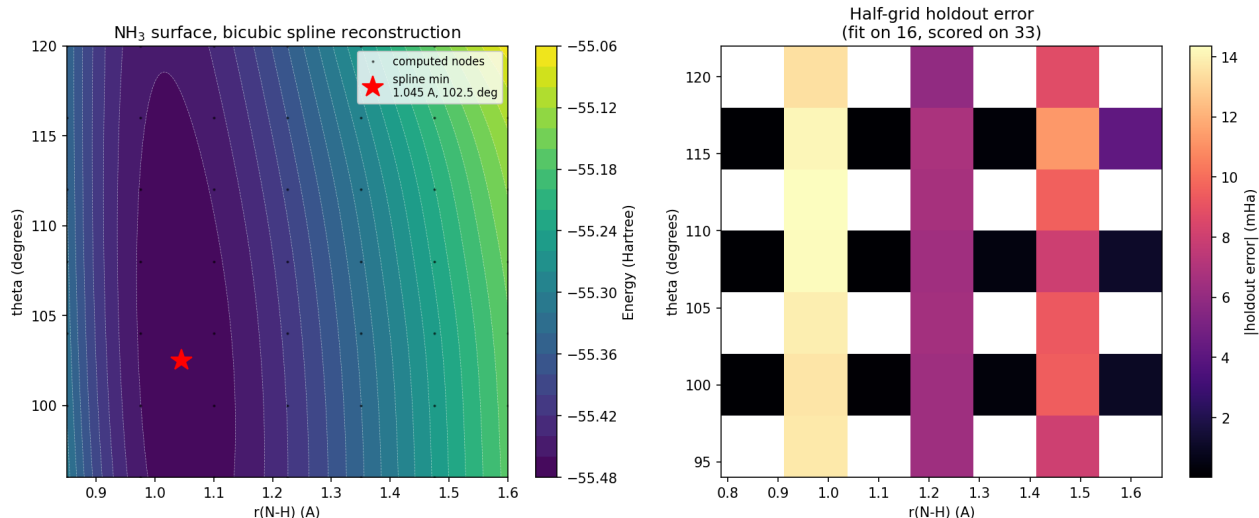
Against *experiment*, 1.0124Å and 106.67°, the gaps are larger, 0.027Å and 2.3°, and again these belong to the model, not to anything interpolation could fix.

4.1.6 N₂: nitrogen.

Setup. N₂ has fourteen electrons in a (6e, 6o) active space, ten qubits, and the largest ansatz in the campaign at 117 parameters. Nine nodes from 0.8 to 2.4Å in 0.2Å steps, optimised with COBYLA under a 1000-evaluation ceiling.

Variational results. This is the campaign's hardest optimisation and its least accurate result: mean error 10.48 mHa, maximum 36.35 mHa, an order of magnitude outside chemical accuracy (Table 16).

The diagnosis is clear. COBYLA spends its full 1000-evaluation budget at every geometry, which on a 117-dimensional landscape is fewer than nine evaluations per parameter, not enough for a derivative-free method to map the surface, let alone converge on it. The variational bound is never violated, so the algorithm behaves correctly; it is simply stopped early. Triple-bond breaking is also

Figure 35: NH_3 surface, total-degree polynomial reconstruction (degree 6).Figure 36: NH_3 surface, bicubic spline reconstruction from the 4×4 half grid.

the most strongly multireference process in the campaign, so the landscape being searched is the roughest one here.

Context matters, though: Hartree–Fock misses 777.0 mHa of correlation energy on this curve, the largest deficit anywhere in the campaign. Missing 10.5 mHa of that still means recovering 98.6% of the correlation the mean-field reference leaves out. The calculation is far from useless; it is simply not converged to spectroscopic standards.

Reconstruction. With only nine nodes at $0.2\text{\AA}$ spacing across a bond-breaking curve, this is the sparsest and most sharply curved 1-D reconstruction problem in the campaign, and all the errors are correspondingly large. Linear interpolation gives 134.18 mHa RMS, a value too large for any chemical purpose. The polynomial at its best degree of 6 reaches 20.34 mHa and the spline 37.59 mHa; the

Table 16: N_2 bond length scan: Hartree–Fock, VQE and exact energies (Ha) at every geometry; $\delta = (E_{\text{VQE}} - E_{\text{exact}}) \times 10^3$ (mHa).

r (Å)	E_{HF}	E_{VQE}	E_{exact}	δ
0.8	-106.680802	-106.713261	-106.714551	1.289
1	-107.419532	-107.519020	-107.520461	1.441
1.2	-107.487784	-107.642667	-107.645024	2.357
1.4	-107.357815	-107.585331	-107.591182	5.851
1.6	-107.184846	-107.510796	-107.513395	2.599
1.8	-107.017327	-107.456615	-107.459506	2.891
2	-106.871504	-107.407238	-107.437024	29.786
2.2	-106.751831	-107.420818	-107.432611	11.793
2.4	-106.656608	-107.397268	-107.433615	36.347

polynomial wins by a factor of 1.8 and improves on the baseline by 6.6 $\times$.

None of these reconstructions reaches chemical accuracy. Nine nodes across a triple-bond dissociation is too coarse a grid for any interpolant to rescue, regardless of which one comes out ahead in the comparison. The degree sweep (Table 17) supports this: even the worst degree reaches only 207.5 mHa, because with seven testable degrees there is no room for the dramatic Runge failures seen elsewhere.

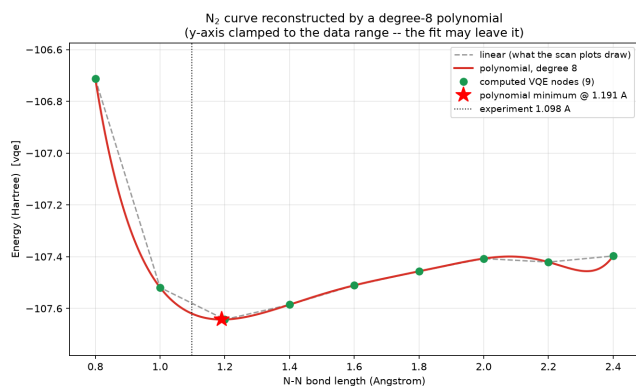


Figure 37: N_2 dissociation curve, polynomial reconstruction.

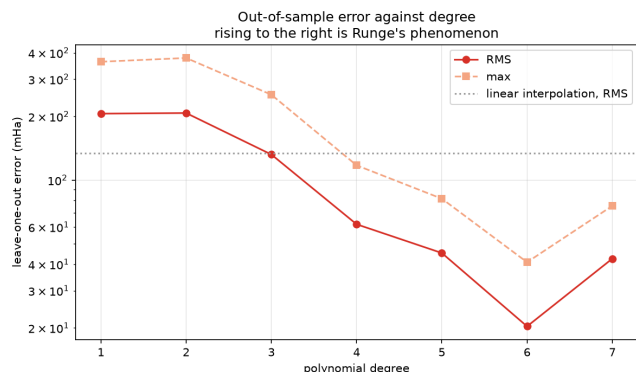


Figure 38: N_2 degree sweep; best degree 6 at 20.34 mHa RMS.

Table 17: N_2 bond length polynomial degree sweep: leave-one-out error (mHa) against fitted degree d ; the best degree is in bold.

Degree d	max	RMS	MAE
1	362.6409	206.1169	171.3931
2	377.8793	207.4911	172.1065
3	253.2753	132.1034	111.9389
4	117.4039	61.8030	53.5987
5	81.6669	45.2077	36.6940
6	40.8968	20.3413	16.3684
7	75.5146	42.4990	32.0552

Geometry. The polynomial places the minimum at 1.191Å and the spline at 1.1685Å, against an experimental bond length of

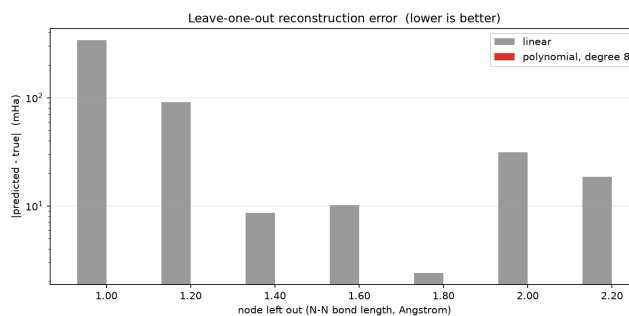


Figure 39: N_2 polynomial versus linear leave-one-out error.

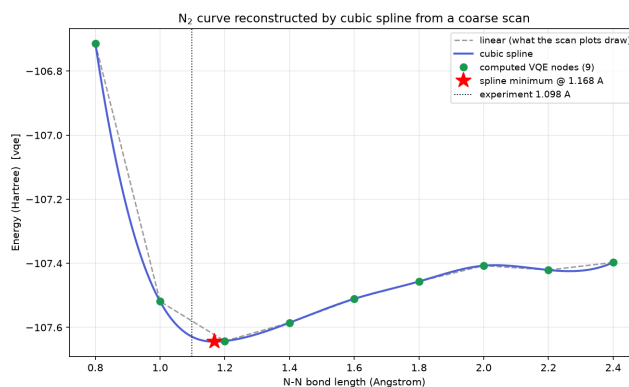


Figure 40: N_2 cubic spline reconstruction.

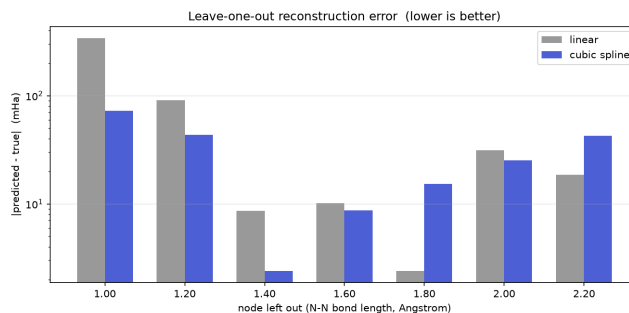


Figure 41: N_2 spline leave-one-out error, RMS 37.59 mHa. Compare the vertical scale with Fig. 12: this is a two-orders-of-magnitude harder problem.

1.098Å. Both overestimate by 0.07–0.09Å. Most of that error belongs to the minimal basis and restricted active space, but with a 10.5 mHa mean variational error sitting on top of a curve whose well depth is being read to sub-milli-Hartree precision, some of it belongs to the incomplete optimisation as well. This is the one molecule in the campaign where solver error and model error cannot be cleanly separated, and we do not claim to separate them.

4.1.7 O₂: oxygen.

Setup. O₂ has sixteen electrons in an (8e, 6o) active space with five α and three β electrons, ten qubits, 68 parameters. O₂ is a ground-state *triplet* ($2S = 2$), which needed a restricted open-shell Hartree–Fock reference rather than the closed-shell RHF used for most of the campaign. Nine nodes from 0.9 to 2.5Å in 0.2Å steps, COBYLA.

Variational results. Accuracy is poor for the same reason as N₂: mean 10.55 mHa, maximum 28.55 mHa, with COBYLA saturating its 1000-evaluation budget at every geometry (Table 18). The open-shell reference adds a further difficulty: the ansatz has to represent a state of definite spin multiplicity, and nothing in the hardware-efficient construction enforces that, so part of the optimiser’s work goes into maintaining a symmetry that a spin-adapted ansatz would get for free. Hartree–Fock misses 576.5 mHa here, so VQE again recovers roughly 98% of the correlation energy despite missing chemical accuracy.

Table 18: O₂ bond length scan: Hartree–Fock, VQE and exact energies (Ha) at every geometry; $\delta = (E_{\text{VQE}} - E_{\text{exact}}) \times 10^3$ (mHa).

r (Å)	E_{HF}	E_{VQE}	E_{exact}	δ
0.9	-147.139439	-147.171132	-147.171162	0.030
1.1	-147.590894	-147.656338	-147.656442	0.103
1.3	-147.618812	-147.735735	-147.736896	1.161
1.5	-147.518566	-147.697642	-147.702491	4.849
1.7	-147.388737	-147.637722	-147.649223	11.501
1.9	-147.268103	-147.601774	-147.616729	14.955
2.1	-147.168643	-147.579407	-147.607961	28.555
2.3	-147.090786	-147.594686	-147.607685	12.999
2.5	-147.032001	-147.587711	-147.608502	20.790

Reconstruction. The polynomial wins clearly: 7.99 mHa RMS at degree 6, against 22.00 mHa for the spline and 80.54 mHa for linear, improvements of 10.1× and 3.7× on the baseline. As with N₂, nine nodes is too sparse for any method to reach chemical accuracy, so read this comparison as a ranking, not an endorsement.

The pattern across N₂, O₂ and F₂ is itself the finding: on sparse, smooth, single-coordinate dissociation curves the global polynomial outperforms the spline, and more decisively as the node count rises. With nine nodes the margin is 2.8× (O₂); with seventeen it becomes 109× (F₂, next section).

Geometry. Polynomial 1.3028Å, spline 1.2754Å, experiment 1.208Å. The spline is the closer of the two, 0.067Å above experiment. Both sit above the true value, consistent with the general tendency of minimal-basis CAS calculations to over-lengthen bonds.

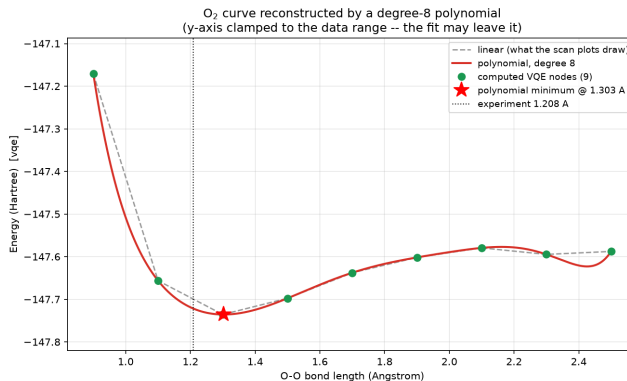


Figure 42: O₂ dissociation curve, polynomial reconstruction.

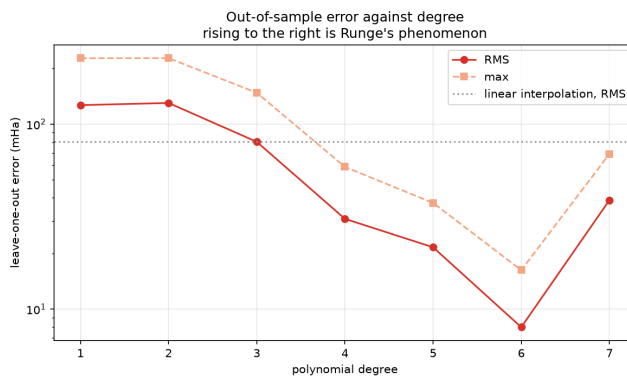
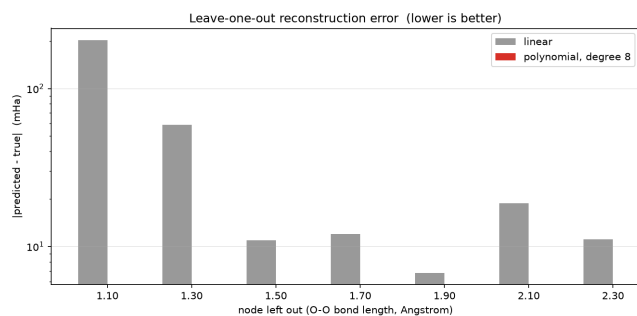
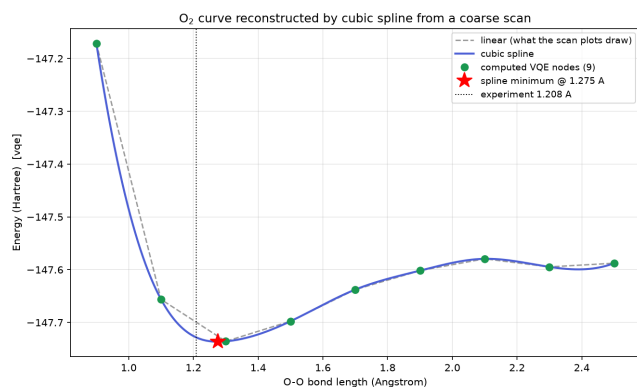
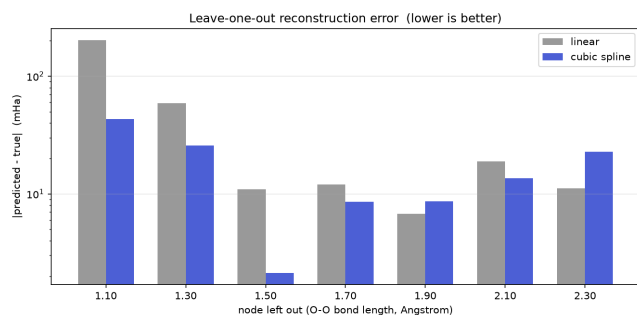


Figure 43: O₂ degree sweep; best degree 6 at 7.99 mHa RMS, worst 130.17 mHa.

Table 19: O₂ bond length polynomial degree sweep: leave-one-out error (mHa) against fitted degree d ; the best degree is in bold.

Degree d	max	RMS	MAE
1	227.2053	126.7282	102.3330
2	227.5155	130.1700	109.7675
3	147.8719	80.2569	70.0870
4	58.7578	30.7725	26.1430
5	37.5126	21.5989	17.5765
6	16.2480	7.9871	6.3197
7	69.0158	38.8415	29.2965

Figure 44: O₂ polynomial versus linear leave-one-out error.Figure 45: O₂ cubic spline reconstruction.Figure 46: O₂ spline leave-one-out error, RMS 22.00 mHa.

4.1.8 F₂: fluorine.

Setup. F₂ has eighteen electrons in a (10e, 6o) active space, the largest electron count in any active space here, ten qubits, but only 35 ansatz parameters. Seventeen nodes from 0.9 to 2.5 Å in 0.1 Å steps.

Variational results. F₂ is the campaign's clearest demonstration that circuit *depth*, not molecular size, governs VQE accuracy. It has more active electrons than N₂ or O₂ and the same qubit count, yet its ansatz carries 35 parameters against their 117 and 68, and it reaches mean error 0.0589 mHa and maximum 0.6568 mHa, comfortably inside chemical accuracy (Table 20). Single-bond dissociation is also far less multireference than breaking a triple bond. Hartree-Fock misses 359.7 mHa on this curve; VQE recovers 99.98% of it.

Table 20: F₂ bond length scan: Hartree-Fock, VQE and exact energies (Ha) at every geometry; $\delta = (E_{\text{VQE}} - E_{\text{exact}}) \times 10^3$ (mHa).

r (Å)	E_{HF}	E_{VQE}	E_{exact}	δ
0.9	-195.219409	-195.228895	-195.228895	0.000
1	-195.638041	-195.652560	-195.652560	0.000
1.1	-195.853469	-195.876050	-195.876050	0.000
1.2	-195.951873	-195.986310	-195.986310	0.000
1.3	-195.981228	-196.031793	-196.031793	0.000
1.4	-195.970461	-196.041633	-196.041633	0.000
1.5	-195.938144	-196.034258	-196.034258	0.000
1.6	-195.896120	-196.020869	-196.020870	0.000
1.7	-195.851442	-196.007365	-196.007366	0.000
1.8	-195.807933	-195.996136	-195.996137	0.001
1.9	-195.767510	-195.987687	-195.987690	0.003
2	-195.731098	-195.981740	-195.981746	0.007
2.1	-195.699107	-195.977800	-195.977801	0.001
2.2	-195.671626	-195.975327	-195.975335	0.007
2.3	-195.648491	-195.973870	-195.973900	0.030
2.4	-195.629339	-195.973075	-195.973370	0.295
2.5	-195.613682	-195.972682	-195.973339	0.657

Reconstruction: the polynomial's best case. This is where global polynomial interpolation is vindicated most clearly. At degree 13 the leave-one-out RMS is 0.0221 mHa, against 2.42 mHa for the cubic spline and 31.28 mHa for the linear baseline. The polynomial is 109× better than the spline and 1415× better than the baseline, and it is the only reconstruction in the campaign to land more than an order of magnitude *below* the variational error of the data it is fitting. At that point the interpolation has stopped being the limiting source of error at all.

The conditions that make this work are the ones identified in Section 4.1.2: a smooth, gently varying, single-minimum curve with no nearby complex singularity, sampled densely enough (17 nodes) to support a high degree. The degree sweep (Table 21) still shows a worst-case RMS of 122.14 mHa, confirming that the good behaviour comes from choosing the right degree, not from high degree in general.

Geometry. Polynomial 1.3932 Å and spline 1.3931 Å agree to four decimal places, against an experimental 1.412 Å. The 0.019 Å gap is the smallest geometry error of any diatomic in the campaign and belongs entirely to the model. That two structurally very different interpolants agree this closely is itself evidence that the reconstruction is well determined here.

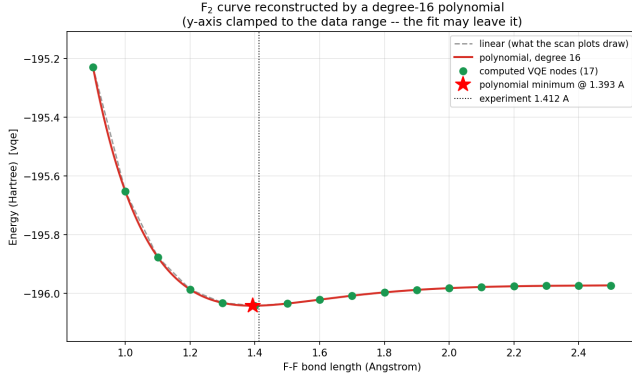


Figure 47: F_2 dissociation curve, polynomial reconstruction at degree 13, the most accurate reconstruction in the campaign.

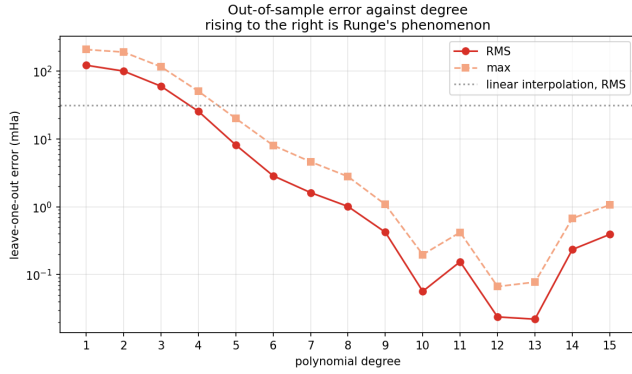


Figure 48: F_2 degree sweep. Best degree 13 at 0.022 mHa RMS.

Table 21: F_2 bond length polynomial degree sweep: leave-one-out error (mHa) against fitted degree d ; the best degree is in bold.

Degree d	max	RMS	MAE
1	208.1189	122.1387	104.6113
2	191.0836	100.2701	84.1729
3	116.2164	60.0132	52.5021
4	50.5991	25.5932	22.4370
5	20.1042	8.1653	6.8857
6	8.0573	2.8694	2.1300
7	4.6188	1.6211	1.1736
8	2.7988	1.0129	0.7095
9	1.0813	0.4218	0.2804
10	0.1979	0.0568	0.0317
11	0.4207	0.1551	0.0822
12	0.0668	0.0238	0.0119
13	0.0777	0.0221	0.0105
14	0.6768	0.2349	0.1033
15	1.0644	0.3923	0.1672

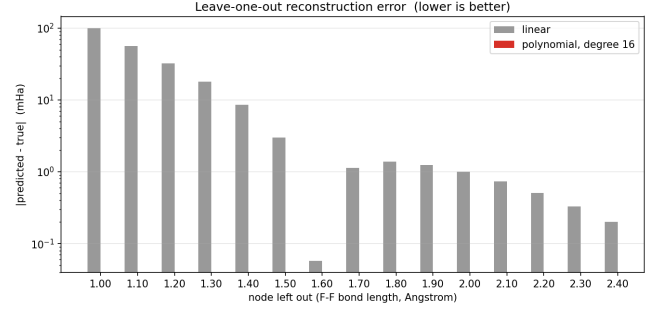


Figure 49: F_2 polynomial versus linear leave-one-out error. Note the logarithmic separation between the two methods.

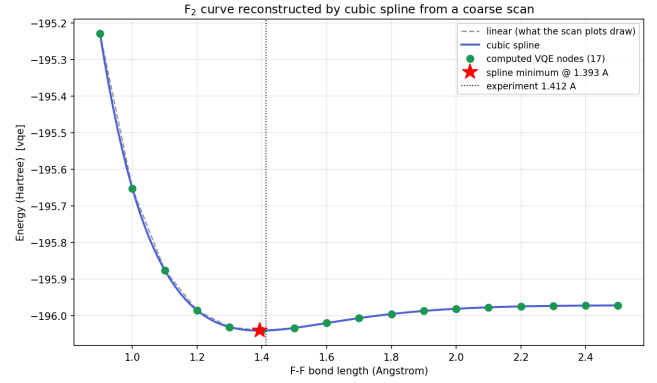


Figure 50: F_2 cubic spline reconstruction, RMS 2.42 mHa.

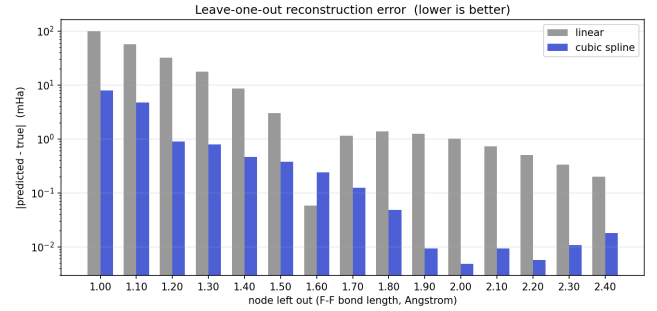


Figure 51: F_2 spline leave-one-out error. A $12.9\times$ improvement on the linear baseline, respectable on its own, but two orders of magnitude behind the polynomial on this curve.

4.1.9 H_2O_2 : hydrogen peroxide.

Setup. H_2O_2 has eighteen electrons in an (8e, 6o) active space, ten qubits, 92 parameters, COBYLA at 1000 evaluations. It is the geometrically richest system in the campaign, the only molecule with a genuine *torsional* degree of freedom. Four scans were run: an O–O stretch (21 nodes, 1.0–3.0 Å), an O–O–H bend (13 nodes, 80–140°), a dihedral twist (19 nodes, 0–180°) and a 10×5 dihedral $\times$ angle surface (50 points).

Variational results. Accuracy is moderate and depends on the scan: twist mean 0.2545 mHa, surface 0.2830 mHa, stretch 0.6311 mHa, bend 0.8882 mHa, all four inside chemical accuracy on the mean, with maxima of 0.45, 0.92, 2.10 and 2.97 mHa respectively. The 92-parameter ansatz with a 1000-evaluation budget is close to the practical limit of what COBYLA can handle, and this system sits right at the boundary between the well-converged and stagnating groups from Section 3.2.

The torsional potential is physically the most interesting curve in the campaign. The computed minimum falls at a dihedral of 90° (Table 24), reflecting the competition between lone-pair repulsion, which favours a perpendicular arrangement, and hyperconjugation, which favours planarity. The experimental gas-phase dihedral is 111.5°; the model’s preference for exactly 90° reflects the minimal basis, and the curve is flat enough near there that the 0.45 mHa variational noise is a non-trivial fraction of the feature being located.

Table 22: H_2O_2 bend angle scan: Hartree–Fock, VQE and exact energies (Ha) at every geometry; $\delta = (E_{\text{VQE}} - E_{\text{exact}}) \times 10^3$ (mHa).

θ (deg)	E_{HF}	E_{VQE}	E_{exact}	δ
80	-148.714504	-148.767895	-148.769078	1.183
85	-148.734474	-148.786281	-148.786483	0.202
90	-148.748650	-148.797650	-148.798071	0.420
95	-148.757081	-148.800381	-148.803354	2.973
100	-148.759960	-148.794902	-148.795601	0.699
105	-148.757557	-148.787872	-148.789150	1.278
110	-148.750190	-148.778444	-148.778812	0.368
115	-148.738201	-148.764620	-148.764906	0.286
120	-148.721952	-148.747332	-148.747617	0.285
125	-148.701825	-148.725833	-148.727161	1.328
130	-148.678225	-148.703047	-148.703855	0.808
135	-148.651589	-148.676904	-148.678126	1.223
140	-148.622398	-148.650035	-148.650529	0.494

Reconstruction. H_2O_2 is the cleanest spline win in the campaign: the spline takes all three 1-D scans.

On the *bend*, spline RMS is 0.776 mHa against 1.252 mHa for the polynomial and 2.440 mHa for linear, a $3.1\times$ improvement on the baseline. On the *stretch*, the margin widens: 1.030 mHa spline, 1.335 mHa polynomial, 14.617 mHa linear, a $14.2\times$ improvement. The stretch degree sweep (Table 26) reaches a worst-case RMS of 1446.8 mHa, the second-largest polynomial failure in the campaign, and the reason is visible in Table 40: the full-degree polynomial places the stretch minimum at 2.975 Å, more than an Ångström past the true value of 1.458 Å and near the far edge of the scanned interval. That is the BeH_2 edge-oscillation failure again, in its most extreme 1-D form.

The *twist* is the one case in the campaign where the global polynomial loses outright to the piecewise-linear baseline, 2.044 mHa

Table 23: H_2O_2 symmetric stretch scan: Hartree–Fock, VQE and exact energies (Ha) at every geometry; $\delta = (E_{\text{VQE}} - E_{\text{exact}}) \times 10^3$ (mHa).

r_{OO} (Å)	E_{HF}	E_{VQE}	E_{exact}	δ
1	-148.356555	-148.375308	-148.375804	0.496
1.1	-148.581109	-148.595731	-148.597497	1.766
1.2	-148.698717	-148.712703	-148.713400	0.697
1.3	-148.751629	-148.765353	-148.765879	0.526
1.4	-148.764399	-148.789524	-148.790117	0.594
1.5	-148.752258	-148.792732	-148.793111	0.379
1.6	-148.725418	-148.785811	-148.786349	0.538
1.7	-148.690988	-148.774269	-148.776365	2.097
1.8	-148.653786	-148.764213	-148.766074	1.860
1.9	-148.616828	-148.756419	-148.756861	0.442
2	-148.581809	-148.748621	-148.749290	0.669
2.1	-148.549568	-148.742545	-148.743419	0.874
2.2	-148.520461	-148.738556	-148.739059	0.502
2.3	-148.494595	-148.735829	-148.735937	0.108
2.4	-148.471936	-148.732292	-148.732553	0.261
2.5	-148.452357	-148.731186	-148.731434	0.248
2.6	-148.435646	-148.730688	-148.730768	0.081
2.7	-148.421533	-148.730267	-148.730403	0.136
2.8	-148.409706	-148.730031	-148.730224	0.193
2.9	-148.399842	-148.729802	-148.730184	0.382
3	-148.391626	-148.729797	-148.730203	0.406

Table 24: H_2O_2 dihedral twist scan: Hartree–Fock, VQE and exact energies (Ha) at every geometry; $\delta = (E_{\text{VQE}} - E_{\text{exact}}) \times 10^3$ (mHa).

ϕ (deg)	E_{HF}	E_{VQE}	E_{exact}	δ
0	-148.746723	-148.784878	-148.785332	0.454
10	-148.747089	-148.785451	-148.785657	0.206
20	-148.748122	-148.786349	-148.786570	0.221
30	-148.749656	-148.787716	-148.787906	0.191
40	-148.751465	-148.789252	-148.789446	0.194
50	-148.753325	-148.790769	-148.790971	0.202
60	-148.755056	-148.792095	-148.792307	0.211
70	-148.756540	-148.793116	-148.793337	0.221
80	-148.757725	-148.793759	-148.794002	0.244
90	-148.758608	-148.794034	-148.794287	0.253
100	-148.759219	-148.793946	-148.794196	0.251
110	-148.759610	-148.793467	-148.793739	0.272
120	-148.759833	-148.792648	-148.792919	0.271
130	-148.759942	-148.791457	-148.791723	0.267
140	-148.759979	-148.789859	-148.790137	0.278
150	-148.759977	-148.787891	-148.788181	0.290
160	-148.759961	-148.785767	-148.786023	0.257
170	-148.759946	-148.783879	-148.784158	0.279
180	-148.759940	-148.783110	-148.783385	0.275

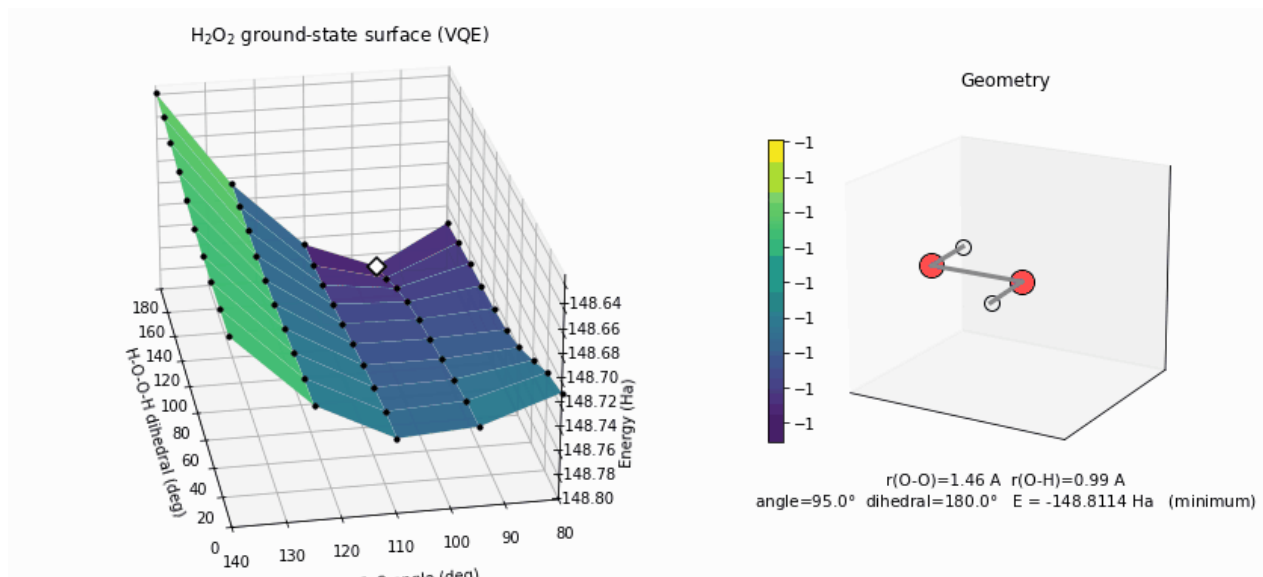


Figure 52: H_2O_2 energy surface over the dihedral $\times$ bend grid (animation frame). The torsional coordinate is the soft direction; the bend is stiffer.

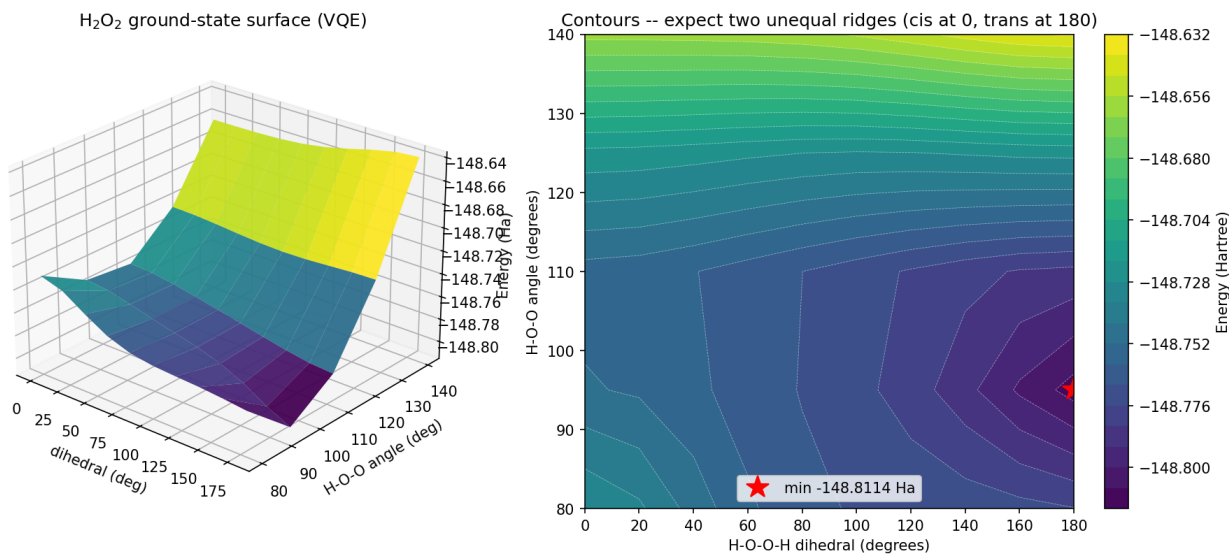


Figure 53: H_2O_2 completed energy surface, static render of the 10×5 grid.

against 1.421 mHa, with the spline narrowly ahead of both at 1.365 mHa. A torsional potential is periodic with a broad flat minimum, exactly the shape a global polynomial handles worst; the degree sweep peaks at 858.5 mHa. The polynomial's interpolated minimum, 167.1° , sits nearly 80° from the computed minimum, and even the spline's 134.3° is off by 44° because the well is so shallow. Where the function is flat, tiny energy errors turn into large positional errors, a statement about conditioning, not about the interpolation method being wrong.

No surface reconstruction. Although the 50-point surface was computed, the repository has no surface entry in either the `h2o2_polynomial/` or `h2o2_spline/` output; only the bend, stretch and twist were post-processed. H_2O_2 therefore contributes no surface row to Table 39. The 10×5 grid would have left only 5×3 points after halving, below the bicubic threshold on one axis, so the omission is defensible, but it is still an omission and we record it as one.

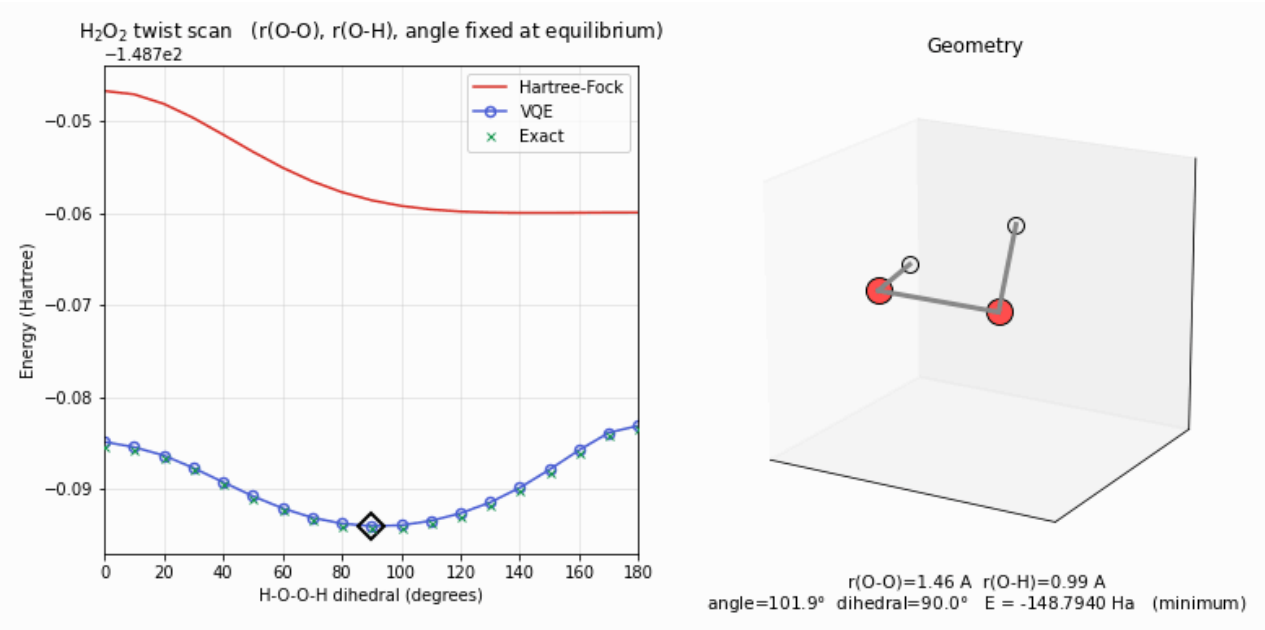


Figure 54: H₂O₂ dihedral twist (animation frame). The molecule rotates about the O–O axis through the full 0–180° range while the energy curve is traced alongside.

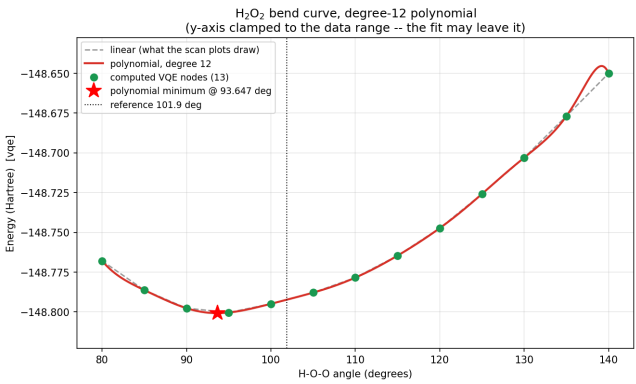


Figure 55: H₂O₂ O–O–H bend, polynomial reconstruction.

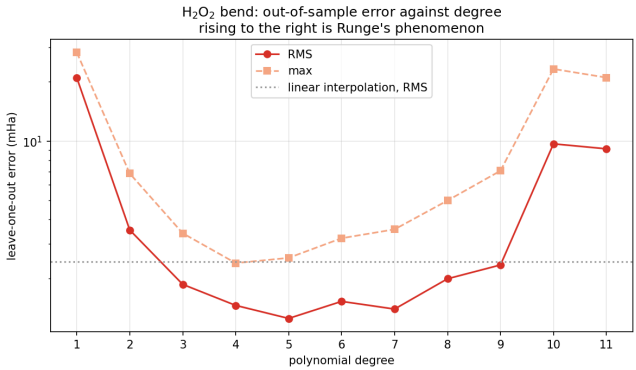


Figure 56: H₂O₂ bend degree sweep; best degree 5.

Table 25: H₂O₂ bend angle polynomial degree sweep: leave-one-out error (mHa) against fitted degree d ; the best degree is in bold.

Degree d	max	RMS	MAE
1	28.2146	20.9762	18.9605
2	6.8346	3.5269	2.7413
3	3.3955	1.8661	1.5398
4	2.3948	1.4589	1.2840
5	2.5453	1.2523	1.0290
6	3.2040	1.5291	1.3109
7	3.5541	1.3976	1.0236
8	4.9817	1.9929	1.4113
9	7.0599	2.3425	1.4044
10	23.2549	9.6742	5.5487
11	21.0222	9.1279	4.9010

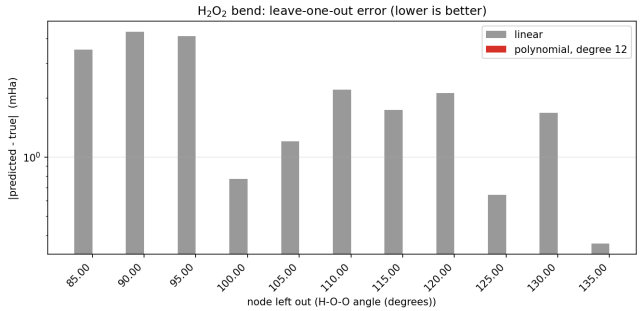


Figure 57: H₂O₂ bend, polynomial versus linear leave-one-out error.

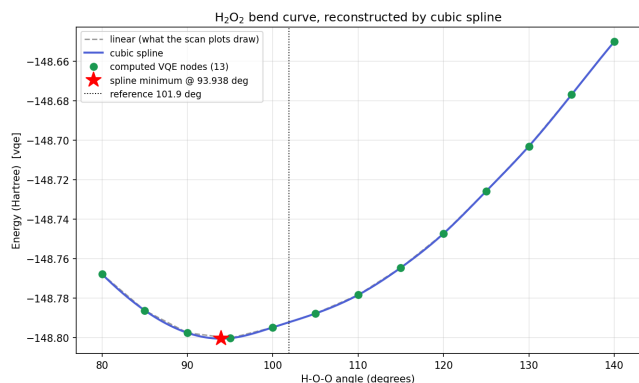


Figure 58: H_2O_2 bend, cubic spline reconstruction, RMS 0.776 mHa.

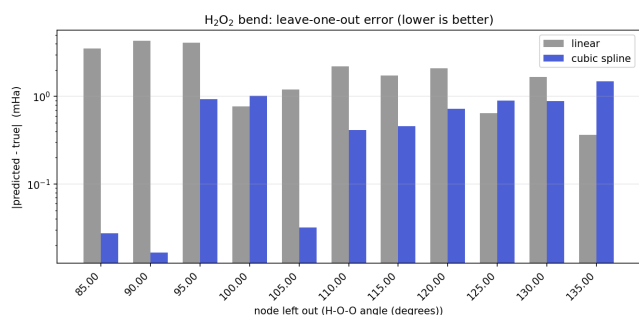


Figure 59: H_2O_2 bend, spline leave-one-out error.

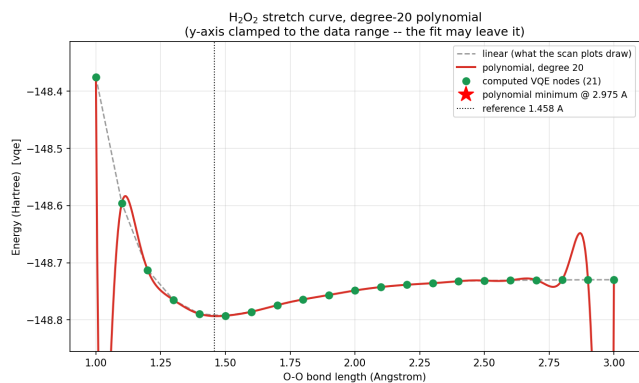


Figure 60: H_2O_2 O–O stretch, polynomial reconstruction. The behaviour near the right edge produces the spurious 2.975Å minimum discussed in the text.

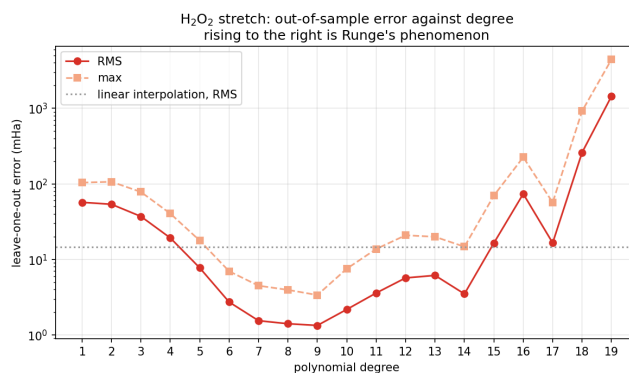


Figure 61: H_2O_2 stretch degree sweep; worst-case RMS 1446.8 mHa.

Table 26: H_2O_2 symmetric stretch polynomial degree sweep: leave-one-out error (mHa) against fitted degree d ; the best degree is in bold.

Degree d	max	RMS	MAE
1	103.8823	56.9259	46.0353
2	106.5620	53.6256	44.8182
3	78.1055	37.0579	32.1022
4	40.4276	19.2324	16.4437
5	17.7383	7.7269	6.6567
6	6.9635	2.7242	2.1064
7	4.4838	1.5437	1.0730
8	3.9464	1.4061	1.1239
9	3.3708	1.3354	1.0558
10	7.5316	2.1751	1.2736
11	13.7322	3.5903	1.8743
12	20.8655	5.6848	2.8409
13	19.9068	6.1381	2.9960
14	14.7073	3.4917	1.4574
15	69.8646	16.4383	5.4986
16	225.2926	73.7564	28.8422
17	56.5335	16.5151	6.4874
18	919.5614	259.8630	94.8950
19	4434.1966	1446.8255	527.1237

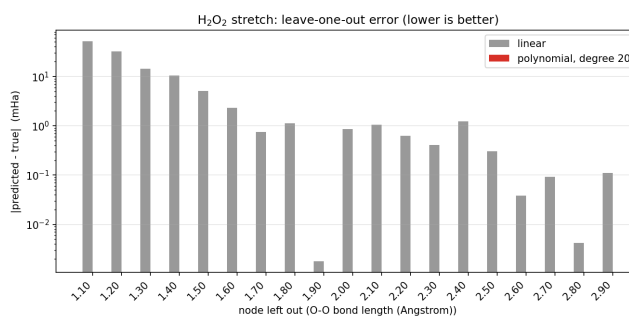


Figure 62: H_2O_2 stretch, polynomial versus linear leave-one-out error.

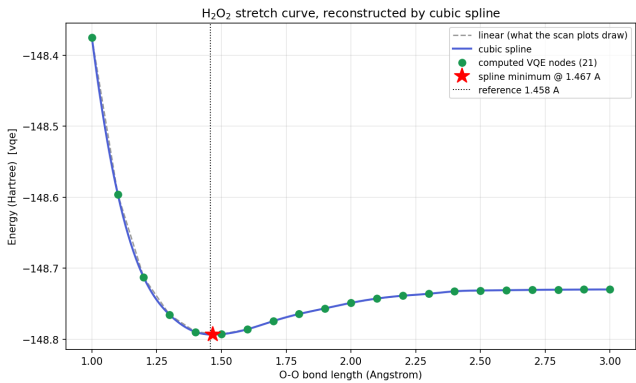


Figure 63: H_2O_2 stretch, cubic spline reconstruction, RMS 1.030 mHa.

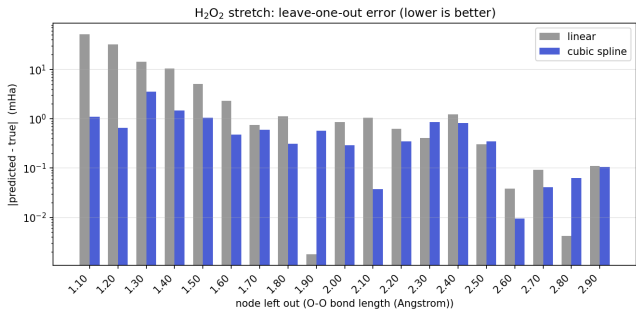


Figure 64: H_2O_2 stretch, spline leave-one-out error.

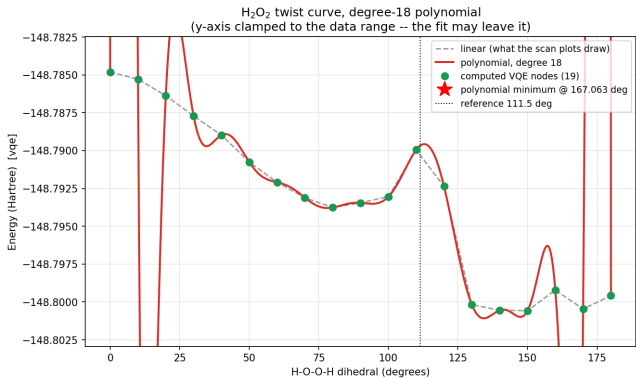


Figure 65: H_2O_2 dihedral twist, polynomial reconstruction. This is the one scan in the campaign where the polynomial is beaten by piecewise-linear interpolation.

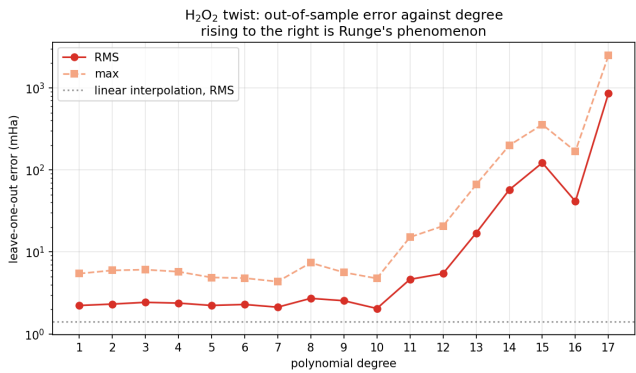


Figure 66: H_2O_2 twist degree sweep; worst-case RMS 858.5 mHa.

Table 27: H_2O_2 dihedral twist polynomial degree sweep: leave-one-out error (mHa) against fitted degree d ; the best degree is in bold.

Degree d	max	RMS	MAE
1	5.4405	2.2236	1.6504
2	5.9582	2.3140	1.6655
3	6.0792	2.4288	1.7216
4	5.7467	2.3789	1.8953
5	4.8826	2.2276	1.5793
6	4.7984	2.2888	1.7817
7	4.3455	2.1211	1.7158
8	7.3751	2.7163	1.9878
9	5.6338	2.5426	1.9562
10	4.7536	2.0442	1.6239
11	15.1352	4.6323	2.8286
12	20.7084	5.4532	2.8669
13	66.1262	16.9995	7.0632
14	198.9544	57.1625	24.0925
15	357.5401	122.0698	50.4368
16	168.6051	41.3574	12.6249
17	2485.2755	858.5445	336.1907

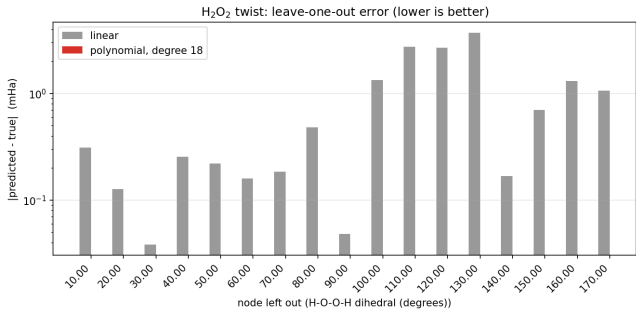


Figure 67: H_2O_2 twist, polynomial versus linear leave-one-out error.

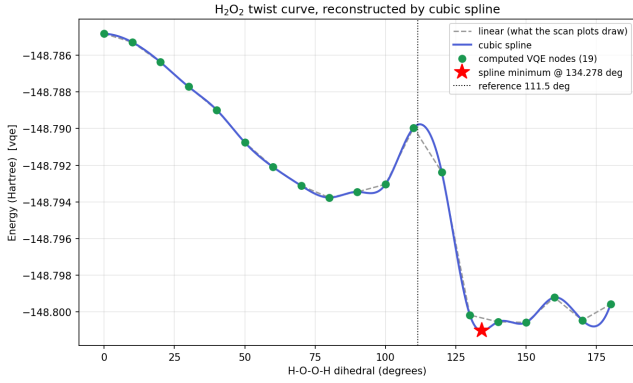


Figure 68: H₂O₂ twist, cubic spline reconstruction of the torsional potential, RMS 1.365 mHa.

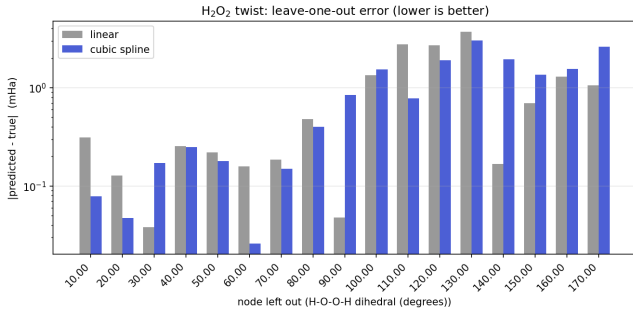


Figure 69: H₂O₂ twist, spline leave-one-out error.

4.1.10 NO₂: nitrogen dioxide.

Setup. NO₂ has twenty-three electrons, an *odd* number, which makes it an open-shell doublet radical ($2S = 1$) needing an ROHF reference. Two active spaces were used: (5e, 4o) on six qubits with 20 parameters for the bend and the surface, and (7e, 5o) on eight qubits with 37 parameters for the stretch. Three scans: bend (19 nodes, 90–180°), symmetric stretch (16 nodes, 0.9–2.4Å) and an 11×10 surface (110 points, joint-largest in the campaign).

Variational results. NO₂ is the campaign’s least accurate system after N₂ and O₂: bend mean 18.93 mHa (maximum 38.58), stretch mean 6.34 mHa (maximum 30.32), surface mean 9.61 mHa (maximum 42.10). All three miss chemical accuracy by a wide margin.

The cause here is different from N₂. The ansatz is small, 20 parameters for the bend, and COBYLA still uses 831 evaluations, plenty for that dimensionality. The difficulty is the open-shell doublet: a 20-parameter hardware-efficient circuit started from an ROHF reference has limited ability to represent the correct spin state, so the variational minimum it can reach genuinely sits above the exact ground state. This is *ansatz-expressivity* error rather than optimiser stagnation, and it is the one place in the campaign where the two causes separate cleanly: N₂ had a large ansatz and too few evaluations; NO₂ has enough evaluations and too small an ansatz.

Reconstruction. The NO₂ bend is the campaign’s only case where the cubic spline is beaten by the *linear* baseline: spline 7.258 mHa

Table 28: NO₂ bend angle scan: Hartree–Fock, VQE and exact energies (Ha) at every geometry; $\delta = (E_{\text{VQE}} - E_{\text{exact}}) \times 10^3$ (mHa).

θ (deg)	E_{HF}	E_{VQE}	E_{exact}	δ
90	-201.192217	-201.249413	-201.249413	0.000
95	-201.202353	-201.263014	-201.263014	0.000
100	-201.162788	-201.248025	-201.249429	1.403
105	-201.197736	-201.253590	-201.264120	10.530
110	-201.223419	-201.252311	-201.288507	36.196
115	-201.241848	-201.268871	-201.305991	37.120
120	-201.254512	-201.280054	-201.317954	37.900
125	-201.262530	-201.286870	-201.325445	38.575
130	-201.266755	-201.291332	-201.329274	37.942
135	-201.267863	-201.294375	-201.330082	35.707
140	-201.266406	-201.301472	-201.328399	26.927
145	-201.262865	-201.301727	-201.324693	22.966
150	-201.257690	-201.299797	-201.319415	19.618
155	-201.251353	-201.296612	-201.313040	16.429
160	-201.244391	-201.292950	-201.306124	13.174
165	-201.237464	-201.289407	-201.299342	9.935
170	-201.231400	-201.286562	-201.293512	6.950
175	-201.227163	-201.284887	-201.289520	4.633
180	-201.225626	-201.284426	-201.288094	3.669

Table 29: NO₂ symmetric stretch scan: Hartree–Fock, VQE and exact energies (Ha) at every geometry; $\delta = (E_{\text{VQE}} - E_{\text{exact}}) \times 10^3$ (mHa).

r (Å)	E_{HF}	E_{VQE}	E_{exact}	δ
0.9	-200.277971	-200.322049	-200.322049	0.000
1	-200.893684	-200.947393	-200.947393	0.000
1.1	-201.174147	-201.236666	-201.236666	0.000
1.2	-201.270208	-201.339202	-201.339202	0.000
1.3	-201.266809	-201.350007	-201.369011	19.004
1.4	-201.212115	-201.305845	-201.324876	19.031
1.5	-201.135184	-201.226384	-201.242355	15.971
1.6	-201.060449	-201.119861	-201.130914	11.053
1.7	-201.003196	-201.034530	-201.037817	3.287
1.8	-200.964938	-200.974190	-200.974660	0.470
1.9	-200.940909	-200.942936	-200.942974	0.039
2	-200.926344	-200.927545	-200.927553	0.009
2.1	-200.914352	-200.914754	-200.914755	0.001
2.2	-200.882229	-200.887391	-200.917712	30.320
2.3	-200.888450	-200.888998	-200.890390	1.392
2.4	-200.886437	-200.887404	-200.888322	0.918

against linear 5.049 mHa, with the polynomial best at 3.924 mHa. The explanation is the 18.93 mHa mean variational error. The spline’s advantage rests on the data being a smooth function sampled exactly; when each node carries an independent error comparable to the curvature being fitted, the C^2 constraint spreads that noise into neighbouring intervals, while piecewise-linear interpolation keeps each error inside its own two subintervals. The general caveat that follows is important: spline superiority assumes accurate data, and that assumption can be tested.

On the stretch, where the variational error is a third as large, normal service resumes: polynomial 5.233 mHa, spline 8.469 mHa, linear 54.194 mHa, the polynomial improving on the baseline by 10.4×.

On the surface the spline wins clearly, 38.28 mHa against 46.23 mHa for the total-degree polynomial and 112.96 mHa for bilinear, and the polynomial’s in-sample residual of 122.5 mHa shows it cannot even represent the fitted points, let alone the held-out ones.

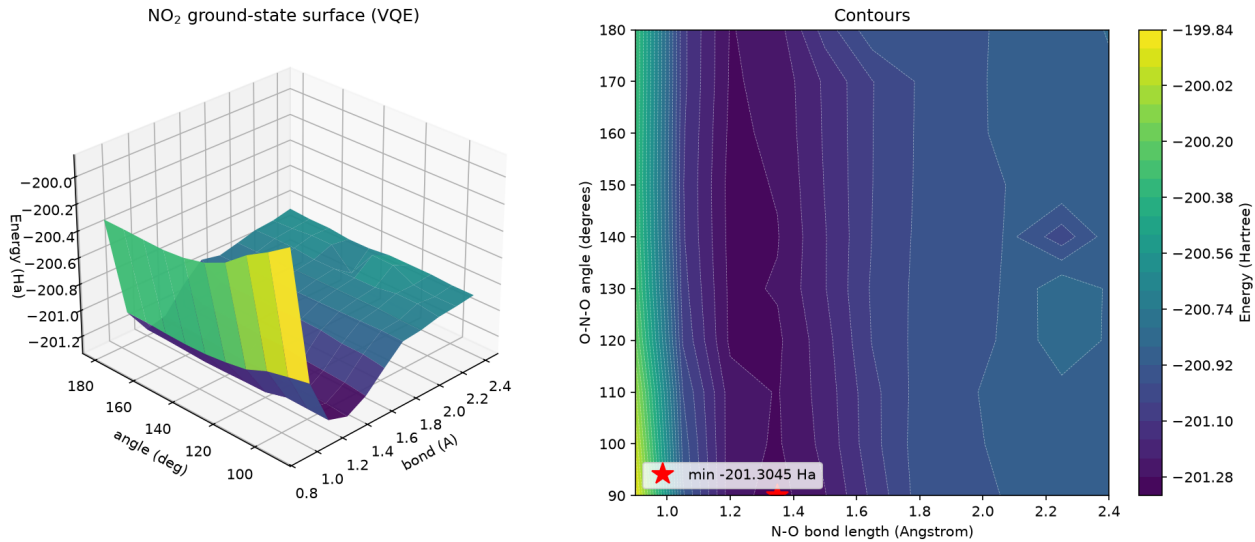


Figure 70: NO₂ energy surface over the bond-length $\times$ bond-angle grid, 110 VQE solves.

Geometry. The bend reconstructions disagree sharply. The spline places the minimum at 142.27° against a reference of 134.3° , an error of 8.0° . The full-degree polynomial places it at 173.43° , an error of 39.1° , close enough to the linear geometry to be qualitatively wrong about the molecule's shape. The stretch shows the same pattern in exaggerated form: spline $1.265\text{\AA}$ versus reference $1.194\text{\AA}$ (error $0.071\text{\AA}$), polynomial $2.373\text{\AA}$ (error $1.179\text{\AA}$, at the far edge of the interval). The degree sweeps reach worst-case RMS values of 2170.2 mHa and 1071.8 mHa respectively, the largest 1-D polynomial failures in the campaign.

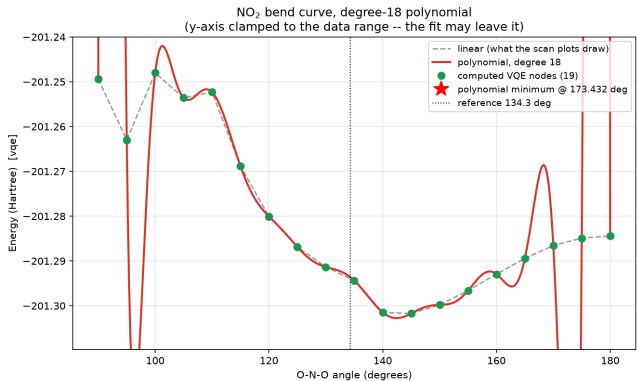


Figure 71: NO₂ bend, polynomial reconstruction.

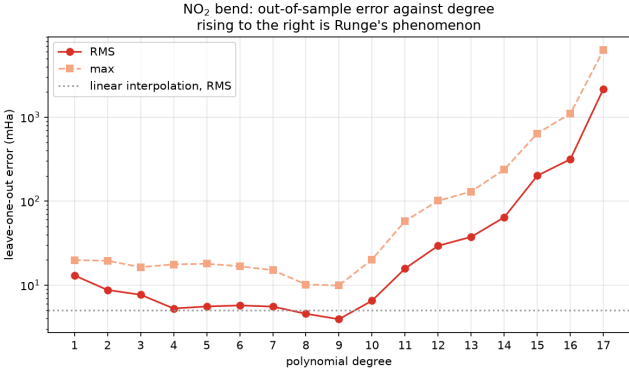


Figure 72: NO₂ bend degree sweep; worst-case RMS 2170.2 mHa .

Table 30: NO₂ bend angle polynomial degree sweep: leave-one-out error (mHa) against fitted degree d ; the best degree is in bold.

Degree d	max	RMS	MAE
1	19.8146	13.0253	11.5843
2	19.5311	8.7503	6.7960
3	16.3382	7.6784	6.3879
4	17.6620	5.2658	2.8707
5	17.9401	5.5656	3.4448
6	16.7375	5.7283	3.8371
7	15.0567	5.5441	4.2174
8	10.1705	4.5566	3.6836
9	9.8915	3.9242	2.6250
10	20.1489	6.5159	4.0064
11	57.8952	15.7515	7.8352
12	100.9952	29.3955	14.3663
13	130.0082	37.4490	17.1481
14	237.0671	64.1154	26.3076
15	639.7099	201.0126	82.6372
16	1101.9309	314.9069	122.6804
17	6282.2910	2170.2329	849.8245

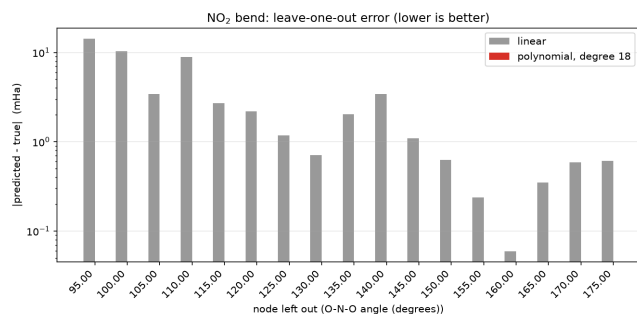


Figure 73: NO₂ bend, polynomial versus linear leave-one-out error.

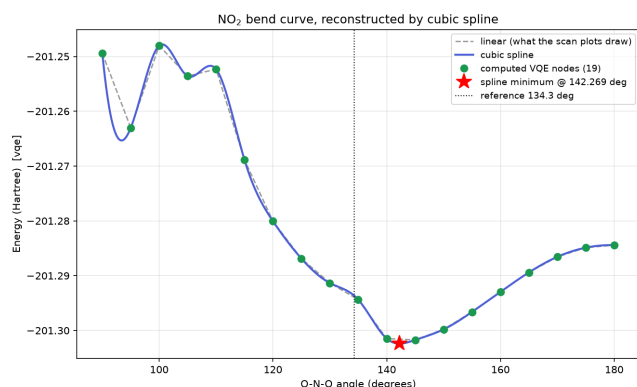


Figure 74: NO₂ bend, cubic spline reconstruction.

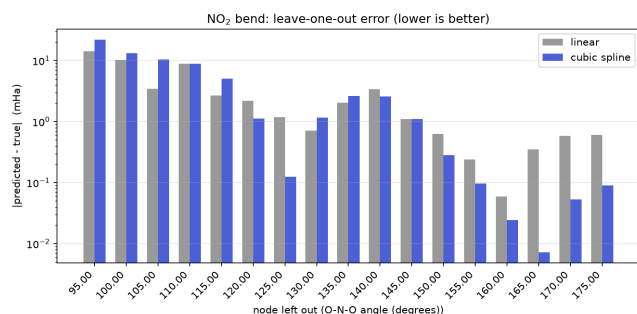


Figure 75: NO₂ bend, spline leave-one-out error. At 7.26 mHa RMS this is worse than the 5.05 mHa linear baseline, the campaign's one spline defeat, caused by the 18.9 mHa variational noise in the underlying data.

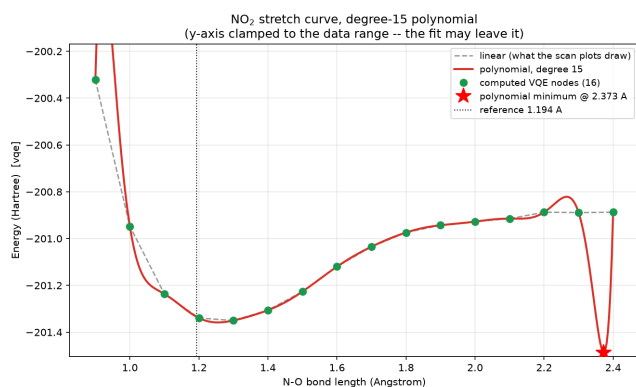


Figure 76: NO₂ symmetric stretch, polynomial reconstruction.

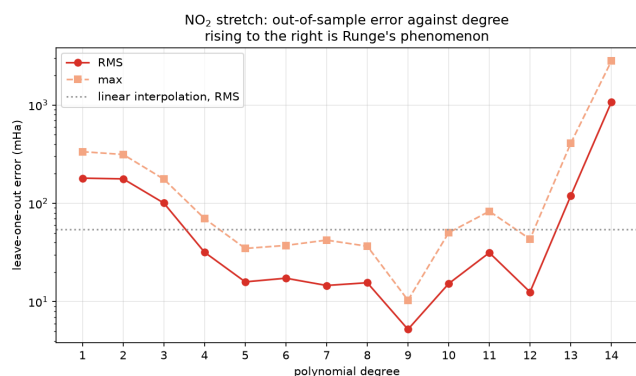


Figure 77: NO₂ stretch degree sweep.

Table 31: NO₂ symmetric stretch polynomial degree sweep: leave-one-out error (mHa) against fitted degree d ; the best degree is in bold.

Degree d	max	RMS	MAE
1	335.3914	180.4174	148.4173
2	314.1513	177.4883	153.4797
3	176.2432	100.7122	89.5247
4	70.0349	31.8396	25.7123
5	34.8118	15.9166	12.7452
6	37.3378	17.3319	14.6221
7	42.2889	14.6234	10.3301
8	36.6522	15.5889	10.4453
9	10.3533	5.2332	4.2782
10	50.3125	15.2589	8.7097
11	83.0536	31.5000	17.0648
12	43.3194	12.5409	6.7376
13	407.8067	119.1430	51.8706
14	2805.3755	1071.7516	479.7738

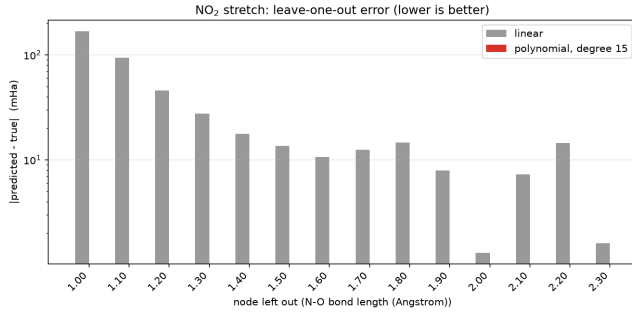


Figure 78: NO₂ stretch, polynomial versus linear leave-one-out error.

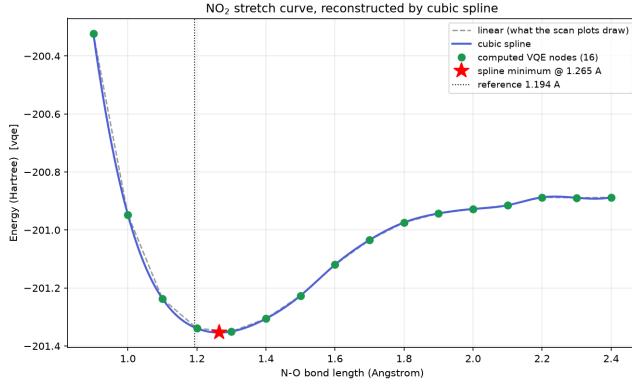


Figure 79: NO₂ stretch, cubic spline reconstruction.

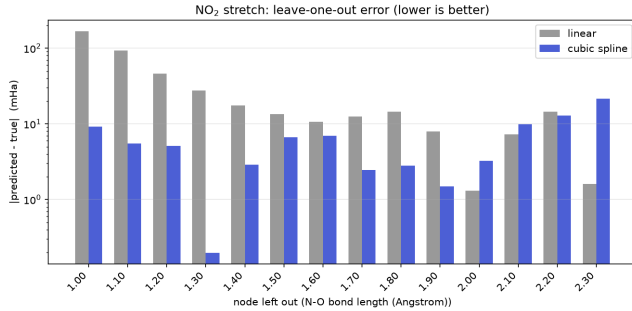


Figure 80: NO₂ stretch, spline leave-one-out error, RMS 8.47 mHa.

4.1.11 N₂O: nitrous oxide.

Setup. N₂O has twenty-two electrons in a linear triatomic with two inequivalent bonds, which makes it the most thoroughly decomposed system in the campaign: the N–N and N–O stretches were scanned *separately*, plus a bend, plus a two-dimensional $r_{\text{NN}} \times r_{\text{NO}}$ surface on a 10×6 grid at fixed linear geometry. The bend and surface used (4e, 4o) on six qubits (26 parameters); the two stretches used (6e, 5o) on eight qubits (54 parameters).

Variational results. Accuracy is poor throughout: bend mean 9.07 mHa, N–N stretch 8.69 mHa, N–O stretch 2.72 mHa, surface 6.98 mHa. None reaches chemical accuracy. The (4e, 4o) active space used for the bend and surface is plainly too small for a 22-electron molecule with two multiple bonds; it correlates only four electrons out of twenty-two. The N–O stretch, which used the larger (6e, 5o) space, is correspondingly the most accurate of the four, and that correlation between active-space size and accuracy is the clearest single-molecule evidence in the campaign that the active space, not the optimiser, is the binding constraint here.

Table 32: N₂O bend angle scan: Hartree–Fock, VQE and exact energies (Ha) at every geometry; $\delta = (E_{\text{VQE}} - E_{\text{exact}}) \times 10^3$ (mHa).

θ (deg)	E_{HF}	E_{VQE}	E_{exact}	δ
90	-181.020241	-181.088133	-181.095279	7.145
95	-181.010381	-181.075638	-181.087151	11.513
100	-181.006934	-181.065925	-181.074079	8.154
105	-181.015267	-181.064291	-181.082783	18.491
110	-181.032663	-181.081917	-181.101767	19.850
115	-181.054324	-181.112432	-181.115480	3.048
120	-181.076910	-181.143284	-181.149129	5.844
125	-181.098411	-181.164259	-181.176590	12.331
130	-181.117703	-181.189586	-181.199002	9.417
135	-181.134324	-181.209673	-181.217386	7.713
140	-181.148277	-181.225636	-181.232551	6.915
145	-181.159787	-181.238507	-181.245117	6.610
150	-181.169136	-181.249004	-181.255556	6.551
155	-181.176593	-181.257649	-181.264224	6.574
160	-181.182389	-181.264675	-181.271365	6.689
165	-181.186709	-181.270273	-181.277101	6.828
170	-181.189696	-181.270574	-181.281411	10.837
175	-181.191449	-181.273158	-181.284138	10.980
180	-181.192027	-181.278247	-181.285079	6.832

Table 33: N₂O N–N stretch scan: Hartree–Fock, VQE and exact energies (Ha) at every geometry; $\delta = (E_{\text{VQE}} - E_{\text{exact}}) \times 10^3$ (mHa).

r_{NN} (Å)	E_{HF}	E_{VQE}	E_{exact}	δ
0.9	-180.847700	-180.894792	-180.903393	8.601
1	-181.093002	-181.155884	-181.165067	9.183
1.1	-181.183836	-181.266783	-181.273831	7.048
1.2	-181.190761	-181.297568	-181.300893	3.325
1.3	-181.155524	-181.287329	-181.288350	1.022
1.4	-181.102893	-181.255671	-181.257457	1.786
1.5	-181.047528	-181.210977	-181.216738	5.761
1.6	-180.997330	-181.159528	-181.174080	14.552
1.7	-180.955065	-181.116350	-181.139861	23.511
1.8	-180.920363	-181.098984	-181.117843	18.859
1.9	-180.891819	-181.092540	-181.104964	12.424
2	-180.868085	-181.089781	-181.097549	7.767
2.1	-180.848130	-181.088461	-181.093260	4.799
2.2	-180.831204	-181.087759	-181.090751	2.992

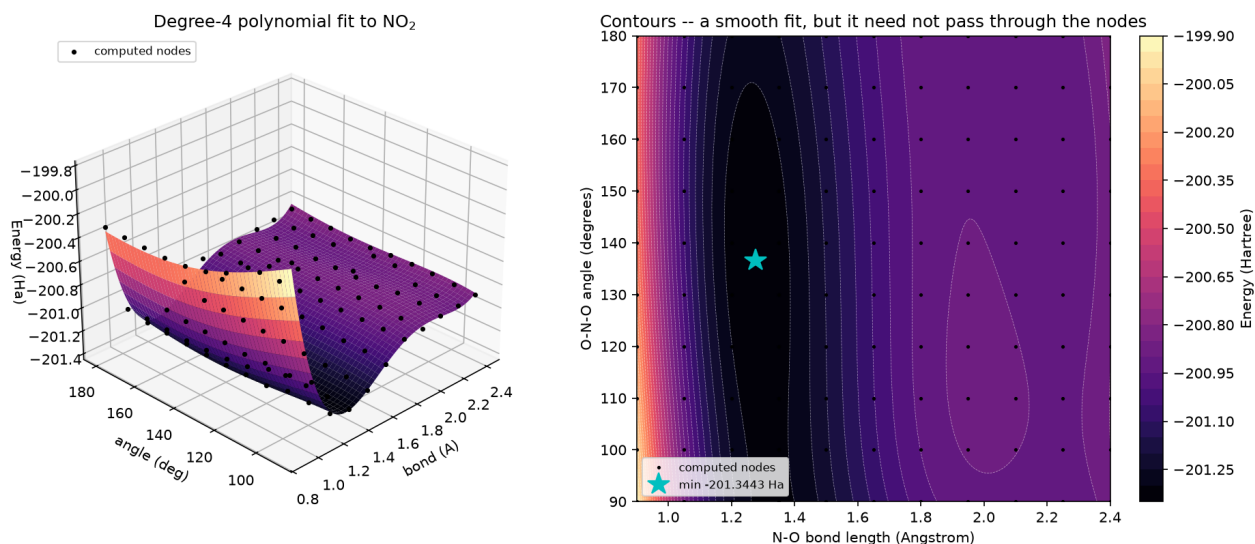


Figure 81: NO₂ surface, total-degree polynomial reconstruction. The in-sample residual alone is 122.5 mHa.

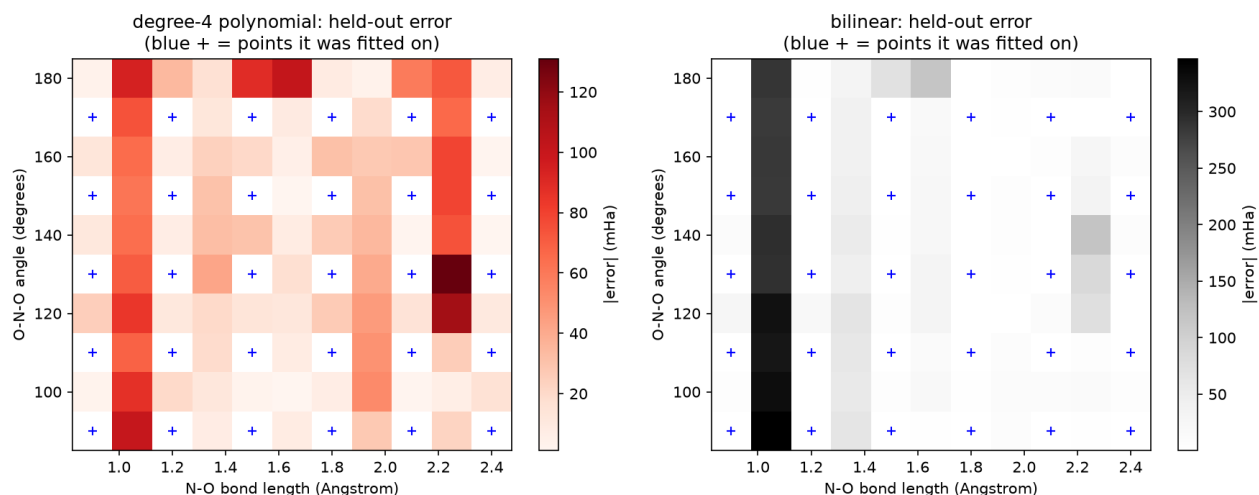


Figure 82: NO₂ surface, polynomial holdout error.

Reconstruction: the campaign's worst failure. The N₂O surface produces the most dramatic failure in this report. The total-degree polynomial has an in-sample residual of 30.7 mHa, already poor, and a *half-grid holdout RMS* of 5186.4 mHa, with a maximum error of 10362.8 mHa. That is an error of more than ten Hartree on a surface whose total energy variation is a fraction of one Hartree. The reconstruction is off by four orders of magnitude, and 69× worse than the trivial bilinear baseline of 74.7 mHa.

The mechanism is the two-dimensional version of edge oscillation. A total-degree fit on a 5 × 3 subgrid, what remains of a 10 × 6 grid after halving, has more coefficients than the data can pin down in the short direction, so the fit is effectively unconstrained along one axis and drifts away from the fitted points. The bicubic spline on the same subgrid achieves 43.2 mHa, comfortably beating

bilinear's 74.7 mHa. This is the strongest argument in the report for using splines by default on surfaces: the spline degrades gracefully where the polynomial fails catastrophically, and on an expensive surface there is no way to find out which regime you are in after the fact.

1-D reconstructions. The one-dimensional scans behave far better, and, unusually, split their verdicts three ways. On the *bend*, the spline wins at 1.895 mHa against 2.627 mHa polynomial and 3.502 mHa linear. On the *N-N stretch*, the spline wins again at 2.526 mHa against 3.812 mHa and 25.902 mHa, a 10.3× improvement on the baseline. But on the *N-O stretch* the polynomial wins decisively: 0.119 mHa at degree 12 against 1.741 mHa spline and 26.824 mHa linear, a factor of 225 over the baseline.

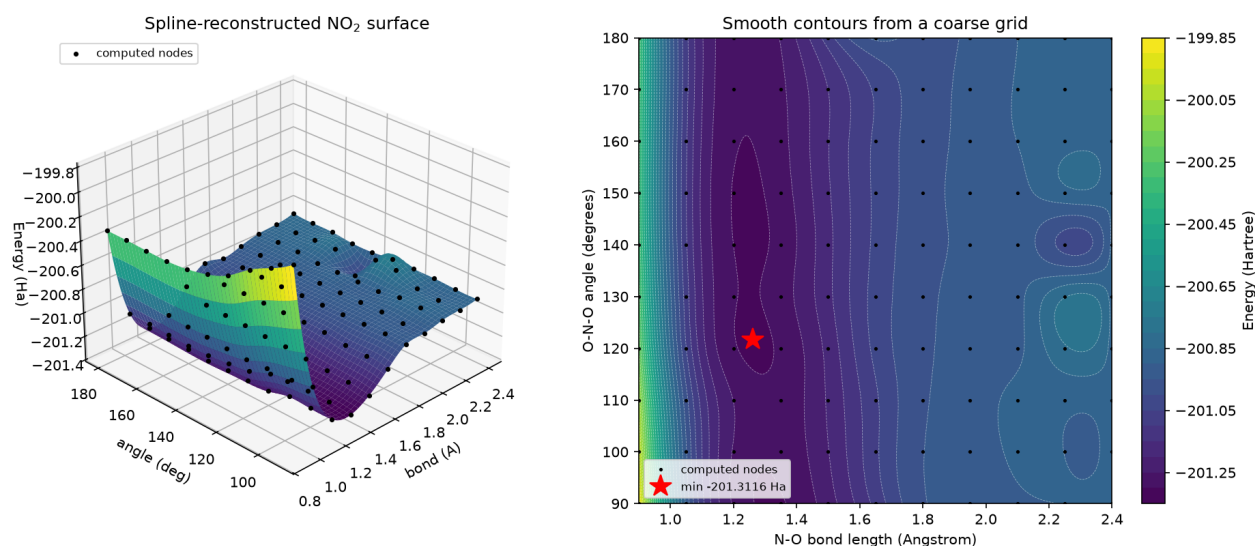


Figure 83: NO₂ surface, bicubic spline reconstruction from the half grid.

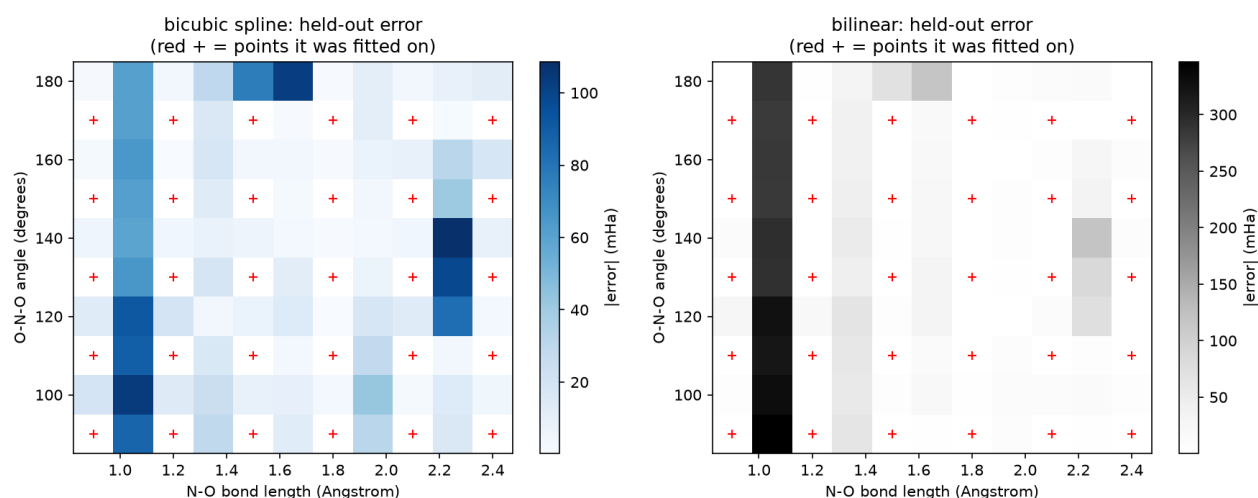


Figure 84: NO₂ surface, spline holdout error, RMS 38.28 mHa against 112.96 mHa bilinear.

The same molecule, scanned along two chemically similar coordinates with the same active space and node spacing, gives opposite verdicts, and that is the sharpest demonstration in the report of its central methodological claim: *which interpolant wins is an empirical property of the particular curve, and has to be measured rather than assumed*. The N–O curve happens to be smooth enough over its interval to support a degree-12 fit; the N–N curve, spanning a stronger bond over a wider effective range, is not.

Geometry. The N–O stretch reconstructions are excellent, placing the minimum at 1.1924 (polynomial) and 1.1927 Å (spline) against a reference of 1.185 Å, errors below 0.008 Å from a 0.15 Å grid. The N–N stretch is worse, 1.213 and 1.216 Å against 1.128 Å, errors of roughly 0.086 Å that reflect the small active space rather than the interpolation. On the bend, the spline correctly returns

the linear geometry 180.0° exactly, while the polynomial reports 173.48°, once again inventing a spurious interior minimum where the true minimum sits at an endpoint.

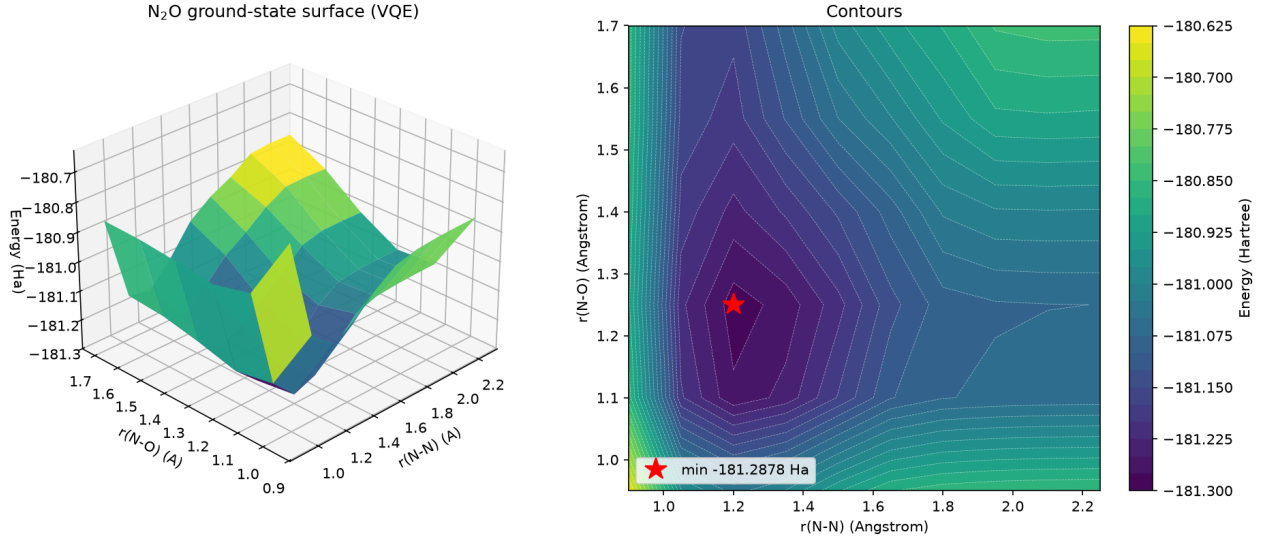

 Figure 85: N₂O energy surface over the two independent bond lengths at fixed linear geometry, 60 VQE solves.

 Table 34: N₂O N–O stretch scan: Hartree–Fock, VQE and exact energies (Ha) at every geometry; $\delta = (E_{\text{VQE}} - E_{\text{exact}}) \times 10^3$ (mHa).

r_{NO} (Å)	E_{HF}	E_{VQE}	E_{exact}	δ
0.9	-180.749006	-180.859944	-180.861662	1.718
1	-181.026828	-181.139886	-181.140782	0.896
1.1	-181.151195	-181.256193	-181.258391	2.198
1.2	-181.195496	-181.281341	-181.288287	6.947
1.3	-181.202329	-181.262330	-181.274458	12.128
1.4	-181.195642	-181.230850	-181.241697	10.847
1.5	-181.187179	-181.204728	-181.209500	4.773
1.6	-181.180884	-181.189138	-181.190220	1.082
1.7	-181.176903	-181.181594	-181.181757	0.163
1.8	-181.174373	-181.178067	-181.178089	0.023
1.9	-181.172581	-181.176094	-181.176097	0.004
2	-181.171143	-181.174678	-181.174679	0.001
2.1	-181.169889	-181.173480	-181.173481	0.001
2.2	-181.168758	-181.172401	-181.172401	0.000
2.3	-181.167736	-181.171419	-181.171419	0.000

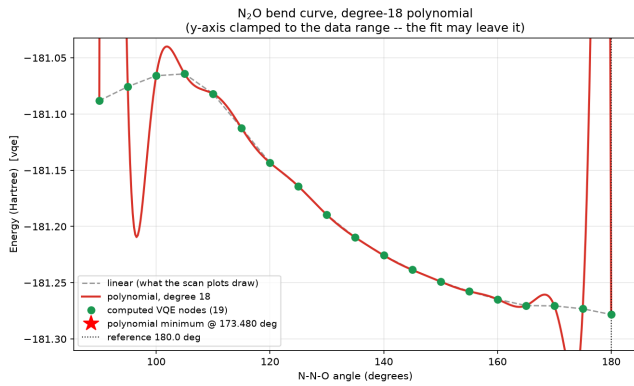
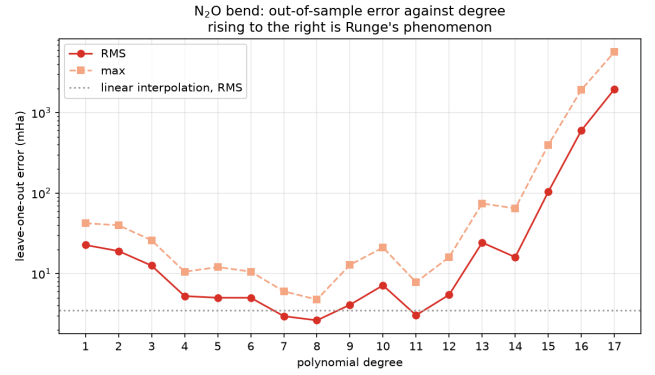

 Figure 86: N₂O bend, polynomial reconstruction.

 Figure 87: N₂O bend degree sweep; worst-case RMS 1959.1 mHa.

 Table 35: N₂O bend angle polynomial degree sweep: leave-one-out error (mHa) against fitted degree d ; the best degree is in bold.

Degree d	max	RMS	MAE
1	42.2695	22.6748	19.3340
2	39.8830	19.0849	16.0842
3	25.9818	12.5738	10.9506
4	10.5461	5.2566	4.3113
5	12.0648	5.0184	3.5877
6	10.5839	5.0119	4.0296
7	6.0250	2.9613	2.4321
8	4.7589	2.6268	2.1837
9	12.8454	4.0973	2.7974
10	21.1114	7.1502	4.5029
11	7.8085	3.0411	2.1155
12	16.0187	5.4564	3.4003
13	74.4241	24.3174	11.3541
14	64.6002	16.0240	5.9935
15	396.7793	103.9210	39.5718
16	1921.8225	600.7182	239.2475
17	5671.2321	1959.1411	767.1648

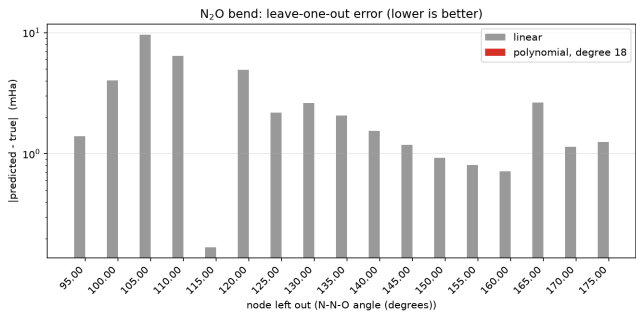


Figure 88: N₂O bend, polynomial versus linear leave-one-out error.

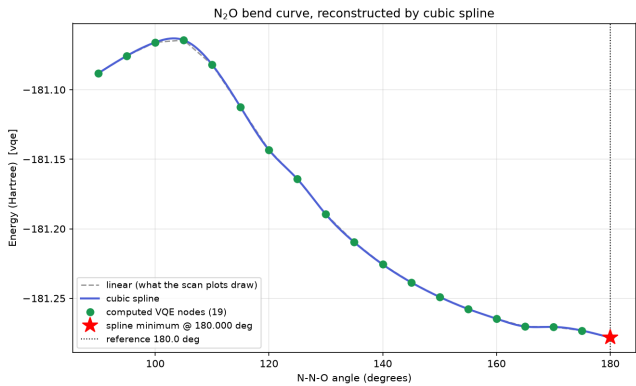


Figure 89: N₂O bend, cubic spline reconstruction.

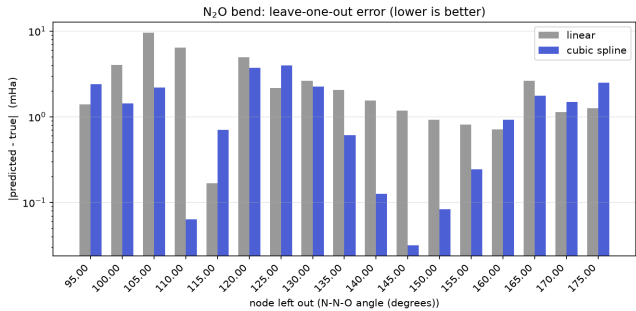


Figure 90: N₂O bend, spline leave-one-out error, RMS 1.895 mHa.

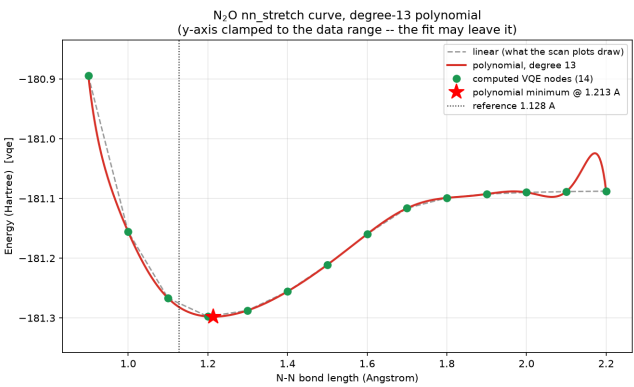


Figure 91: N₂O N–N stretch, polynomial reconstruction.

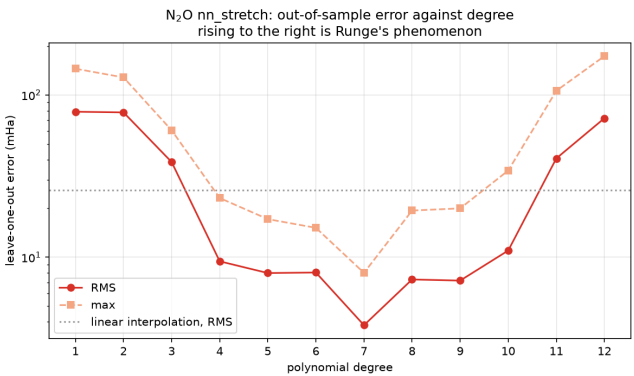


Figure 92: N₂O N–N stretch degree sweep; best degree 7.

Table 36: N₂O N–N stretch polynomial degree sweep: leave-one-out error (mHa) against fitted degree d ; the best degree is in bold.

Degree d	max	RMS	MAE
1	145.0041	78.7828	65.8487
2	128.1584	78.0175	68.5992
3	60.5396	38.5869	34.7940
4	23.1560	9.4348	7.3997
5	17.1673	7.9891	6.3352
6	15.1735	8.0507	6.6646
7	8.0099	3.8120	3.1057
8	19.3879	7.2945	5.0476
9	19.9984	7.1784	4.7093
10	34.2069	10.9983	6.0242
11	106.0466	40.6286	21.5088
12	173.3713	71.8440	36.0606

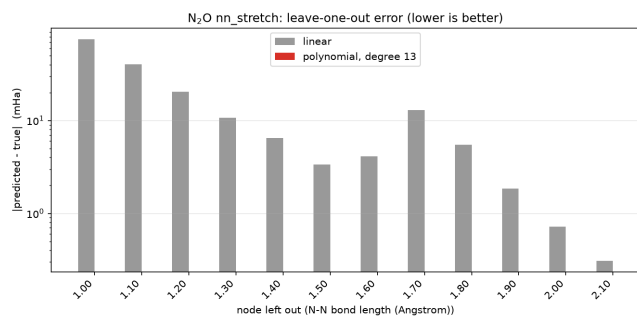


Figure 93: N_2O N–N stretch, polynomial versus linear leave-one-out error.

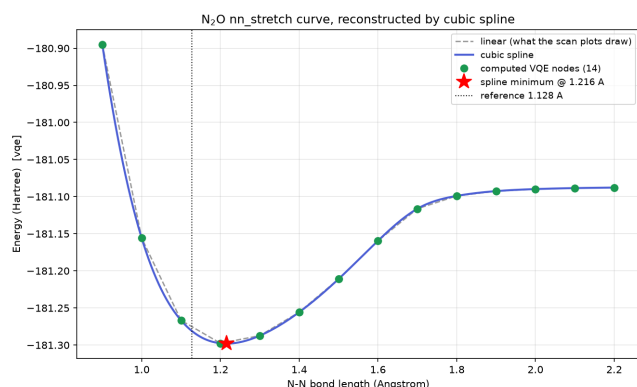


Figure 94: N_2O N–N stretch, cubic spline reconstruction, RMS 2.526 mHa.

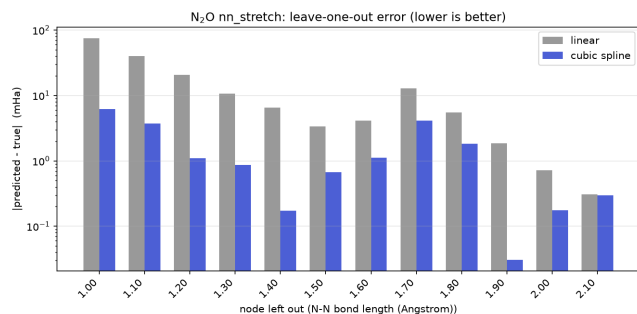


Figure 95: N_2O N–N stretch, spline leave-one-out error.

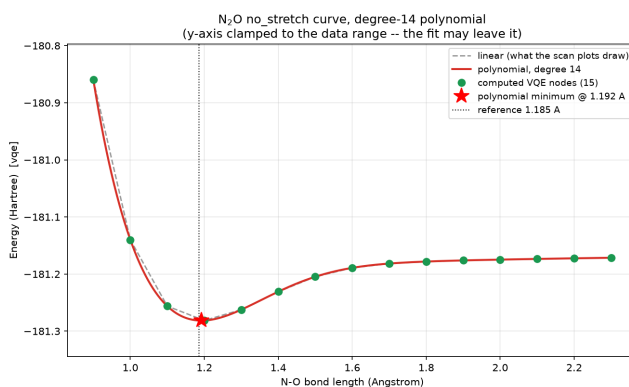


Figure 96: N_2O N–O stretch, polynomial reconstruction at degree 12, RMS 0.119 mHa, the second-best reconstruction in the campaign.

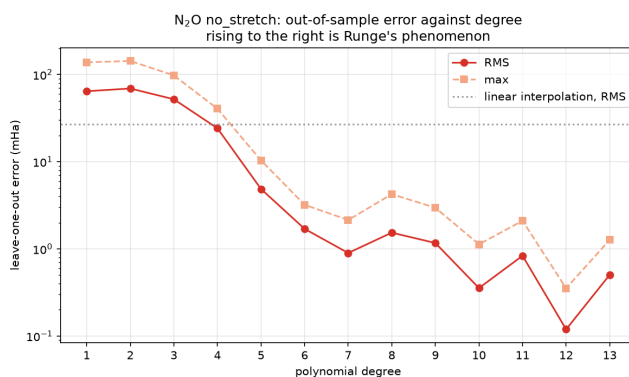


Figure 97: N_2O N–O stretch degree sweep. Contrast the shape with the N–N sweep (Fig. 92), the same molecule, opposite verdicts.

Table 37: N_2O N–O stretch polynomial degree sweep: leave-one-out error (mHa) against fitted degree d ; the best degree is in bold.

Degree d	max	RMS	MAE
1	138.0157	64.3447	46.4402
2	143.1219	69.1727	57.3741
3	97.8883	52.0650	44.7846
4	40.5530	24.1991	21.4597
5	10.3508	4.8643	4.0952
6	3.2071	1.6978	1.4323
7	2.1444	0.8976	0.7764
8	4.2406	1.5355	1.0663
9	2.9648	1.1689	0.7307
10	1.1288	0.3554	0.1969
11	2.1007	0.8345	0.4423
12	0.3536	0.1194	0.0584
13	1.2711	0.5049	0.2386

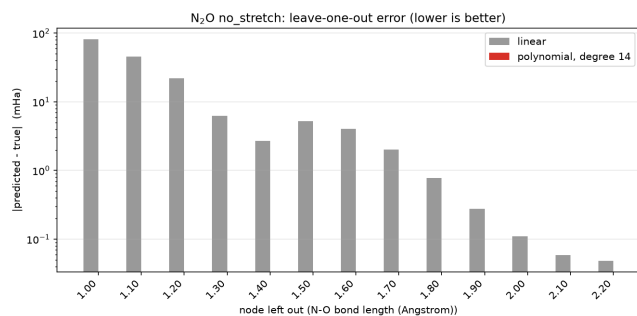


Figure 98: N_2O N–O stretch, polynomial versus linear leave-one-out error.

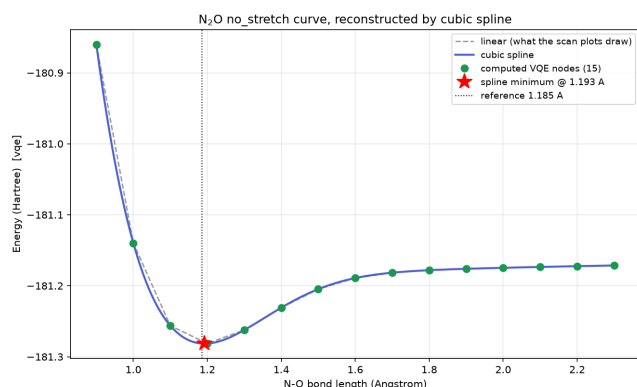


Figure 99: N_2O N–O stretch, cubic spline reconstruction, RMS 1.741 mHa.

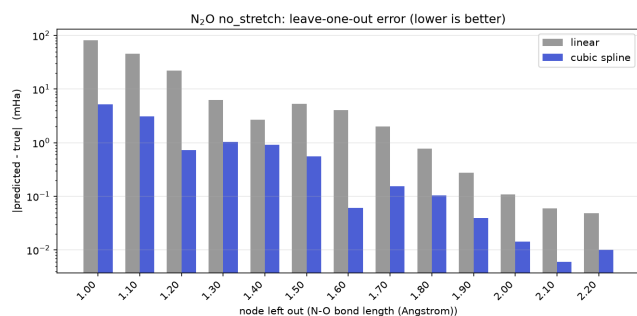


Figure 100: N_2O N–O stretch, spline leave-one-out error.

4.2 Campaign-wide accuracy

Table 38 gives the variational error against exact diagonalisation for every scan. Three things stand out.

4.2.1 The variational bound holds everywhere. Across all 692 optimisations, $E_{\text{VQE}} \geq E_{\text{exact}}$ with no exception. This is the correctness check Section 2.1 promised: an energy below the exact reference would point to a Hamiltonian construction error or a mismatched active space, and we found none.

4.2.2 Accuracy tracks ansatz depth, not molecule size. The scans split cleanly into two groups. Systems with modest parameter counts, H_2 ($\bar{\Delta} < 0.001$ mHa), H_2O (0.0007 mHa), LiH (0.0001 mHa), the BeH_2 bend (0.0044 mHa), the NH_3 bend (0.0158 mHa), reach chemical accuracy by three or four orders of magnitude. Systems with the deepest circuits, N_2 ($p = 117$, $\bar{\Delta} = 10.48$ mHa), O_2 ($p = 68$, 10.55 mHa), the NO_2 bend (18.93 mHa), fall short of it.

This is not a failure of the variational principle. It is optimiser stagnation: COBYLA gets 1000 evaluations, and on a 117-parameter landscape that budget runs out before convergence. The evidence is direct: the scans that miss chemical accuracy are exactly the ones whose evaluation counts saturate at the 1000-evaluation ceiling, while SLSQP on LiH converges in 167 evaluations and on NH_3 in 255–348, comfortably inside budget, and both reach chemical accuracy. The fix is more evaluations or a better-conditioned ansatz, not a different algorithm.

4.2.3 Even the worst variational errors are small next to the correlation energy. The $\Delta_{\text{max}}^{\text{HF}}$ column puts the errors in context. For N_2 , Hartree–Fock misses 777 mHa of correlation energy, while VQE misses 10.5 mHa on average, so the variational calculation still recovers about 98.6% of what mean-field theory left out. For F_2 the numbers are 359.7 mHa missed by Hartree–Fock against 0.059 mHa by VQE, a recovery of 99.98%. The method is doing real work even where it misses the chemical-accuracy threshold.

4.3 Campaign-wide reconstruction

Table 39 collects the out-of-sample reconstruction errors for every post-processed scan, and Table 40 the recovered equilibrium geometries; Section 4.1 already interpreted these numbers molecule by molecule, and Section 4.4 draws the conclusions that only make sense across all of them.

Two things worth restating here. First, H_2 and H_2O were scanned but never post-processed: the repository has no `h2_polynomial/`, `h2_spline/`, `h2o_polynomial/` or `h2o_spline/` directory, and correspondingly no interpolation figures for either. Their sections above present scan data only, and we said so there rather than quietly skipping the molecules. Second, the *Winner* column of Table 39 is decided on out-of-sample RMS alone; where the margin is narrow, the per-molecule discussion already said so.

4.4 Discussion

4.4.1 When does the variational solver succeed? Across 692 optimisations, accuracy tracks ansatz conditioning rather than molecular size. Table 38 splits cleanly into three groups.

Well-converged (14 of 23 scans). Small to moderate parameter counts with an optimiser that converges inside budget. H_2 , H_2O ,

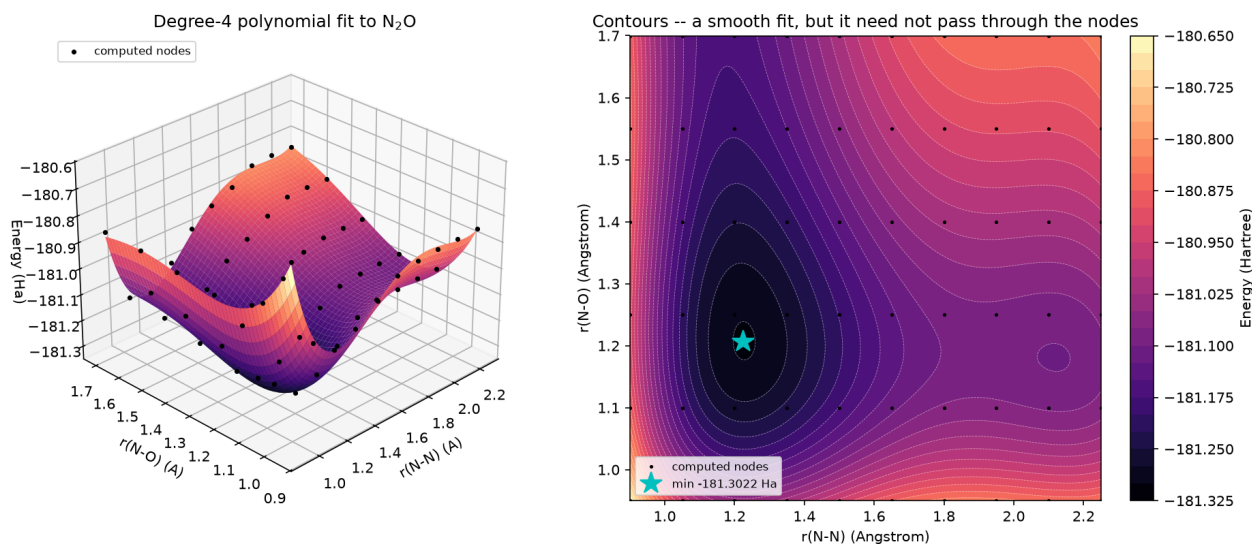


Figure 101: N₂O surface, total-degree polynomial reconstruction. Holdout RMS 5186.4 mHa, the campaign’s worst failure, 69× worse than bilinear interpolation.

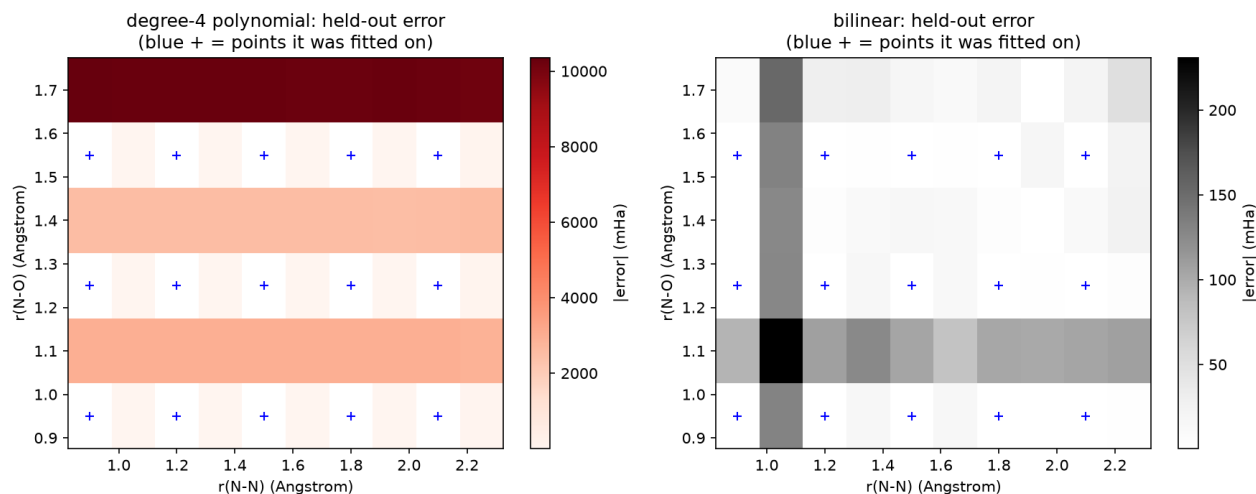


Figure 102: N₂O surface, polynomial holdout error. Note the scale.

LiH, F₂, the BeH₂ bend and all three NH₃ scans reach mean errors between 10^{-4} and 0.2 mHa, three to four orders of magnitude inside chemical accuracy. F₂ makes the point most sharply: ten qubits and ten active electrons, yet only 35 parameters, and a mean error of 0.059 mHa.

Optimiser-limited. N₂ ($p = 117$) and O₂ ($p = 68$) both saturate COBYLA’s 1000-evaluation ceiling at every geometry and land at 10.5 mHa mean error. Fewer than nine evaluations per parameter is not enough for a derivative-free method. The fix is more budget, a gradient-based optimiser (the parameter-shift rule of Section 2.1 exists for exactly this), or, as the O₂ anecdote in Section 2.1 shows, simply warm-starting from a neighbouring geometry. It is not a different algorithm.

Ansatz-limited. NO₂ has only 20 parameters and uses 831 evaluations, so it is not starved of budget; its 18.9 mHa bend error is a genuine limit of what that circuit can represent for an open-shell doublet. N₂O is limited the same way by a (4e, 4o) active space over 22 electrons, and the fact that its one scan with a larger active space, the N–O stretch, (6e, 5o), is also its most accurate, makes the diagnosis concrete.

The distinction matters because the two failure modes have opposite remedies. Adding evaluations helps an optimiser-limited run and does nothing for an ansatz-limited one; enlarging the active space helps the latter and makes the former worse.

4.4.2 When does each interpolant win? Counting the winners in Table 39: the global polynomial takes 9 of 18 scored scans, the spline

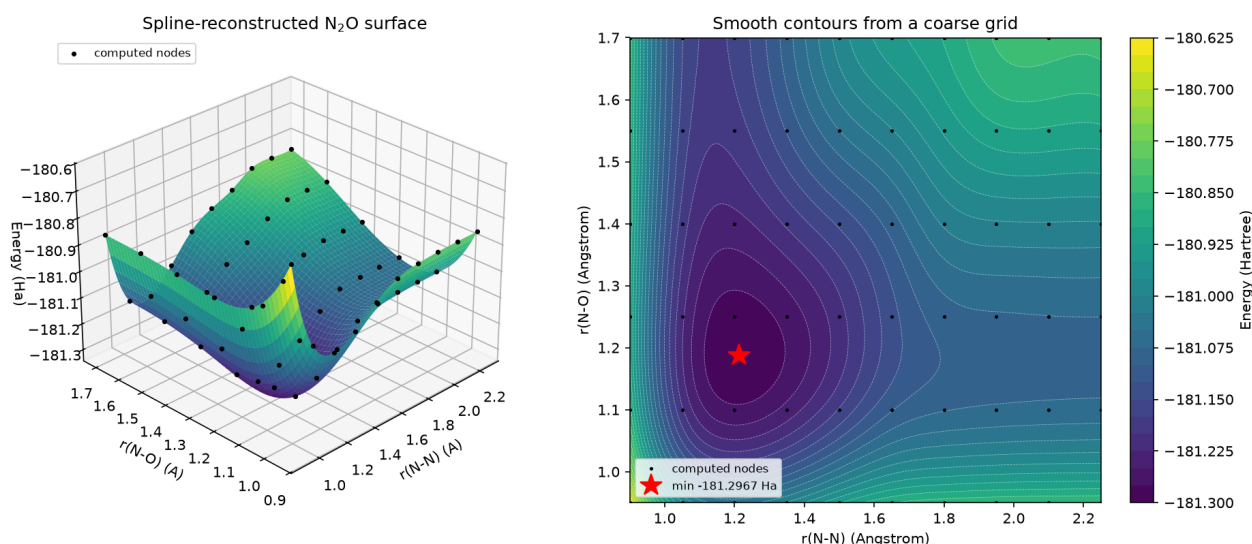


Figure 103: N_2O surface, bicubic spline reconstruction from the 5×3 half grid, the same data on which the polynomial fails.

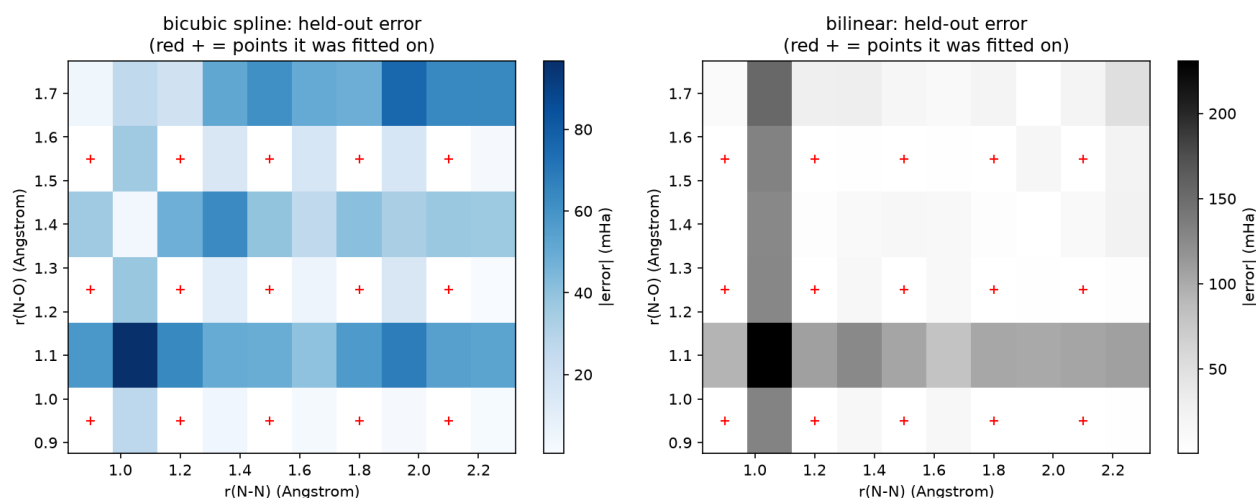


Figure 104: N_2O surface, spline holdout error, RMS 43.2 mHa against 74.7 mHa bilinear.

8, and piecewise-linear interpolation 1 (the NO_2 bend). No method dominates, and the exceptions are informative.

The polynomial wins on smooth, densely sampled 1-D curves. F_2 (0.022 mHa at 17 nodes), LiH (0.047 mHa at 13 nodes) and the N_2O N–O stretch (0.119 mHa at 15 nodes) are the three best reconstructions in the campaign, and all three are global polynomials. The common features are a single smooth minimum, no nearby complex singularity, and enough nodes to support a high degree. Where these hold, the polynomial wins by one to two orders of magnitude.

The spline wins where the curve is shaped awkwardly. All three H_2O_2 scans, the N_2O N–N stretch, the N_2O bend, and every surface but one go to the spline. Torsional potentials, flat wells and

functions with an endpoint minimum all defeat a global polynomial, and the spline’s locality handles them without incident.

The spline wins decisively on surfaces. Four of five surfaces go to the spline, and the fifth, NH_3 , is a statistical tie. The N_2O surface is the crucial data point: the total-degree polynomial reaches a holdout RMS of 5186 mHa, 69 $\times$ worse than doing nothing clever at all, while the bicubic spline on the same subgrid achieves 43.2 mHa. Downside risk of that size settles the default choice on its own.

Linear interpolation wins when the data are noisy. The NO_2 bend, carrying 18.9 mHa of variational error per node, is the one scan where the spline loses to the baseline. Spline superiority assumes the samples are accurate; when node error approaches the curvature being fitted, the C^2 constraint spreads noise rather than

Table 38: Accuracy of the variational solution against exact diagonalisation of the same active-space Hamiltonian, in milli-Hartree. $\bar{\Delta}$ and $\Delta_{\max}$ are the mean and maximum of $|E_{\text{VQE}} - E_{\text{exact}}|$ over the scan; $\Delta_{\max}^{\text{HF}}$ is the largest correlation energy the Hartree-Fock reference misses on the same points, i.e. the error the variational calculation is trying to recover. The chemical-accuracy threshold is 1.6 mHa.

Molecule	Coordinate	$\bar{\Delta}$	$\Delta_{\max}$	$\Delta_{\max}^{\text{HF}}$	$E_{\min}$ (Ha)	evals	Chem. acc.?
H ₂	bond length	0.0000	0.0003	277.58	-1.137117	–	yes
LiH	bond length	0.0001	0.0006	153.21	-7.880947	167	yes
BeH ₂	bend angle	0.0044	0.0062	5.90	-15.566211	–	yes
BeH ₂	symmetric stretch	0.5595	3.4778	68.08	-15.567233	–	yes
BeH ₂	2-D surface	0.4541	4.6537	68.08	-15.564426	–	yes
H ₂ O	bend angle	0.0007	0.0015	11.13	-74.971353	–	yes
NH ₃	bend angle	0.0158	0.0494	21.13	-55.473791	255	yes
NH ₃	symmetric stretch	0.1937	0.7605	105.97	-55.472133	348	yes
NH ₃	2-D surface	0.3343	4.5362	108.27	-55.470597	347	yes
N ₂	bond length	10.4839	36.3471	777.01	-107.642667	1000	no
O ₂	bond length	10.5491	28.5546	576.50	-147.735735	1000	no
F ₂	bond length	0.0589	0.6568	359.66	-196.041633	–	yes
H ₂ O ₂	bend angle	0.8882	2.9732	54.57	-148.800381	1000	yes
H ₂ O ₂	symmetric stretch	0.6311	2.0967	338.58	-148.792732	1000	yes
H ₂ O ₂	2-D surface	0.2830	0.9229	62.30	-148.811427	1000	yes
H ₂ O ₂	dihedral twist	0.2545	0.4539	38.61	-148.794034	1000	yes
NO ₂	bend angle	18.9302	38.5753	86.64	-201.301727	831	no
NO ₂	symmetric stretch	6.3435	30.3202	112.76	-201.350007	955	no
NO ₂	2-D surface	9.6097	42.0953	188.72	-201.304513	777	no
N ₂ O	bend angle	9.0698	19.8504	93.05	-181.278247	–	no
N ₂ O	N–N stretch	8.6878	23.5106	259.55	-181.297568	–	no
N ₂ O	N–O stretch	2.7185	12.1282	113.95	-181.281341	–	no
N ₂ O	2-D surface	6.9779	27.0626	265.91	-181.287840	–	no

suppressing it. That this happened exactly once, in exactly the scan with the largest variational error, is a reassuring consistency check.

4.4.3 What interpolation buys, quantitatively. Two concrete returns justify the post-processing layer.

Sub-grid geometry resolution. Table 40 shows equilibrium geometries recovered to 0.0007–0.09 Å from grids spaced at 0.075–0.25 Å. The best cases stand out: NH₃’s stretch minimum lands within 0.0007 Å of the model reference from a 0.075 Å grid, a hundredfold gain; LiH reproduces its model bond length to four decimals from a 0.25 Å grid. Since the piecewise-linear minimum is always a grid node by construction, none of these gains would exist without a smooth reconstruction.

Avoiding quadratic cost on surfaces. The half-grid holdout answers a question with a direct budget attached. On BeH₂, fitting a bicubic spline to one quarter of the 110-point surface reconstructs the remaining three quarters at 3.10 mHa RMS. On NO₂, 38.3 mHa; on N₂O, 43.2 mHa; on NH₃, 8.28 mHa. At two to four minutes per variational solve, running quarter grids would have saved several hours per surface.

The caveat is the floor. On NH₃, halving a 7 × 7 grid leaves 4 × 4, the bare minimum for a bicubic spline, and the holdout error of 8.28 mHa sits well outside chemical accuracy. Below roughly five points per axis there is not enough information left, whatever method is used. The practical rule that follows: sample at least ten points per axis if the surface is going to be halved, and use a spline.

4.4.4 On reporting the LiH result. Section 4.1.2 documents an experiment written to demonstrate Runge’s phenomenon that failed to demonstrate it. We report this as the headline finding for that molecule rather than bury it, for three reasons.

First, it is true, and the measurement is unambiguous: leave-one-out RMS falls monotonically to the highest testable degree, and the full interpolant has exactly one interior minimum.

Second, the reason is more instructive than the expected result would have been. Runge’s phenomenon is often taught as a consequence of high degree on equally spaced nodes, which oversimplifies it: it is a consequence of the interpolated function having a singularity near the real interval. A numerical-analysis course is better served by a worked example that exposes that distinction than by one more copy of the standard picture.

Third, the campaign has plenty of genuine Runge failures to balance it: the BeH₂ stretch, where the degree-15 interpolant invents a minimum at the interval edge and misplaces the bond length by 0.40 Å; the H₂O₂ stretch, misplacing it by 1.52 Å; the NO₂ stretch, by 1.18 Å; and the N₂O surface, failing by four orders of magnitude. The phenomenon is real. It just is not automatic, and only measuring out of sample tells the two cases apart.

4.4.5 Supplementary: execution on IBM Quantum hardware. A separate Qiskit implementation queues the same LiH calculation on real IBM Quantum hardware, using PySCFDriver, a UCCSD ansatz mapped with a ParityMapper, and SLSQP. The optimisation loop itself always runs against a free, exact statevector estimator; only

Table 39: Out-of-sample reconstruction error for every scan that was post-processed, in milli-Hartree. One-dimensional scans are scored by leave-one-out at the interior nodes; surfaces by the half-grid holdout of Section 2.2.5. *Linear* is the piecewise-linear baseline implied by every raw scan plot. A dash means the method was not applied to that scan; *n/a* means the full interpolating degree left too few points to score.

Molecule	Coordinate	Linear baseline		best d	Global polynomial		Cubic spline		Winner
		RMS	max		RMS	worst RMS	RMS	max	
LiH	bond length	13.373	39.643	11	0.047	31.64	1.815	5.147	polynomial
BeH ₂	symmetric stretch	10.119	25.310	7	4.496	572.51	4.647	10.984	polynomial
BeH ₂	bend angle	0.344	0.561	9	0.000	7.59	0.001	0.002	polynomial
BeH ₂	2-D surface	17.969	48.857	–	3.966	–	3.097	10.052	spline
NH ₃	symmetric stretch	12.787	29.486	6	0.466	53.42	0.920	1.608	polynomial
NH ₃	bend angle	0.653	0.792	4	0.060	3.36	0.067	0.140	polynomial
NH ₃	2-D surface	30.108	65.067	6	8.493	74.48	8.284	14.359	spline
N ₂	bond length	134.178	341.056	6	20.341	207.49	37.592	72.518	polynomial
O ₂	bond length	80.541	202.904	6	7.987	130.17	22.002	43.203	polynomial
F ₂	bond length	31.283	100.088	13	0.022	122.14	2.420	7.959	polynomial
H ₂ O ₂	symmetric stretch	14.617	51.725	9	1.335	1446.83	1.030	3.538	spline
H ₂ O ₂	bend angle	2.440	4.319	5	1.252	20.98	0.776	1.487	spline
H ₂ O ₂	dihedral twist	1.421	3.720	10	2.044	858.54	1.365	3.044	spline
NO ₂	symmetric stretch	54.194	168.036	9	5.233	1071.75	8.469	21.643	polynomial
NO ₂	bend angle	5.049	14.295	9	3.924	2170.23	7.258	22.116	polynomial
NO ₂	2-D surface	112.955	346.984	–	46.230	–	38.278	108.663	spline
N ₂ O	N–N stretch	25.902	75.096	7	3.812	78.78	2.525	6.210	spline
N ₂ O	N–O stretch	26.824	81.818	12	0.119	69.17	1.741	5.193	polynomial
N ₂ O	bend angle	3.502	9.629	8	2.627	1959.14	1.895	3.988	spline
N ₂ O	2-D surface	74.718	231.025	–	5186.448	–	43.208	96.800	spline

Table 40: Equilibrium geometries recovered from the reconstructed curves and surfaces, compared with the best node actually computed and with the reference value (experimental where quoted in the scan metadata, otherwise the model’s own optimised geometry). Lengths in Å, angles in degrees. The gap between the *best node* and the *interpolated* column is the resolution the interpolant buys over the raw grid.

Molecule	Coordinate	Method	Best node	Interpolated	Reference	[Interp.–Ref.]	Molecule	Coordinate	Method	Best node	Interpolated	Reference	[Interp.–Ref.]
LiH	bond length	polynomial	1.65	1.5474	1.5949	0.0475	H ₂ O ₂	bend angle	polynomial	95	93.6465	101.9	8.2535
LiH	bond length	spline	1.65	1.5496	1.5949	0.0453	H ₂ O ₂	bend angle	spline	95	93.938	101.9	7.962
BeH ₂	bend angle	polynomial	180	180	180	0	H ₂ O ₂	symmetric stretch	polynomial	1.5	2.9748	1.458	1.5168
BeH ₂	bend angle	spline	180	180	180	0	H ₂ O ₂	symmetric stretch	spline	1.5	1.4671	1.458	0.0091
BeH ₂	symmetric stretch	polynomial	1.3	0.9305	1.3264	0.3959	H ₂ O ₂	dihedral twist	polynomial	150	167.0633	111.5	55.5633
BeH ₂	symmetric stretch	spline	1.3	1.289	1.3264	0.0374	H ₂ O ₂	dihedral twist	spline	150	134.278	111.5	22.778
NH ₃	bend angle	polynomial	105	104.3755	104.272	0.1035	NO ₂	bend angle	polynomial	145	173.4323	134.3	39.1323
NH ₃	bend angle	spline	105	104.3845	104.272	0.1125	NO ₂	bend angle	spline	145	142.269	134.3	7.969
NH ₃	symmetric stretch	polynomial	1.075	1.0391	1.0401	0.001	NO ₂	symmetric stretch	polynomial	1.3	2.3728	1.194	1.1788
NH ₃	symmetric stretch	spline	1.075	1.0394	1.0401	0.0007	NO ₂	symmetric stretch	spline	1.3	1.2649	1.194	0.0709
N ₂	bond length	polynomial	1.2	1.191	1.098	0.093	N ₂ O	bend angle	polynomial	180	173.4797	180	6.5203
N ₂	bond length	spline	1.2	1.1685	1.098	0.0705	N ₂ O	bend angle	spline	180	180	180	0
O ₂	bond length	polynomial	1.3	1.3028	1.208	0.0948	N ₂ O	N–N stretch	polynomial	1.2	1.2132	1.128	0.0852
O ₂	bond length	spline	1.3	1.2754	1.208	0.0674	N ₂ O	N–N stretch	spline	1.2	1.2157	1.128	0.0877
F ₂	bond length	polynomial	1.4	1.3932	1.412	0.0188	N ₂ O	N–O stretch	polynomial	1.2	1.1924	1.185	0.0074
F ₂	bond length	spline	1.4	1.3931	1.412	0.0189	N ₂ O	N–O stretch	spline	1.2	1.1927	1.185	0.0077

the already-optimised circuit for each bond length is sent to real hardware for one final energy evaluation. Reproducing this part of the project needs an IBM Quantum account, whose credentials (a Cloud Resource Name and an API key) go in a `.env` file copied from `.env.example`.

Running real hardware costs real quota, and the script is written around that constraint: it picks whichever backend service `least_busy` reports as least busy, submits every bond length as a single batched job rather than one job per point, asks for explicit confirmation before submitting, and fixes the shot

count at 2000 rather than leaving it at the service default. The shot-count choice has a documented reason: an earlier run left the shot count unset and burned about four minutes of a ten-minute account quota on a single seven-point job before it had to be cancelled. Error mitigation is switched off (`resilience_level = 0`) to keep runtime and cost down. That last choice matters for reading Table 41: the huge hardware error below is device noise measured with mitigation deliberately turned off to fit inside a small quota, not a measurement of the hardware at its best.

Table 41: LiH on an exact statevector simulator versus IBM Quantum hardware, from VQE demo on IBM Quantum Computer/. Errors are $|E_{\text{VQE}} - E_{\text{exact}}|$ in mHa. The simulator is accurate to sub-micro-Hartree; the hardware, run with error mitigation off to control cost, is off by more than two Hartree.

r (Å)	E_{exact}	E_{sim}	δ_{sim}	δ_{hw}
0.50	-7.047910	-7.047910	0.000287	2143.85
0.75	-7.574579	-7.574579	0.000110	2127.61
1.00	-7.784021	-7.784021	0.000028	—
1.25	-7.861612	-7.861612	0.000043	—
1.50	-7.882140	-7.882140	0.000105	—
1.75	-7.876947	-7.876947	0.000075	—
2.00	-7.860828	-7.860828	0.000049	—

Only two of the seven bond lengths finished on hardware; the rest were left out to stay inside the account’s quota. The simulator reproduces exact diagonalisation to better than 3×10^{-4} mHa at every point, four orders of magnitude inside chemical accuracy. The two completed hardware runs are off by about 2.14 Ha, roughly $1340\times$ the chemical-accuracy threshold, and larger than the entire correlation energy being sought. Gate errors, readout errors, decoherence, and the decision to skip error mitigation to save quota, together overwhelm the quantity of interest.

This is not a criticism of the hardware, it is a statement of where the technology stands today with mitigation off and a tight quota, and it justifies the architectural choice made throughout the rest of this project: the physics and the numerical analysis are both better served by exact classical simulation of the quantum circuits. Every result in Sections 3.2 and 4.1 is a statement about the VQE *algorithm*, not about a particular noisy device.

4.4.6 Comparison with the base paper. The base paper [13] established that a variational loop of short quantum circuits and a classical optimiser can recover a molecular dissociation curve, demonstrated on one two-qubit molecule. Our campaign confirms its two load-bearing claims at a much larger scale. The variational bound it relies on held in all 692 optimisations, and the algorithm reaches chemical accuracy on 14 of 23 scans with up to ten qubits and 117 parameters. Where it falls short, the cause is the classical half of the loop the base paper introduced, the optimiser and the ansatz, not the variational principle itself (Section 4.4). Our supplementary hardware run (Section 4.4.5) also echoes the base paper’s motivation from the other side: on present-day hardware, without error mitigation, device noise is larger than the entire correlation energy being sought. Finally, where the base paper reports energies only at the sampled bond lengths, our reconstruction results show that the curve between those samples can be recovered to sub-grid accuracy, provided the reconstruction method is chosen by out-of-sample measurement rather than assumed.

5 Conclusion and Future Work

5.1 Key takeaways

This project treated building molecular potential energy surfaces as what it is from a numerical-analysis standpoint: reconstructing a smooth function from a small number of very expensive samples. We implemented the Variational Quantum Eigensolver over

CUDA-Q and PySCF, ran 23 scans across 11 molecules totalling 692 variational optimisations, checked every energy against exact diagonalisation of the identical active-space Hamiltonian, and reconstructed each scan by three interpolation methods scored strictly out of sample.

The main findings are these.

- (1) **VQE accuracy tracks ansatz conditioning, not molecular size.** Fourteen of 23 scans reach chemical accuracy, several by three to four orders of magnitude. F_2 , with ten active electrons and only 35 parameters, outperforms N_2 , with six active electrons and 117. The failures split cleanly into optimiser-limited and ansatz-limited cases with opposite remedies.
- (2) **The variational bound held in all 692 optimisations**, the strongest evidence available that the Hamiltonian construction and active-space bookkeeping are correct throughout.
- (3) **Splines are the safer default; polynomials are the better gamble when the curve allows it.** The polynomial takes 9 of 18 scored scans and produces the three best reconstructions in the campaign (0.022, 0.047 and 0.119 mHa). But it also produces every catastrophic failure, including a surface reconstruction $69\times$ worse than doing nothing. The spline never fails badly. On an expensive surface, that asymmetry is worth more than the polynomial’s occasional win.
- (4) **Interpolation buys roughly an order of magnitude of effective grid resolution for no extra quantum cost.** Equilibrium geometries were located to $0.0007\text{--}0.09\text{\AA}$ from grids spaced at $0.075\text{--}0.25\text{\AA}$, and the half-grid holdout shows that quarter-cost surfaces reconstruct to $3\text{--}43$ mHa RMS, provided at least about ten points per axis are sampled.
- (5) **Runge’s phenomenon is a property of the function, not of the degree.** On LiH the global polynomial outperformed the cubic spline by $38\times$ with no oscillation at any testable degree, and we found the node density at which the expected failure does appear. The campaign still contains four clear Runge failures, so the phenomenon is real, it is simply not automatic, and only out-of-sample measurement tells the two cases apart.

The idea that ties these together, and the one we would most want carried forward, is that in-sample residuals carry no information for an interpolant. Our polynomial fits reach residuals as low as 1.4×10^{-11} mHa while being wrong by four orders of magnitude between nodes. Only leave-one-out cross-validation and the half-grid holdout tell a reconstruction that works apart from one that merely fits.

5.2 Limitations of the current approach

We state the limitations of this work plainly.

Minimal basis throughout. Every calculation uses STO-3G. Absolute energies are far from the basis-set limit, and equilibrium geometries carry model errors of up to $0.09\text{\AA}$ and several degrees (Table 40). No conclusion about *chemistry* should be drawn from these numbers; the conclusions here are about *numerical methods*, for which internal consistency against exact diagonalisation is what matters.

Small active spaces. Active spaces range from (2e, 5o) to (10e, 6o). For N_2O , (4e, 4o) correlates four electrons out of twenty-two, inadequate by any chemical standard, and that shows up in the results.

Optimiser budgets not converged for the largest ansätze. N_2 , O_2 , NO_2 and N_2O did not reach chemical accuracy. For the first two the cause is a 1000-evaluation ceiling, and those scans should be rerun with a larger budget, or with warm-starting, before their numbers are quoted as converged results.

Coverage gaps. H_2 and H_2O were scanned but never post-processed, so the campaign’s densest scan and its canonical example molecule contribute no interpolation evidence. The H_2O_2 surface was computed but not reconstructed. The H_2O symmetric stretch is implemented in the driver and documented in the project README, but no saved scan exists in the repository. These are gaps in the campaign, not choices we can justify after the fact.

Half-grid holdout is a proxy. The 2-D validation withholds three quarters of the points from a rectangular subgrid. That is the right question for cost planning, but it is not leave-one-out, and its error estimates run pessimistic relative to what a well-chosen adaptive sampling scheme would achieve.

No treatment of noise or error mitigation in the main campaign. The main campaign simulates circuits exactly. Shot noise, gate error, and the error-mitigation techniques that would matter on hardware sit outside its scope, as Section 4.4.5 makes clear, and the one hardware run we do report was taken with error mitigation switched off to control cost.

Reference values are heterogeneous. Table 40 compares some molecules against experimental values and others against the model’s own optimised geometry, depending on what the scan metadata recorded. The column is labelled accordingly, but cross-molecule comparison of the final column should keep that in mind.

5.3 Future work

The clearest next steps: rerun N_2 , O_2 and NO_2 with gradient-based optimisation and the parameter-shift rule, to separate optimiser error from ansatz error definitively; close the coverage gaps by post-processing H_2 , H_2O and the H_2O_2 surface; replace the uniform grids with Chebyshev-distributed nodes, which would remove the edge-oscillation failures behind every bad polynomial geometry in Table 40; and extend the half-grid study into adaptive sampling, placing new expensive points where the current reconstruction is least certain instead of on a fixed lattice.

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6 Code and Installation

CSE 402: Numerical Analysis, Simulation and Modeling • Group 09 •
Section A • Supervisor: Ishrat Jahan

 <https://github.com/QuitTTT/CSE-402-Numerical-Analysis-Simulation-and-Modeling-Project>

The complete implementation is in a public GitHub repository [18]. It contains the VQE core, every per-molecule experiment driver, the numerical post-processing scripts, the saved scan files from which every table and figure in this report was generated, the supplementary IBM Quantum script and the LaTeX source of this report. The code is Python and builds on CUDA-Q Solvers [12], PySCF [19], Qiskit [15], NumPy [4], SciPy [23] and Matplotlib [6].

Installation. Clone the repository, create a virtual environment (Python 3.10 or newer) and install the pinned requirements, as in Listing 5.

```
1 git clone https://github.com/QuitTTT/CSE-402-Numerical-Analysis-
  ↳ Simulation-and-Modeling-Project.git
2 cd CSE-402-Numerical-Analysis-Simulation-and-Modeling-Project
3 python -m venv .venv
4 source .venv/bin/activate # Windows: .venv\Scripts\activate
5 pip install -r requirements.txt
```

Listing 5: Installation.

The per-molecule experiment drivers need only Qiskit, Qiskit Nature and PySCF. `cudaq-solvers`, used by the `src/` core and the demo scripts, is declared in `requirements.txt` as a Linux/WSL-only dependency (`sys_platform != "win32"`), so those parts need WSL or a Linux machine, while the post-processing scripts run anywhere Python and SciPy do. The supplementary IBM Quantum run additionally needs an IBM Quantum account, whose credentials go in a `.env` file copied from `.env.example` (Section 4.4.5).

6.1 Reproduction

Every figure and table in this report can be regenerated without re-running any quantum simulation, because every scan is saved with its metadata. Note that `cudaq-solvers` is a Linux/WSL-only dependency, so the VQE scans themselves need a Linux environment even though the post-processing scripts run anywhere Python and SciPy do.

```
1 # environment
2 python -m venv .venv && source .venv/bin/activate
3 pip install -r requirements.txt
4
5 # --- re-run a scan from scratch (expensive) ---
6 python experiment/h2/h2_ground_state_estimation.py \
7     --start 0.3 --stop 2.5 --step 0.1
8 python experiment/h2o/h2o_ground_state_estimation.py \
9     --coordinate bend --start 70 --stop 160 --step 5
10
11 # --- reopen a saved scan interactively (seconds) ---
12 python experiment/beh2/beh2_ground_state_estimation.py \
13     --load beh2_surface_scan.json
14 python experiment/h2o/h2o_ground_state_estimation.py \
15     --load h2o_bend_scan.json
16
17 # --- numerical post-processing (runs no chemistry) ---
18 python experiment/beh2_spline/beh2_spline_interpolation.py
19 python experiment/lih_polynomial/lih_polynomial_interpolation.py
20
21 # --- reproduce the LiH Runge finding ---
22 python experiment/lih/lih_ground_state_estimation.py \
23     --step 0.125 --save lih_fine
24 python experiment/lih_polynomial/lih_polynomial_interpolation.py \
25     --load lih_fine.json
26
27 # --- quick tour of every ansatz, optimiser and visualisation
28 # backend, without running a full molecular scan ---
29 python examples/molecule_demo.py
```

```

30 python examples/vqe_demo.py --quantum
31
32 # --- regenerate this report's tables from the saved scans ---
33 python CSE402_report_acm/gen_tables.py

```

Listing 6: Reproducing the campaign.

6.2 Repository structure

```

1 src/
2 molecule.py # CUDA-Q / PySCF wrapper + 3D visualisation (1887 lines)
3 vqe.py # ansatze, optimisers, parameter-shift rule (740 lines)
4 examples/
5 molecule_demo.py, vqe_demo.py # runnable tour of src/
6 out/
7 *.html # visualisation backends exercised by molecule_demo.py
8 experiment/
9 scan_io.py # shared save/load contract, format_version guard
10 <mol>/ # ground-state estimation driver + saved scans
11 <mol>_polynomial/ # global polynomial reconstruction + L00
12 <mol>_spline/ # cubic/bicubic spline reconstruction + L00
13 NEW_MOLECULE_PROMPT.md
14 # the project's own notes for adding a new molecule: DOF-counting,
15 # reusable code patterns, and a list of traps found the hard way
16 VQE demo on IBM Quantum Computer/
17 qiskit_vqe.py, lih_energies.csv, lih_energies_simulator.csv
18 Presentation slides/
19 Molecular Simulation - Proposal Presentation.pptx, Project Plan.pdf
20 CSE402_report_acm/
21 report.tex, refs.bib, gen_tables.py, figures/, tables/ # this report

```

Listing 7: Layout of the project repository.

Molecules with an experiment/ directory: H_2 , H_2O , LiH , BeH_2 , N_2 , O_2 , F_2 , NH_3 , H_2O_2 , NO_2 , N_2O . Of these, all but H_2 and H_2O additionally have `_polynomial` and `_spline` post-processing directories.

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